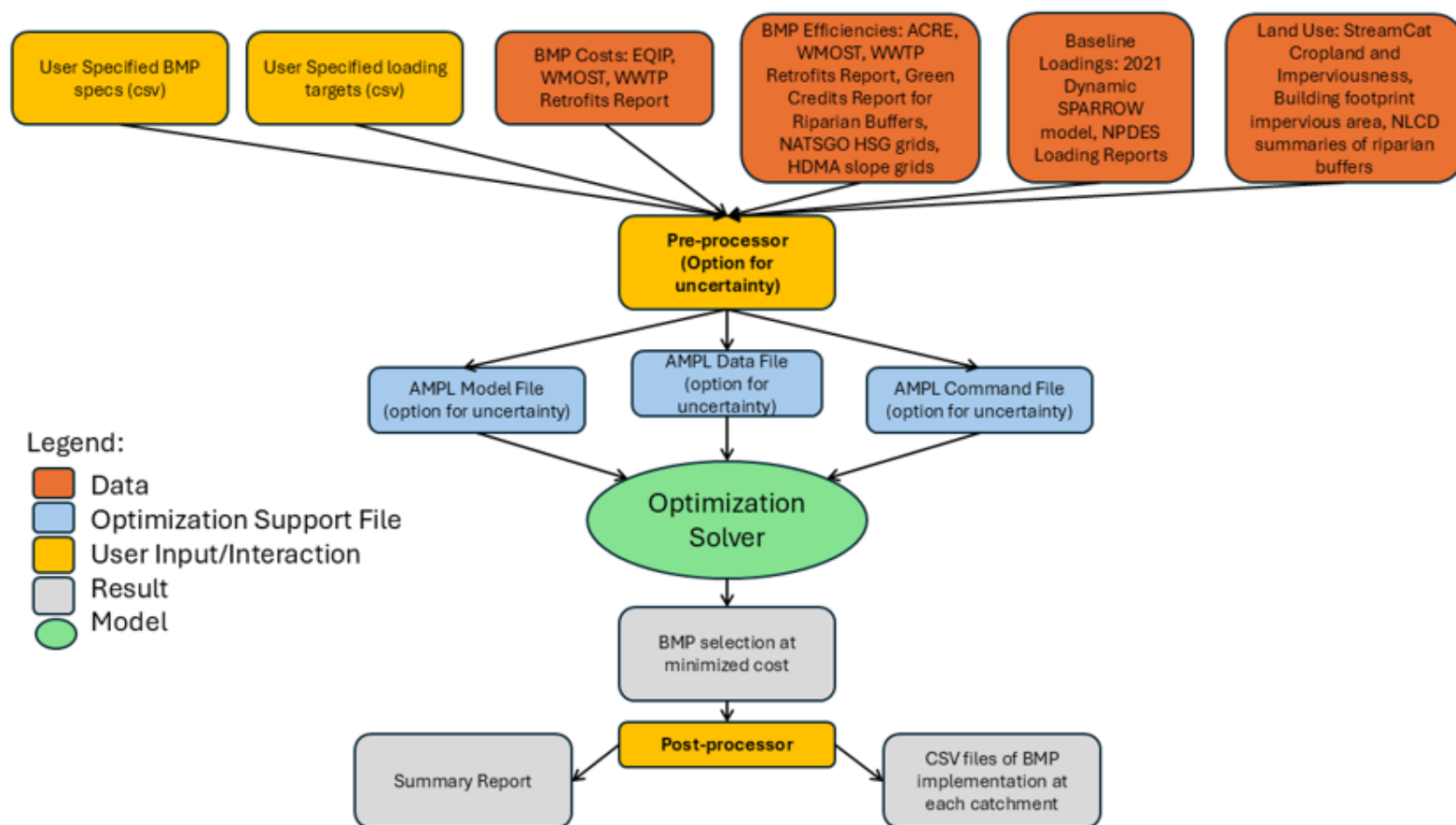


Long Island Sound Dynamic River Basin Export Reduction Optimization Support Tool (RBEROST) User Guide



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by
Naomi E Detenbeck
Craig Connolly
Coastal Science Solutions Division
Narragansett, RI 02882

Alyssa Le
Colin Guider
Sam Ennett
Kelly-Anne Moffa
Hunter Parker
Samuel Kovach

ICF
Waltham, MA 02451

Contract: 68HERC24A0011/CO 68HERC25F0073

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Acknowledgments

This report is the result of the support of numerous individuals whose contribution is briefly described and acknowledged below.

Alternate Contracts Office Representative (altCOR)

Steven Rego (EPA/OASES, Narragansett, RI)

Contributors to previous versions of RBEROST

Marilyn ten Brink (former EPA/ORD, former altCOR)

Catherine Chamberlin (former EPA ORISE, current US Geological Survey, Northborough, MA)

Kate Munson (ICF, Waltham, MA)

Isabelle Morin (ICF, Waltham, MA)

Miranda Marks (ICF, Waltham, MA)

Yishen Li (ICF, Waltham, MA)

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Abbreviations

ACSB	Atlantic Coastal Science Branch
ACRE	Agricultural Conservation Reduction Estimator
AMPL	A Mathematical Programming Language
API	Application programming interface
BMP	Best Management Practice
BNR	Biological Nutrient Removal
BUFF	Buffer
CEMM	Center for Environmental Measurement and Modeling
CEQ	Council on Environmental Quality
COMID	Common Identifier
CPI	Consumer Price Index
CPLEX	Optimizer based on the simplex method as implemented in the C programming language
CSSD	Coastal Science Solutions Division
CSV	Comma Separated Values
CTRL	control
DEC	Department of Environmental Conservation
DEL_FRAC	Delivery Fraction
EPA	Environmental Protection Agency
EQIP	Environmental Quality Incentives Program
EROS	Earth Resources Observation and Science
FRAC	fraction
HDMA	Hydrologic Derivatives for Modeling and Analysis
HRU	Hydrologic Response Unit
HSG	Hydrologic Soil Group
HUC	Hydrologic Unit Code
HUC8	8 digit HUC
HUC10	10 digit HUC
HUC12	12 digit HUC
IBM	International Business Machine
ID	Identification
ISPRS	International Society for Photogrammetry and Remote Sensing
ISR	Internal Storage Reservoir
JAWRA	Journal of the American Water Resources Association
KM	Kilometer

KM2	Square Kilometers
LFS	Large File Storage
LIS	Long Island Sound
LOESS	Locally Estimated Scatterplot Smoothing
MA	Massachusetts
MIN	minimum
MS4	Municipal Separate Storm Sewer System
N	Nitrogen
NATSGO	National Soil Survey Geographic Database
NEIWPCC	New England Interstate Water Pollution Control Commission
NEOS	Network
NH	New Hampshire
NHD	National Hydrography Dataset
NHDPlus	National Hydrography Dataset Plus
NLCD	National Land Cover Dataset
NPDES	National Pollutant Discharge Elimination System
NRCS	Natural Resource Conservation Service
NYDEC	New York State Department of Environmental Conservation
NYS	New York State
OASES	Office of Applied Sciences and Environmental Solutions
ORISE	Oak Ridge Institute for Science and Education
I/A OWTS	Innovative/Alternative Onsite Wastewater Treatment System
P	Phosphorus
PI	Principal Investigator
PPI	Producer Price Index
RBEROST	River Basin Export Reduction Optimization Support Tool
Rshiny	R package providing graphical user interface for inputs and outputs
RMD	R Markdown
Se	Standard error
SPARROW	Spatially Referenced Regressions On Watershed attributes
SPDES	State Pollutant Discharge Elimination System
SSWR	Safe and Sustainable Water Research Program
SWP	Subwatersheds Wastewater Plan
TDEP	Total Mass Deposition
TMDL	Total Maximum Daily Load
TN	Total Nitrogen

TP	Total Phosphorus
URL	Uniform Resource Locator
US	United States
USDA	U.S. Department of Agriculture
USGS	U.S. Geological Survey
UVM	University of Vermont
VT	Vermont
WBD	National Watershed Boundary Dataset
WEDB	Watershed and Estuarine Diagnostics Section
WMOST	Watershed Management Optimization Support Tool
WQv	Water Quality Volume
WWTP	Wastewater Treatment Plant
XML	Extensible Markup Language

1 Introduction

The River Basin Export Reduction Optimization Support Tool (RBEROST) is a decision support tool designed to support integrated, regional, watershed planning. RBEROST allows users to screen which locations and choices of Best Management Practices (BMPs) will meet annual or seasonal loading targets for the least financial cost. There are three main steps within RBEROST including a preprocessing step, an interaction with an online solver and a postprocessing step (Figure 1-1).

1. The preprocessing step combines medium-resolution National Hydrography Dataset Plus (NHDPlus v2; McKay et al. (2012)) reach lengths, NHDPlus v2 catchment-level seasonal nutrient loading, land use data, hydrologic soil group data, user-defined loading targets, user-defined agricultural, urban, point source, septic and riparian buffer BMPs, and data on BMP-specific costs and nutrient removal efficiencies incorporated in RBEROST input files. The preprocessing step then uses this information to write three program files in A Mathematical Programming Language (AMPL), including a model, data and command file (Figure 1-1). There are two options available for the preprocessing step (i.e., creating AMPL files with or without uncertainty information included). The AMPL files describe a model that defines a cost-minimization optimization problem subject to meeting downstream annual loading targets.
2. These AMPL files are then sent to a free online CPLEX server hosted by the Network-Enabled Optimization System (NEOS) (Czyzyk et al., 1998; Dolan, 2001; Gropp and Moré, 1997). CPLEX is a linear solver that will solve the optimization problem defined by the user inputs of loading targets and selected BMPs for the least cost. Once the solution is optimized, the model outputs its decisions of which BMPs to implement and where to implement them.
3. The final step of RBEROST is the postprocessing step (Figure 1-1). This step parses the output from NEOS into a summary report describing which BMPs were implemented as well as the total cost. It also provides .csv data files for download that describe the BMP choices at the NHDPlus v2 catchment-level scale. If the user chooses to run the model with uncertainty, the postprocessor will provide information on the expected cost range and the likelihood of meeting each loading target. When run with uncertainty, the model displays multiple scenarios with increasing cost and likelihood for meeting targets. Figure 1-1 shows a diagram of the model framework.

Additional work may be necessary before beginning RBEROST to collect and format the necessary data. This version of RBEROST was built for use in Long Island Sound (this model implementation is referred to herein as “The Long Island Sound Dynamic model”); however, use of RBEROST isn’t limited to this area. Again, additional work may be necessary to extend RBEROST to other study areas.

The Long Island Sound Dynamic model is built upon the framework of the earlier Upper Connecticut RBEROST model but has been updated to account for seasonal changes in nutrient loading using output from the Long Island Sound dynamic SPARROW model (Hayes et al., 2026) for North Shore watersheds and the Vaudrey et al. (2017) Long Island Sound Nitrogen Loading Model (with inputs updated by US EPA) for South Shore watersheds. In contrast, the earlier Upper Connecticut RBEROST model used estimates of long-term average annual loading centered on 2012 conditions from the Northeast SPARROW model (Ator 2019) as inputs. . Inclusion of seasonal and annual variability in source load

estimates in the dynamic version allows users to evaluate the effect of weather variability on the choice of least-cost management strategies and the probability of achieving nutrient load targets. The underlying dynamic SPARROW model also incorporates a new storage loading term that reflects the contribution of legacy nutrients stored in the system that can introduce lags in the system response to both inputs and effect of management practices (Van Meter et al., 2018, Ilampooranan et al., 2019, Schmadel et al. 2024).

The watersheds incorporated into the model extend into the states of New York, Connecticut, Massachusetts, Vermont and New Hampshire. The tool is designed to help reduce nutrient loading to targeted waterbodies for the least financial cost. This tool optimizes costs for meeting targets for nutrient export at the annual or seasonal scale and is designed to be used as a screening tool for large watersheds (e.g., HUC 2-HUC 8 scale). The tool is mathematically similar to the Watershed Management Optimization Support Tool, or WMOST (Detenbeck et al., 2018a; Detenbeck et al., 2018b). WMOST is built for optimization on the HUC10 or HUC12 scale and can be run on daily or monthly timesteps. In contrast, RBEROST was developed as a regional screening tool to overcome computational challenges with running WMOST at larger spatial watershed scales. All RBEROST scripts were initially written in R versions 4.3.1 and 4.3.2 and updated under R version 4.4. with R package versions enforced with the R package renv (Ushey and Wickham 2026). RBEROST may display errors with future R versions beyond 4.4. Execution of the RBEROST application is recommended with RStudio version 2025.05.1+ Build 513 and may display incompatibilities with other versions. The R code and instructions on how to execute the code necessary for these installations are included in the documentation below.

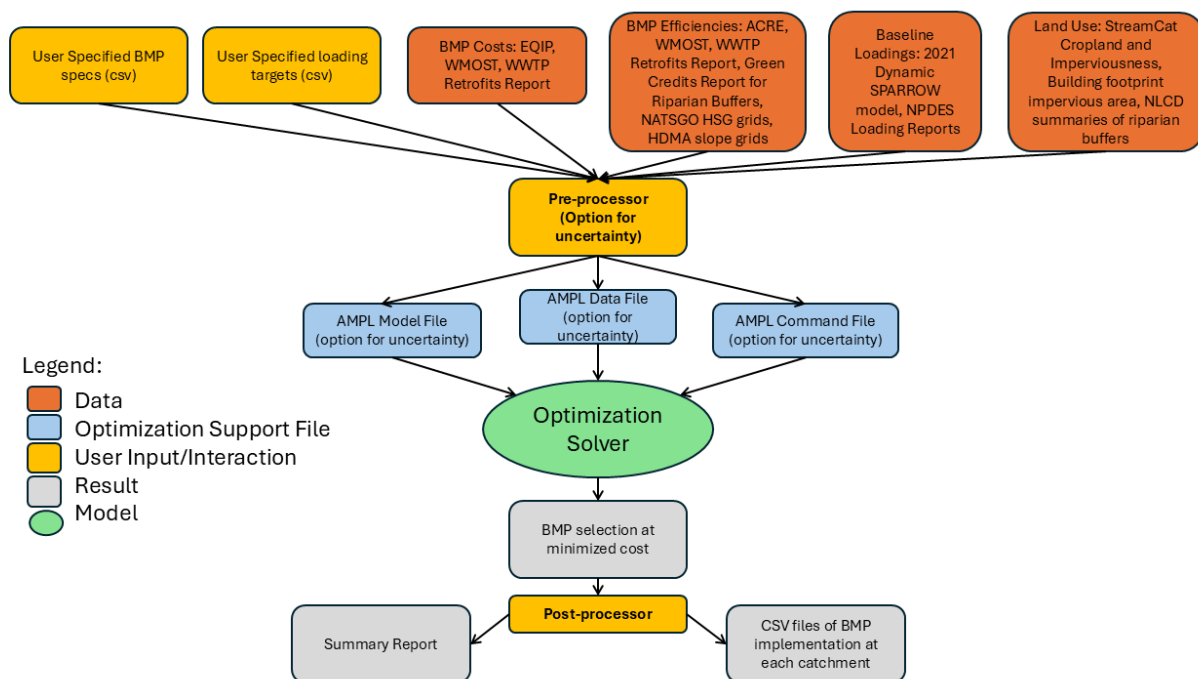


Figure 1-1. Schematic overview of RBEROST, including input data, support files, user determined inputs, result files and other model components. The preprocessing R script incorporates data from compiled data sets of nutrient loads, land-use, BMP costs and efficiencies, point source loads, user-selected BMP options and user-specified loading targets to create AMPL input files for upload to the NEOS server. The CPLEX optimization solver produces output which describes the least cost management solutions, which are formatted by a postprocessing R script to create a summary report and downloaded .csv files of BMP.

2 Model Framework

2.1 Optimization Method

An optimization problem includes the following components: an objective (the variable to be maximized or minimized), constraints on the solution and decision variables. RBEROST is designed to minimize the total annualized cost of management practices. Constraints on the solution include target loads for N and/or P that must be achieved and the total urban or agricultural area, unbuffered stream/river length and number of parcels with septic systems that can be treated with management actions. Users can specify additional constraints by setting upper and/or lower limits on the fraction of areas that can be treated by each BMP. Finally, decision variables describe parameters that are adjusted during the optimization process to minimize costs while meeting loading and other constraints (e.g., the type, placement and magnitude of various management practices). Together these features and relationships are described as a mathematical problem coded in the AMPL language which is designed to describe an optimization problem.

RBEROST uses the IBM ILOG CPLEX Optimizer for linear programming that solves a mathematical problem written in AMPL to minimize the total annualized cost of selected BMPs. RBEROST interacts with this solver through the NEOS server, hosted by the University of Wisconsin-Madison. The CPLEX Optimizer was chosen because it accepts XML calls so that RBEROST can interact with it directly, but users also have the option of manually interacting with the server through the webpage by submitting input files through the webpage at <https://neos-server.org/neos/solvers/lp:CPLEX/AMPL.html>. (See Appendix B).

2.2 Optimization Variables and Constraints

RBEROST includes five categories of management practices, all of which are listed in the **01_UserSpecs_BMPs.csv** file.:

1. **Point Source Controls.** Point source BMP options include water treatment plant low-cost upgrades to facilitate nutrient removal in wastewater effluent (JJ Environmental 2015) as well as the addition of Biological Nutrient Removal (BNR) to all wastewater treatment plants at which BNR has not yet been implemented. Low-cost upgrades considered in the former study (mostly Modified Ludzak-Ettinger-type processes) are practical and have proven successful at many treatment facilities. They are cost effective and are relatively easy to operate and control. Depending on the existing plant configuration and process, the designs might include addition of nitrate recycle pumps, creation of anoxic and or swing zones (swing zones can be used as aerobic or anoxic zones depending on either seasonal or flow changes at a particular treatment plant) and/or mixers. The effectiveness of adding BNR to existing plants is estimated by comparing the loads from specific WWTPs to loads estimated for systems with comparable discharge volumes for which BNR has been implemented (US EPA 2015). In the **01_UserSpecs_BMPs.csv** file, users should either choose to implement the set of low-cost retrofits (WWTP = Claremont, Hanover, Hinsdale, Keene, Ludlow, or Springfield) for the Upper Connecticut River or the BNR option (remaining WWTP list). In the current version of RBEROSTv1d, users could incorporate alternative upgrade treatment options if desired if cost and efficiency data are available through substituting appropriate data in the **01_UserSpecs_BMPs.csv** file for specific WWTPs. Future versions of RBEROSTv1d could incorporate multiple simultaneous upgrade options by adopting data file and code changes available in the version of RBEROST developed for the Puget Sound (RBEROST2d, Detenbeck et

al., in review).

2. **Urban BMPs.** Urban BMP options include practices applied to developed land and may serve functions such as increasing evaporation from standing surface water, filtration, infiltration of ponded water into soil media, percolation of infiltrated water into groundwater, or other denitrification mechanisms. Urban BMPs accomplish these functions through options such as biofiltration with underdrain, bioretention basins, enhanced biofiltration with internal storage reservoir (ISR), extended dry detention basin, grass swale with detention, gravel wetlands, infiltration basins, chambers or trenches, porous pavement with either subsurface infiltration or underdrains, sand filters with underdrains and wet ponds.
3. **Agricultural BMPs.** Agricultural BMP options include practices applied to agricultural land and may serve functions such as slowing runoff flow velocities from cropland areas, increasing infiltration into underlying soils, adjusting fertilizer application, routing runoff through pools and basins and other farming practices to slow and reduce nutrient transport to waterbodies. RBEROST only treats row crop area with agricultural BMPs and 'ag' throughout the code and documentation refers only to row crop area. Agricultural BMPs include options such as the use of filter strips, contour farming and manure injection, amongst others.
4. **Riparian Buffer BMPs.** Riparian Buffer BMP options include the conversion of land within riparian areas to either herbaceous/grassed or forested land. Such management practices slow water as it approaches the stream or river and increase infiltration. Nutrients may then be removed by soil processes. Buffers of different widths can be implemented to provide varying degrees of runoff velocity reduction.
5. **Septic BMPs.** Septic BMP options include septic tank upgrades from traditional septic systems to innovative/alternative onsite wastewater treatment systems (I/A OWTS) and conversions from the use of traditional septic systems to new sewer line connections. Such practices prevent additional overland nutrient runoff originating in septic leaks and inefficiencies. Furthermore, sewer conversions prevent the displacement of groundwater already saturated with nitrogen.

RBEROST can consider removal of both total nitrogen (TN) and total phosphorus (TP) simultaneously. Removal of TN and TP contribute to achieving user-defined loading targets, which are applied at the terminal, or furthest downstream, stream reach. Stream reaches are tracked by NHDPlusV2 common identifiers (COMIDs) that are unique to individual stream reaches throughout the model. A single watershed may have one or multiple terminal COMIDs, and each terminal COMID receives water from the reaches upstream from it. Not all terminal COMIDs have targets for both phosphorus and nitrogen, and the number of upstream COMIDs varies.

Table 3-1 summarizes the optimization variables included in the model and their associated constraints. Constraints on the point source BMPs are binary, such that the model chooses to implement or not to implement point source BMPs (in this case, either low-cost wastewater treatment plant retrofits (JJ Environmental 2015) or addition of BNR based on the associated cost and removal efficiency at each individual wastewater treatment facility (US EPA 2015). The cost of implementing retrofits or initiating BNR capabilities varies for each individual WWTP depending on volume of treated water. Constraints on the urban and agricultural BMPs relate to the fraction of each

land area that is treated. The model chooses to treat a fraction of urban or agricultural land area with each BMP based on the associated cost and removal efficiency; however, the sum of all fractions must be between 0 and 1. No agricultural or urban land can be treated simultaneously by two BMPs. Additional constraints exist on infiltration-based BMPs, as the model will not implement them in catchments with very low infiltration values and on porous pavement BMPs, as the model will only implement these on roadways and parking lots. Constraints on riparian buffer BMPs relate to the length (in ft) of stream bank available to be treated, where riparian BMPs can only be implemented on currently un-buffered stream lengths. Initial conditions for the optimization model reflect current practices, where the selected point source, urban and agricultural BMPs have not yet been implemented, and the load is set to the baseline annual nutrient load delivered by the upstream reaches to the target waterbodies. Septic BMPs are implemented differently based on which BMPs are chosen. Septic upgrades to I/A OWTS type systems decrease the septic baseload within a COMID by increasing the efficiency of systems *in situ*. The amount of nitrogen load attributable to each COMID is calculated during preprocessing so that RBEORST may choose varying levels of septic BMP implementation. Septic BMP choices are constrained by the number of parcels within each class category per COMID. Estimated costs per parcel vary depending on land-use (residential vs. commercial), lot size and depth to groundwater (Suffolk County Department of Health Services, 2020).

Maximum removal is achieved by applying upgrades to all available parcels within all septic classes. The connection to sewer BMP decreases the septic load within a COMID and adds the corresponding load reduction to the point source category where it can be reduced through point source BMPs. The implementation of sewer connections alone does not remove nitrogen from the system, instead it shifts the availability of effective BMPs from the septic category to the point category. Therefore, for true reductions in overall loadings to occur, RBEROST would need to select both sewer connections *and* point source retrofits. It is unlikely for the model to select sewer connections without point source retrofits as doing so would incur additional costs while providing no meaningful reduction toward the target goals.

Table 3-1. A summary of RBEROST model variables, constraints, and initial conditions.

Optimization Variable Description	Optimization Variable Name in AMPL Model File	Constraint Description	Constraint	Initial Conditions
Per-catchment agricultural BMP selection	agBMP_bin	Choose whether to implement	Binary (0 or 1)	0
Per-catchment urban BMP selection	urbanBMP_bin	Choose whether to implement	Binary (0 or 1)	0
Point Source BMP	point_dec	Choose whether to implement	Binary (0 or 1)	0
Urban BMP	urban_frac	Fraction of urban land treated	Fraction $\geq$ User Specified Min * Fraction urban area that is suitable. Fraction $\leq$ User Specified Max * Fraction urban area that is suitable. Sum Fractions ≤ 1. Sum Fractions for pavement BMPs $\leq$ Fraction urban area that is roads	0
Agricultural BMP	ag_frac	Fraction of row crop land treated	Fraction $\geq$ User Specified Min. Fraction $\leq$ User Specified Max. Sum Fractions ≤ 1	0
Septic BMP	septic_frac	Number of parcels upgraded or converted to sewer connections	Number of parcels chosen ≥ 0 Number of parcels chosen $\leq$ maximum number of available parcels for each upgrade class	0
Riparian Buffer BMP	ripbuf_length	Length of stream reach treated	Length $\leq$ Unbuffered Stream length * User specified max fraction. Removal along all Length $\leq$ Riparian loads. Sum all lengths $\leq$ total unbuffered bank length	0

2.3 Objective Function

The objective function minimizes BMP implementation costs while achieving a reduction in baseline nutrient loadings to the specified targets. Minimized costs are a function of the costs to implement point source, urban, agricultural, riparian buffer and septic BMPs in the catchments included in the optimization model (Eq. 3.1). All costs are in 2025/26 US dollars. (Adjustment factors used to update costs to 2020 – 2025 dollars are included in Appendix E along with methods used to allow users to update costs in later years.)

$$Cost_{Minimized} = \sum_{i=1}^n \left(\sum_{j=1}^{j=n} (Cost_{Agricultural,i,j}) + \sum_{k=1}^{k=n} (Cost_{Urban,i,k}) + Cost_{PointSource,i} \right. \\ \left. + \sum_{l=1}^{l=n} (Cost_{Riparian\ Buffers,i,l}) + \sum_{m=1}^{q=n} (Cost_{Septic,i,q}) \right) \quad (3.1)$$

In Eq 3.1 and following equations:

i	=	NHDPlus catchments
j	=	agricultural BMPs
k	=	urban BMPs
l	=	riparian buffer BMPs
q	=	septic upgrade/conversion BMPs

Individual parts of Eq. 3.1 are defined below in Eq. 3.2, Eq. 3.3, Eq. 3.4, Eq. 3.5, 3.6 and Eq. 3.7.

$$Cost_{Agricultural,i,j} = (Fraction_{TreatedLand,i,j}) * (Agricultural\ Land\ Area_i) * Ag\ Cost\ Adjustment \\ * (Capital\ Costs_{i,j} + O\&M\ Costs_{i,j}) \quad (3.2)$$

$Cost_{Agricultural,i,j}$ describes the costs of implementing BMP j in row crop agricultural fields. $Fraction_{TreatedLand,i,j}$ ranges 0-1 and is treated as a variable in the optimization problem, $Agricultural\ Land\ Area_i$ is the amount of row crop area in catchment i , $Ag\ Cost\ Adjustment$ reflects the difference in Environmental Quality Incentives Program (EQIP) base payments and actual costs of agricultural BMPs, $Capital\ Costs_{i,j}$ represent the annualized capital costs of BMP j , and $O\&M\ Costs_{i,j}$ represent the annual operations and maintenance costs of BMP j . Costs may differ by catchment i .

$$Cost_{Urban,i,k} = (Capital\ Costs_k + O\&M\ Costs_k) * Urban\ Cost\ Adjustment_i * WQv_k \quad (3.3)$$

$Cost_{Urban,i,k}$ describes the cost of treating stormwater runoff from urbanized land in catchment i with stormwater BMP k . $Capital\ Costs_k$ and $O\&M\ Costs_k$ are base costs for urban BMPs and do not differ by catchment. The $Urban\ Cost\ Adjustment_i$ value scales urban costs between one- and three-times base costs based on the expected amount of retrofitting as determined by the intensity of urban development in catchment i (Hill, et al., 2015; Voorhees, 2016; Yang, et al., 2018). WQv_k is the water quality treatment volume (in cubic feet) of runoff that can be treated by BMP k as defined below.

$$WQv_k = Urban\ design\ depth_k * Rv_i * (Urban\ Land\ Area_i) * (Fraction_{TreatedLand,i,k}) \quad (3.4)$$

$Urban\ design\ depth_k$ is user-specified and determines the treatment capacity of stormwater BMP k . $Urban\ design\ depth_k$ is defined by the user in inches and is converted to feet within RBEROST. Rv_i is the runoff coefficient for catchment i calculated as $0.05 + 0.009 * Percent\ Site\ Imperviousness_i$ (Schueler, 1987; Vermont Agency of Natural Resources, 2017). $Urban\ Land\ Area_i$ is the area of urbanized

land in catchment i (in ft^2), and $Fraction_{TreatedLand,i,k}$ ranges 0-1 and is treated as a variable in the optimization problem.

$Cost_{PointSource,i} = (0,1)_i * (Capital\ Costs_i + O\&M\ Costs_i)$	(3.5)
---	-------

$Cost_{PointSource,i}$ describes the total costs of retrofitting WWTPs given a binary decision $(0,1)_i$ of whether to implement within each COMID. This binary decision is treated as a variable in the optimization model. *Capital Costs* and *O&M Costs* refer to annual operations and maintenance costs, and differ by WWTP in catchment i . In addition to reducing the point source baseload for a given COMID, implementing point source BMPs will also reduce any load transferred to the point source category through the selection of sewer conversions in the septic category.

$Cost_{Riparian\ Buffers,i,l} = Length_{Treated\ Bank,i,l} * Ag\ Cost\ Adjustment * (Capital\ Costs_{i,l} + O\&M\ Costs_{i,l})$	(3.6)
---	-------

$Cost_{Riparian\ Buffers,i,l}$ describes the costs of converting riparian zones into forested or grassed buffers, $Length_{TreatedBank,i,l}$ is a variable in the optimization and ranges from 0 ft to twice the length of stream reach (reflecting total bank length in ft) in catchment i . $Capital\ Costs_{i,l}$ and $O\&M\ Costs_{i,l}$ are base costs from EQIP, and are adjusted with *Ag Cost Adjustment* as in Eq. 3.2.

$Cost_{Septic,i,q} = Septic_frac_{i,q} * (Capital\ Costs_{i,q} + O\&M\ Costs_{i,q})$	(3.7)
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$Cost_{Septic,i,q}$ describes the costs of upgrading current septic systems to more efficient I/A OWTS septic systems or converting them to sewer connections so that waste is transferred to WWTP. *Septic_frac* is a variable in the optimization and ranges from 0 to the total number of parcels available for each category of upgrades q within each COMID i . $Capital_{i,q}$ and $O\&M_{i,q}$ Costs reflect the cost per individual parcel for a given upgrade or conversion class and COMID.

Sources for BMP costs (and other parameters) are given in Table 3-2. The water quality volume (WQ_v , treated volume), reflects Vermont Water Quality Treatment Standards (Vermont Agency of Natural Resources, 2017). Table 3-2 summarizes parameters in Long Island Sound Dynamic RBEROST and their sources for the case studies described later.

Table 3-3 summarizes additional parameters used for uncertainty analysis and their data sources.

Table 3-2. A summary of RBEROST parameters.

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Nitrogen baseline annual average loading data per catchment to each TN target	baseloads_N1...baseloads_Nn	Long Island Sound Dynamic SPARROW model seasonal loadings output (Moore et al., 2026) supplemented by outputs from updated Long Island Sound Nitrogen Loading Model (Vaudrey et al. 2017, updated by US EPA with input data from (PredictN_NoBFlow.csv) modified to incorporate near-surface groundwater flow.
Phosphorus baseline annual average loading data per catchment to each TP	baseloads_P1...baseloads_Pn	Northeastern Regional SPARROW model (Moore et al., 2026) annual average loadings output (Predict_P.csv)
Nitrogen baseline annual riparian loading data per catchment to each TN target	riparianload_N1...riparianload_Nn	Methodology from Pollutant Removal Credits for Buffer Restoration in MS4 Permits Final Panel Report (Houle et al. 2019) using land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018) and river reach shape files from NHDPlus V2 (McKay et al. 2012)
Phosphorus baseline annual riparian loading data per catchment to each TP target	riparianload_P1...riparianload_Pn	Methodology from Pollutant Removal Credits for Buffer Restoration in MS4 Permits Final Panel Report (Houle et al. 2019) using land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018) and river reach shape files from NHDPlus V2 (McKay et al. 2012)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Available urban and row crop areas per catchment	area	Urban area: Percent developed area from the 2019 NLCD (catchment-specific data per state downloaded from StreamCat; Hill et al., 2015) – “PctUrbHi2019Cat”, “PctUrbLo2019Cat”, “PctUrbMd2019Cat”, and “PctUrbOp2019Cat” fields. Summed to give total percent of area that is developed. Row crop/Agricultural area: Percent cropland data from the 2019 NLCD database (catchment-specific data per state downloaded from StreamCat; Hill et al. 2015) - “PctCrop2019Cat” field
Percent of Urban land that is suitable for BMP implementation	urban_bmp_implementationpotential	Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b) and USDA NRCS 2009. River Reach shape files from NHD plus v2 (McKay et al. 2012). Building footprints from a national dataset of rasterized building footprints for the U.S. (Heris et al., 2020). High resolution impervious area from the Vermont Base Landcover 2015 database (UVM Spatial Analysis Lab, 2019)
Length of streambank that is not already buffered per catchment	unbuffered_banklength	Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018) and river reach shape files from NHDPlus V2 (McKay et al. 2012)
Total stream bank length (2 x reach length) per catchment	total_banklength	River reach shape files from NHDPlus V2 (McKay et al. 2012)
Capital costs associated with implementing agricultural BMPs	ag_costs_capital	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2020. For the case study, Massachusetts costs were used when New Hampshire or Vermont costs were not available (U.S. Department of Agriculture Staff, 2021)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Operations and maintenance costs associated with implementing agricultural BMPs	ag_costs_operations	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2020. (U.S. Department of Agriculture Staff, 2021)
Costs associated with implementing wastewater treatment plant retrofits for nutrient removal	point_costs	NEIWPCC Final Report - Low-Cost Retrofits for Nitrogen Removal at Wastewater Treatment Plants in the Upper Long Island Sound Watershed (JJ Environmental, 2015)
Costs associated with implementing urban BMPs	urban_costs	WMOST Theoretical Documentation (Detenbeck, ten Brink et al., 2018; Table 7-2)
Capital costs associated with implementing riparian buffer BMPs	ripbuf_costs_capital	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2020.
Operations and maintenance costs associated with implementing riparian buffer BMPs	ripbuf_costs_operations	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2020.
Calculated urban volumetric runoff coefficient, based on percent impervious area per catchment	runoff_coeff_urban	Imperviousness data from National Land Cover Database, 2019 (catchment-specific data per state downloaded from StreamCat; Hill et al. 2015) - "PctImp2019Cat" field. Data used to calculate urban runoff coefficient, based on Vermont Stormwater Manual equation ($0.05 + 0.009 * \text{PctImp2019Cat}$; Vermont Agency of Natural Resources, 2017)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Adjustments to urban BMP costs based on the expected amount of retrofitting (proportional to relative intensity of urban land per catchment)	urban_cost_adjustment_coef	Development intensity data by catchment from the National Land Cover Database, 2019 (catchment-specific data per state downloaded from StreamCat; Hill et al. 2015) - “PctUrbOp2019Cat,” “PctUrbLo2019Cat,” “PctUrbMd2019Cat,” and “PctUrbHi2019Cat” fields. Cost adjustment factors from Opti-Tool documentation (Voorhees, 2016)
Capital costs associated with implementing septic system upgrades and conversions	septic_costs_capital	Cost data by septic BMP class from the EPA, Suffolk County SWP, Forge River Project Documentation, and Bureau of Labor Statistics datasets. (Forge River Watershed Sewer Project, 2022; Suffolk County Department of Health Services, 2020; U.S. Bureau of Labor Statistics, 2023, 2024a, 2024b)
Operations and maintenance costs associated with implementing septic system upgrades and conversions	septic_costs_operations	Cost data by septic BMP class from the EPA, Suffolk County SWP, Forge River Project Documentation, and Bureau of Labor Statistics datasets. (Forge River Watershed Sewer Project, 2022; Suffolk County Department of Health Services, 2020; U.S. Bureau of Labor Statistics, 2023, 2024a, 2024b)
Agricultural nitrogen removal efficiency per BMP option and HUC12	ag_effic_N	“Fert_20” and “Manure_Injection” BMPs: Efficiencies provided by EPA via NRCS. Remaining agricultural BMPs: Efficiencies summarized by HUC12 (or if unavailable, HUC10 and/or HUC8) based on ACRE database (White et al., 2019)
WWTP-specific nitrogen removal efficiency	point_effic_N	NEIWPCC Final Report - Low Cost Retrofits for Nitrogen Removal at Wastewater Treatment Plants in the Upper Long Island Sound Watershed (JJ Environmental, 2015), Efficiencies for N removal calculated for biological nutrient reduction calculated from data in US EPA (2025b)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Urban nitrogen removal efficiency per BMP option and catchment	urban_effic_N	New Hampshire MS4 Permit BMP Performance Curves (New Hampshire Department of Environmental Services, 2020b). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b; USDA NRCS 2009). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018)
Septic upgrade and conversion nitrogen removal efficiency per parcel	septic_effic_N	Removal efficiencies are a reduction in the nonpoint loadings results from traditional septic systems to newer I/A OWTS septic systems, or reductions results from connecting to sewer systems in which the reduction results in a simultaneous increase in waste load at the nearest WWTP (Suffolk County Department of Health Services, 2020)
Agricultural phosphorus removal efficiency per BMP option and HUC12	ag_effic_P	“Fert_20” and “Manure_Injection” BMPs: Efficiencies provided by EPA via NRCS. Remaining agricultural BMPs: Efficiencies summarized by HUC12 (or if unavailable, HUC10 and/or HUC8) based on ACRE database (White et al., 2019)
WWTP-specific phosphorus removal efficiency	point_effic_P	Efficiencies for P removal calculated for biological nutrient reduction calculated from data in JJ Environmental (2015) and US EPA (2015).
Urban phosphorus removal efficiency per BMP option and catchment	urban_effic_P	New Hampshire MS4 Permit BMP Performance Curves (New Hampshire Department of Environmental Services, 2020b). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b; USDA NRCS 2009). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Total nitrogen removal by riparian buffers per catchment for each TN target	riparianremoval_N1...riparianremoval_Nn	Annual removal rates are a function of riparian loading and nutrient removal efficiency of riparian buffers. Loading is dependent on the amount of urbanized land within a 400 ft riparian zone, and efficiency is dependent on slope and hydrologic soil group (Houle et al. 2019). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018). River reach shape files from NHDPlus V2 (McKay et al. 2012). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b). Slope data from Hydrologic Derivatives for Modeling and Analysis by the USGS (Verdin, 2017)
Total phosphorus removal by riparian buffers per catchment for each TP target	riparianremoval_P1...riparianremoval_Pn	Annual removal rates are a function of riparian loading and nutrient removal efficiency of riparian buffers. Loading is dependent on the amount of urbanized land within a 400 ft riparian zone, and efficiency is dependent on slope and hydrologic soil group (Houle et al. 2019). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018). River reach shape files from NHDPlus V2 (McKay et al. 2012). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a, 2020b). Slope data from Hydrologic Derivatives for Modeling and Analysis by the USGS (Verdin, 2017)
Total nitrogen load allowed after reduction for all TN targets	loads_lim_N1...loads_lim_Nn	User supplied
Total phosphorus load allowed after reduction for all TP targets	loads_lim_P1...loads_lim_Pn	User supplied

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Agricultural BMP costs reflected in ag_costs represent base payment costs (75 percent of actual costs). The agcost_frac parameter reflects 100 percent of costs ($100/75 = 1.33$).	agcost_frac	Correspondence with EPA and NRCS state conservationist for Vermont
Conversion from acre-ft to cubic ft	acfttoft3	Not applicable
Conversion for precipitation: inches to ft	pcp	Not applicable
Parameter to ensure that agricultural BMPs are implemented on at least some minimum acreage. Currently set to 0.	agBMP_minarea	Not applicable
Minimum allowed percent implementation of urban BMPs per catchment	urban_frac_min	User supplied
Maximum allowed percent implementation of urban BMPs per catchment	urban_frac_max	User supplied

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Minimum allowed percent implementation of agricultural BMPs per catchment	ag_frac_min	User supplied
Maximum allowed percent implementation of agricultural BMPs per catchment	ag_frac_max	User supplied
Minimum allowed percent implementation of riparian buffer BMPs per catchment	ripbuf_frac_min	User supplied
Maximum allowed percent implementation of riparian buffer BMPs per catchment	ripbuf_frac_max	User supplied
The specified design depth for each urban BMP	urban_design_depth	User supplied

Table 3-3: A summary of RBEROST parameters used for uncertainty analyses.

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Standard error of nitrogen baseline annual average loading data per catchment to each TN target	baseloads_N1_se...baseloadsNn_se	Dynamic SPARROW model for Long Island Sound (Moore et al. 2026), Updated Long Island Sound Nitrogen Loading Model (Vaudrey et al. 2017, updated by US EPA using data from Jahn and Walter 2025), Northeastern Regional SPARROW model (Schmadel et al. 2021) seasonal loadings output (Predict_NoBFlowN_WSeptic.csv)
Standard error of phosphorus baseline annual average loading data per catchment to each TP target	baseloads_P1_se...baseloadsPn_se	Dynamic SPARROW model for Long Island Sound (Moore et al. 2026), Northeastern Regional SPARROW model (Schmadel et al. 2021) annual average loadings output (Predict_P.csv)
Standard error of available urban and row crop areas per catchment	area_se	Coefficient of variation for NLCD data and StreamCat values from Wickham et al., 2017 and Hill et al., 2015. Uncertainty in Incremental area from U.S. Geological Survey and U.S. Department of Agriculture, Natural Resources Conservation Service, 2013, McKay et al., 2012, and U.S. Geological Survey, National Geospatial Program, 2020.
Standard error of capital costs associated with implementing agricultural BMPs	ag_costs_capital_se	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2018-2021. For the case study, Massachusetts costs were used when New Hampshire or Vermont costs were not available (U.S. Department of Agriculture Staff, 2021)
Standard error of operations and maintenance costs associated with implementing agricultural BMPs	ag_costs_operations_se	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2018-2021. For the case study, Massachusetts costs were used when New Hampshire or Vermont costs were not available (U.S. Department of Agriculture Staff, 2021)
Standard error of capital costs associated with implementing riparian buffer BMPs	ripbuf_costs_capital_se	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2018-2021 (U.S. Department of Agriculture Staff, 2021)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Standard error of operations and maintenance costs associated with implementing riparian buffer BMPs	ripbuf_costs_operations_se	U.S. Department of Agriculture; Environmental Quality Incentives Program state-by-state cost sheets from 2018-2021 (U.S. Department of Agriculture Staff, 2021)
Standard error of the calculated urban volumetric runoff coefficient, based on percent impervious area per catchment	runoff_coeff_urban_se	Standard error around the slope and intercept of the equation presented by Vermont Agency of Natural Resources, 2017 calculated from the data originally published by Schueler, 1987.
Standard error of adjustments to urban BMP costs based on the expected amount of retrofitting (proportional to relative intensity of urban land per catchment)	urban_cost_adjustment_coef_se	Sampled from development intensity data by catchment from the National Land Cover Database, 2019 (catchment-specific data per state downloaded from StreamCat; Hill et al. 2015) -- "PctUrbOp2019Cat," "PctUrbLo2019Cat," "PctUrbMd2019Cat," and "PctUrbHi2019Cat" fields. Cost adjustment factors from Opti-Tool documentation (Voorhees, 2016)
Standard error of agricultural nitrogen removal efficiency per BMP option and HUC12	ag_effic_N_se	"Manure_Injection" BMP: Dell et al. 2016. Remaining agricultural BMPs: Standard error of efficiencies summarized by HUC12 (or if unavailable, HUC10 and/or HUC8) based on ACRE database (White et al., 2019)
Standard error of urban nitrogen removal efficiency per BMP option and catchment	urban_effic_N_se	Sampled from values calculated according to New Hampshire MS4 Permit BMP Performance Curves (New Hampshire Department of Environmental Services, 2020b). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018)
Standard error of agricultural phosphorus removal efficiency per BMP option and HUC12	ag_effic_P_se	"Manure_Injection" BMP: Dell et al. 2016. Remaining agricultural BMPs: Standard error of efficiencies summarized by HUC12 (or if unavailable, HUC10 and/or HUC8) based on ACRE database (White et al., 2019)

Optimization Parameter Description	Optimization Parameter Name in AMPL Model and Data Files	Data Source
Standard error of urban phosphorus removal efficiency per BMP option and catchment	urban_effic_P_se	Sampled from values calculated according to New Hampshire MS4 Permit BMP Performance Curves (New Hampshire Department of Environmental Services, 2020b). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018)
Standard error of total nitrogen removal by riparian buffers per catchment for each TN target	riparianremoval_N1_se...riparianremoval_Nn_se	Sampled values of annual removal rates. Annual removal rates are a function of riparian loading and nutrient removal efficiency of riparian buffers. Loading is dependent on the amount of urbanized land within a 400 ft riparian zone and efficiency is dependent on slope and hydrologic soil group (Houle et al. 2019). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018). River reach shape files from NHDPlus V2 (McKay et al. 2012). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b). Slope data from Hydrologic Derivatives for Modeling and Analysis by the USGS (Verdin, 2017)
Standard error of total nitrogen removal by riparian buffers per catchment for each TP target	riparianremoval_P1_se...riparianremoval_Pn_se	Sampled values of annual removal rates. Annual removal rates are a function of riparian loading and nutrient removal efficiency of riparian buffers. Loading is dependent on the amount of urbanized land within a 400 ft riparian zone, and efficiency is dependent on slope and hydrologic soil group (Houle et al. 2019). Land cover data from the National Landcover Database (Jin et al. 2019, Yang et al. 2018). River reach shape files from NHDPlus V2 (McKay et al. 2012). Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a, 2020b). Slope data from Hydrologic Derivatives for Modeling and Analysis by the USGS (Verdin, 2017)

2.4 Optimization Parameters

The dynamic SPARROW model predicts not only seasonal N and P loads from various current sources such as agricultural fertilizer, but also a “storage” load representing legacy nutrients that are stored in the system and released in subsequent seasons. Thus, storage loads are not reduced immediately by the action of management practices. In the SPARROW model, the storage load is assumed to be proportional to the previous seasonal load.

The various optimization parameters come together in two sets of equations that must be true for costs to be calculated and optimized. These equations, known as the incremental storage load and total load equations, must meet their constraints to provide a feasible response to return. Incremental storage load equations represent the remaining baseload loadings following BMP implementation as well as the partially transferred portion of the prior seasons load for a given NHDPlus catchment (COMID) in a given season and given year. Total loads represent the accumulated loads from upstream NHDPlus catchments at the base of a watershed where target loads are established.

RBEROST includes several constraints on the optimization function, including constraints that ensure loading reduction targets are met. For watersheds without septic systems, these constraints are defined for each specified loading target as:

$$\sum_{s=1}^4 \sum_{c=1}^d (\text{baseloads}_{c,'ag'} \times [1 - \sum_{a=1}^b (\text{ageff}_{c,a} \times \text{agfrac}_{c,a})] + \text{baseloads}_{c,'urban'} \times [1 - \sum_{u=1}^v (\text{urbaneff}_{c,u} \times \text{urbanfrac}_{c,u})] + \text{baseloads}_{c,'point'} \times [1 - \text{pointeff}_{c,'point'} \times \text{pointdec}_c] - \sum_{r=1}^s [\text{riparianremoval}_{c,r} \times \text{ripbuflength}_{c,r}]) + \text{otherloads} \leq \text{loadslim},$$

where $\text{baseloads}_{c,'ag'}$, $\text{baseloads}_{c,'urban'}$ and $\text{baseloads}_{c,'point'}$ refer to loading estimated from the SPARROW model of either TN or TP depending on the target, and $\text{ageff}_{c,a}$, $\text{urbaneff}_{c,u}$, and $\text{pointeff}_{c,'point'}$ are the nutrient removal efficiencies of agricultural or urban BMPs, or WWTP upgrades for either N or P depending on the target. otherloads is the sum of all untreated loads of either TN or TP being contributed to the loading target and loadslim is the mass limit of either TN or TP that corresponds to the percent reduction specified by the user. When septic systems are present the point source load component above is replaced by:

$$(\text{baseloads}_{c,'point'} + ((\text{baseloads}_{c,'septic'} / \text{total_parcels}) \times \text{septic_frac} \times \text{septic_effic_N})) \times (1 - (\text{point_effic_N} \times \text{point_dec})) + (\text{baseloads}_{c,'septic'} - \sum \text{baseloads}_{c,'septic'} / \text{total_parcels}) \times \text{septic_frac} \times \text{septic_effic_N}$$

where $\text{baseloads}_{c,'septic'}$ refers to N septic loads estimated by the Vaudrey model, septic_frac represents the fraction of septic systems upgraded, septic_effic_N represents the nutrient removal efficiency of upgraded septic. Conversion of septic systems is reflected in a loss of septic loads and a partially offsetting gain in point source loads.

It is important to note that the transferred portion of the prior season load is calculated differently based on the given season. Figure 2-1 shows the source of the seasonal transfer component. The initial model season (year = 1; season = 1) is based on the SPARROW storage component. All subsequent seasons in year 1 and following years (year = n; season > 1) calculate transferred load based on the previous season within the same year (Figure 2-1). This is because in

RBEROST, the seasonal loads are being reduced by management practices, and this will affect the amount stored in the system. Lastly, initial seasons for all subsequent years (year > 1; season = 1) are calculated from season 4 loads of the prior year. Overall, the variation in the transferred component allows a seamless transition between all seasons in the model. The incremental loads account for the transferred seasonal load component and the reduced septic, ag, urban, riparian and point source baseloads. A diagram of the incremental storage load equations is shown in Figure 2-2. Incremental load calculations may also contain “other” load components sourced from SPARROW data, depending on user parameter choices. Other loads are loads for which there are no applicable BMPs, such as atmospheric deposition and snowmelt, which will be added to the remaining load without reduction. Users can determine whether to include these components in the model. More information on this functionality is discussed in section 3.2. The total load equation is simply the sum of all upstream incremental storage loads for a given watershed (collection of COMIDs, light orange box in Figure 2-2) for a given season and year. The total load equation must meet the reduction limits set by the user and specified within the load reduction targets.

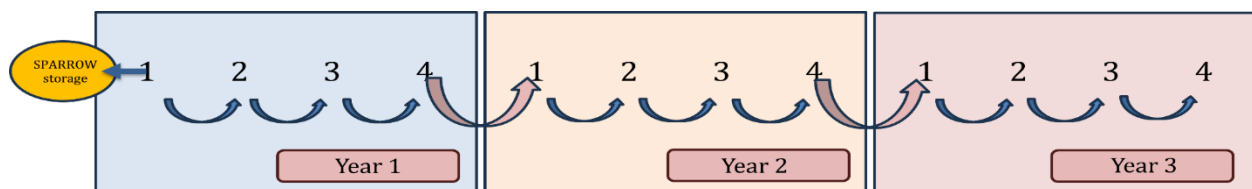


Figure 2-1. RBEROST sources partial transfer loads from the prior season using modified equations to create a multi-year model with continuity between all years and seasons.

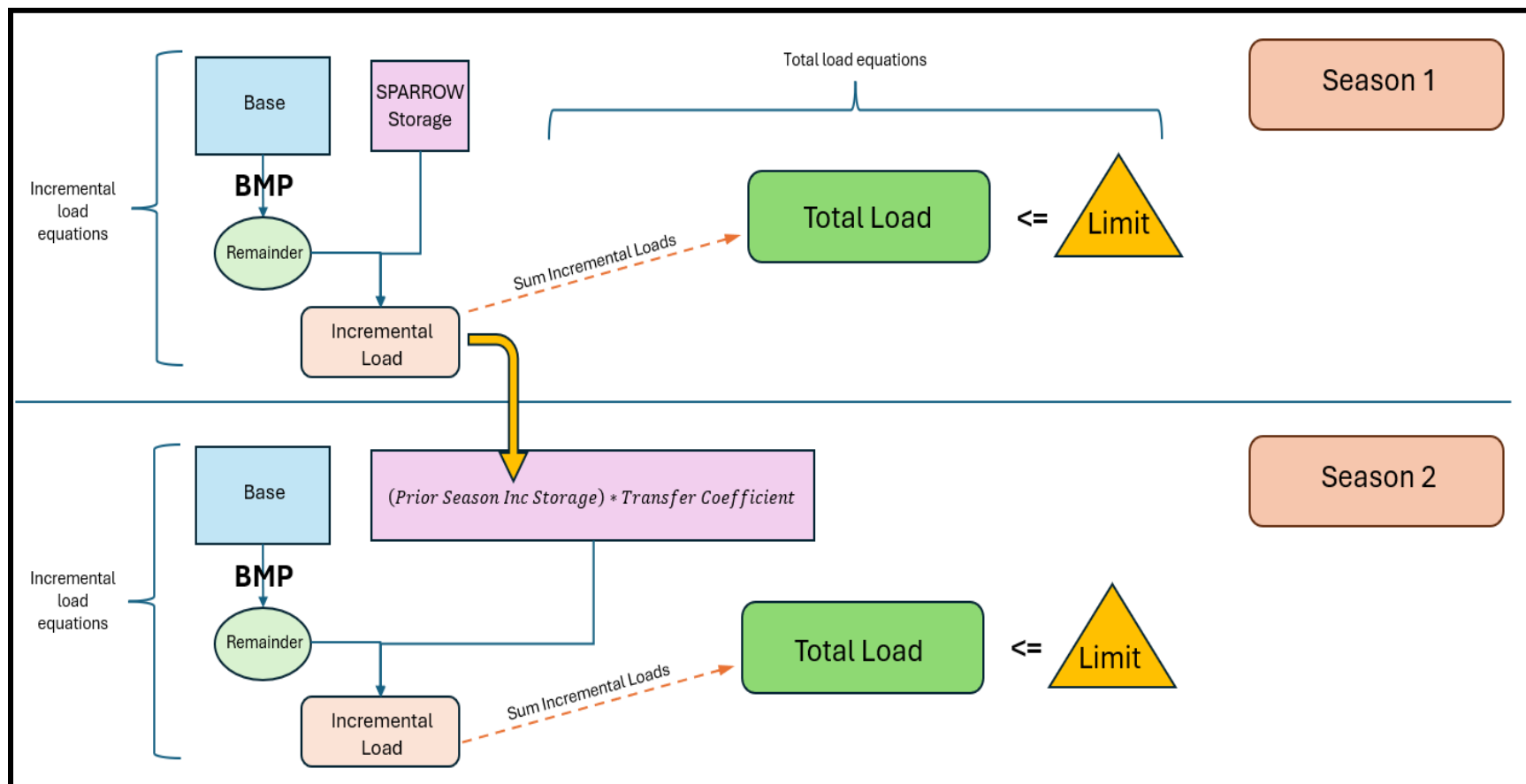


Figure 2-2. Diagram showing a generic depiction of the incremental storage load calculations for the first two years of the model. Note that all seasons after season 2 would be calculated similarly to season 2.

3 Model Use

To run RBEROST, the user first defines specifications, including the load reduction goals and the BMPs to be considered for the optimization model. The user specifications are then run through the R preprocessing code, which develops and formats the AMPL files required for the NEOS server optimization run. The user can choose to generate AMPL files that either include uncertainty or do not include uncertainty analyses. The user uploads the AMPL files to the NEOS server and accesses the optimization results using information e-mailed to the user. RBEROST was recently updated to submit AMPL files to NEOS directly in RStudio, without the need to visit the server website. Finally, the user runs the R postprocessing code to summarize results of the NEOS optimization. Figure 3-1 summarizes this workflow.

The dynamic (i.e., seasonal) RBEROST model differs from the past static (i.e., annual) versions in that the model is now capable of iterating overtime as well as providing a single snapshot of management decisions. Preprocessing data, which typically varies by COMID and BMP in the static model, now varies by COMID, BMP, year and season (quarterly, 1-4). The resulting .dat file generated by the preprocessing code is generally larger than the corresponding static model version, given the same watershed choices.

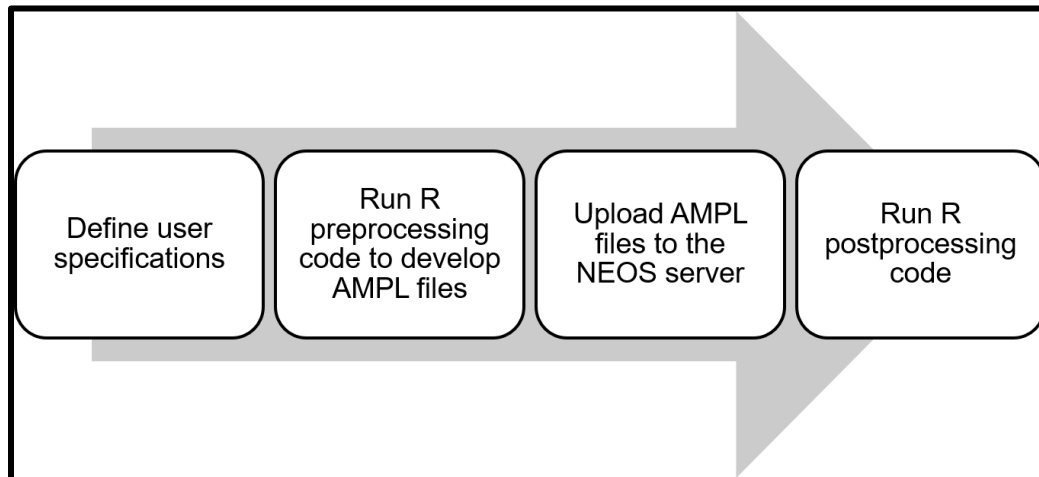


Figure 3- 1. User workflow of RBEROST.

3.1 Getting Started

Users download a .zip file that contains RBEROST program files and data required for preprocessing (creation of AMPL files) and postprocessing (viewing of NEOS optimization results). It is necessary for the file structure of the zip file as well as naming conventions for Preprocessing inputs and R files to be maintained. The full repository including complete versions of lfs files (see below) will occupy 6.12 GB.

3.1.1 Option 1: Downloading repository using git software

Users can install git software for managing git repositories and then use the following sequence at the command line:

```
# Initialize Git LFS globally
git lfs install --skip-repo
```

```
git clone https://github.com/USEPA/RBEROSTV1D.git
```

Since the repository is configured with `.gitattributes` for LFS, pointer files will be replaced automatically with actual large files during checkout. If after cloning you see placeholder pointer files instead of actual content, run:

```
cd local-repo
git lfs pull
```

This downloads all large files for the current commit.

3.1.2 Option 2: Downloading remote repository as zip file

Program files are typically downloaded from the RBEROST repository where they are published. Standard download methods suffice for most RBEROST program files. The user should go to the location of the EPA repository (<https://github.com/USEPA/RBEROSTV1D>), then click on the green <> Code button and select Download zip (Figure 3-2).

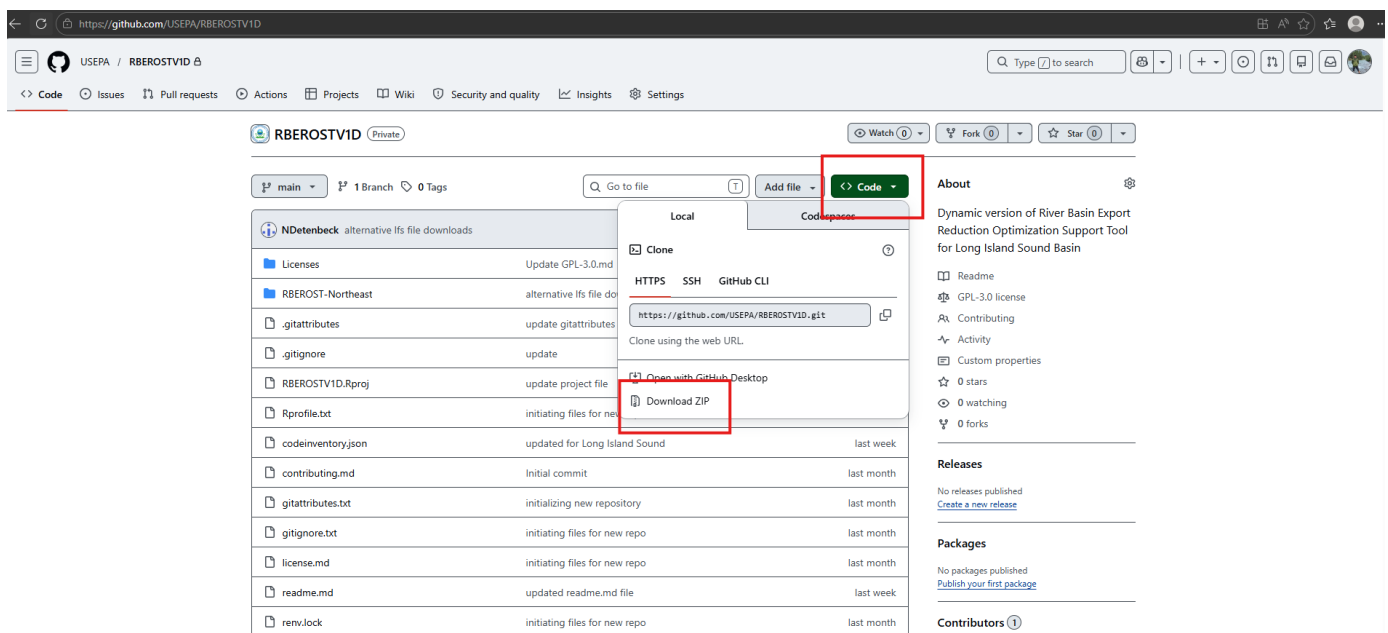


Figure 3- 2. Screenshot of EPA github site for RBEROST showing selection of code button then download zip option.

When the downloaded file is unzipped it will retain the directory structure necessary for the code to work, but only placeholders will be downloaded for the large files (lfs, > 100 MB) designated in the .gitattributes file in the repository. An optional code chunk to automate download of large (lfs) files to the Inputs subdirectory is included in the R Markdown file, RunRBEROST-Northeast-Dynamic.Rmd (see below).

3.1.3 Option 3: Downloading remote repository from within RStudio

Alternatively, to clone the Northeast RBEROST GitHub repository from within RStudio, navigate to the GitHub webpage for the model (<https://github.com/USEPA/RBEROSTV1D>). On the GitHub page, find the green “< > Code” button and select it as shown in Figure 3-3. A small window will appear and display a URL as well as other information. Select the icon to the right of the URL to copy the URL (Figure 3-4). Next, open RStudio and select “New Project” under the “File” tab at the top left portion of the window as shown in Figure 3-5. A window will appear that shows several ways to create a new project. Select the “Version Control” option (Figure 3-6). Next, select the “Git” option that appears in the subsequent window (Figure 3-7). The next window that appears will contain a space to insert the URL copied from the GitHub webpage. Next, name the file as you wish and select a workspace on your device to save the cloned version of the repository. This will create a cloned version of the repository in the selected workspace on your device. Figure 3-8 shows where to paste the GitHub repository URL, as well as where to add a name for your repository and select a location on your device for the RBEROST files.

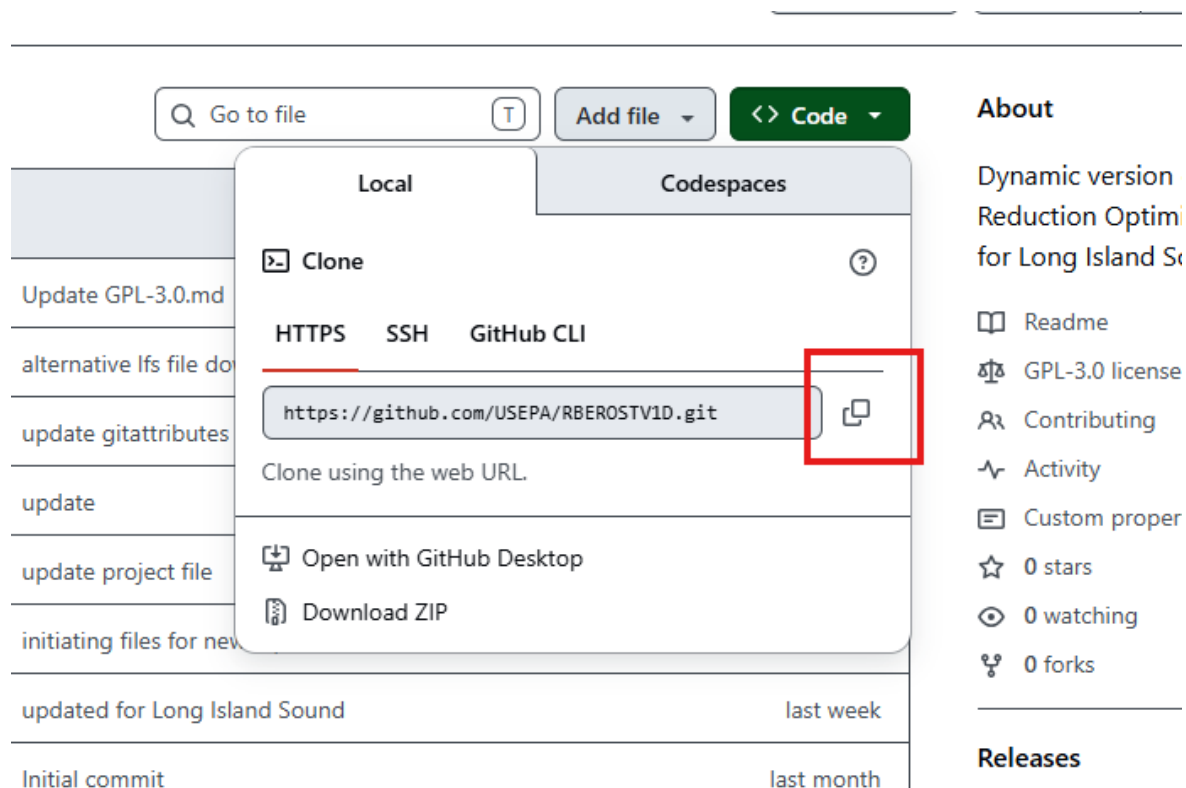


Figure 3-3. Initial step to clone RBEROST repository from GitHub webpage.

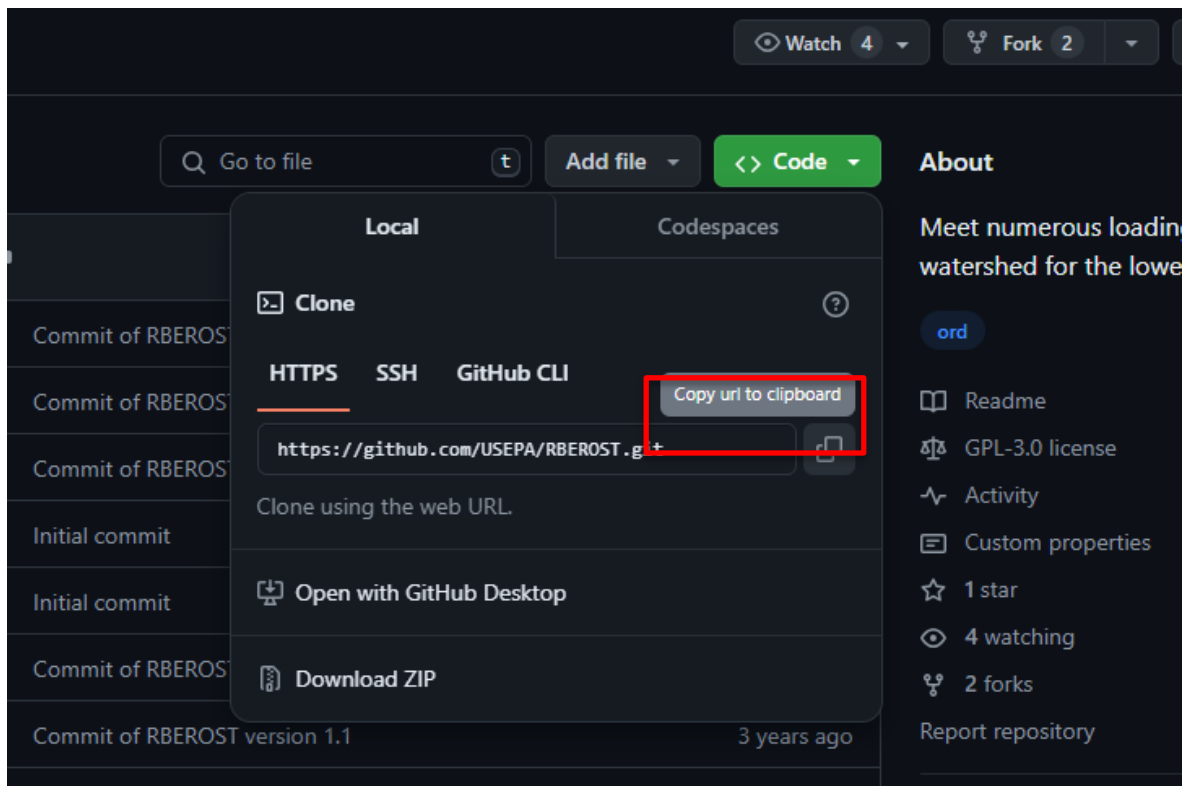


Figure 3- 4. Subsequent step to clone RBEROST repository from GitHub webpage.

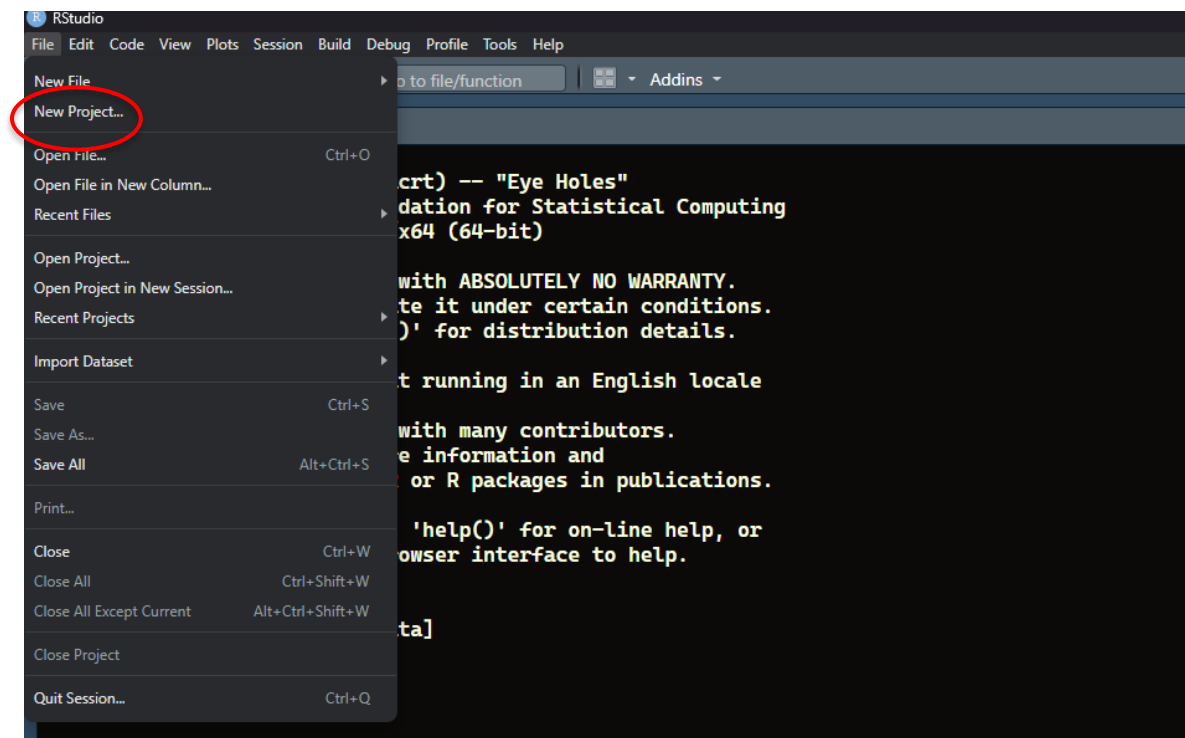


Figure 3- 5. Initial step to clone RBEROST repository in RStudio.

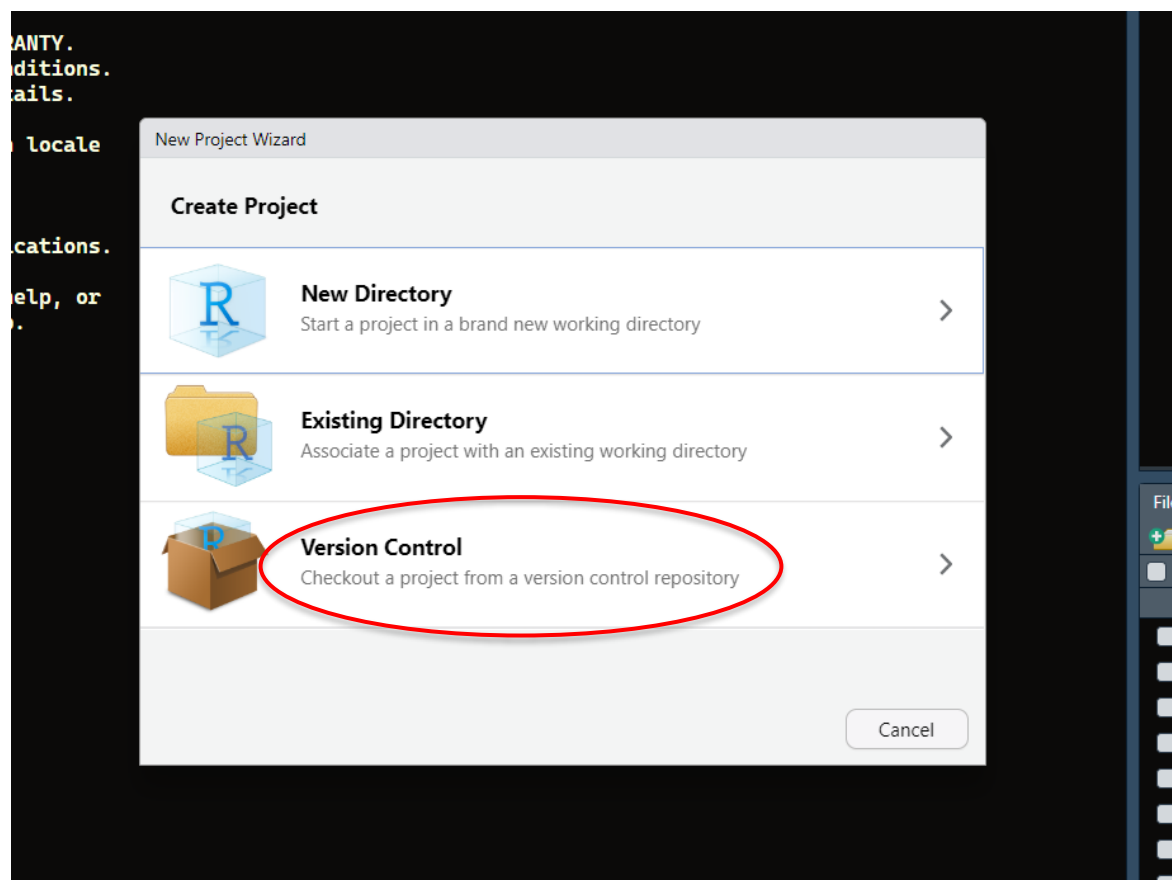


Figure 3- 6. Subsequent step to clone RBEROST repository in RStudio.

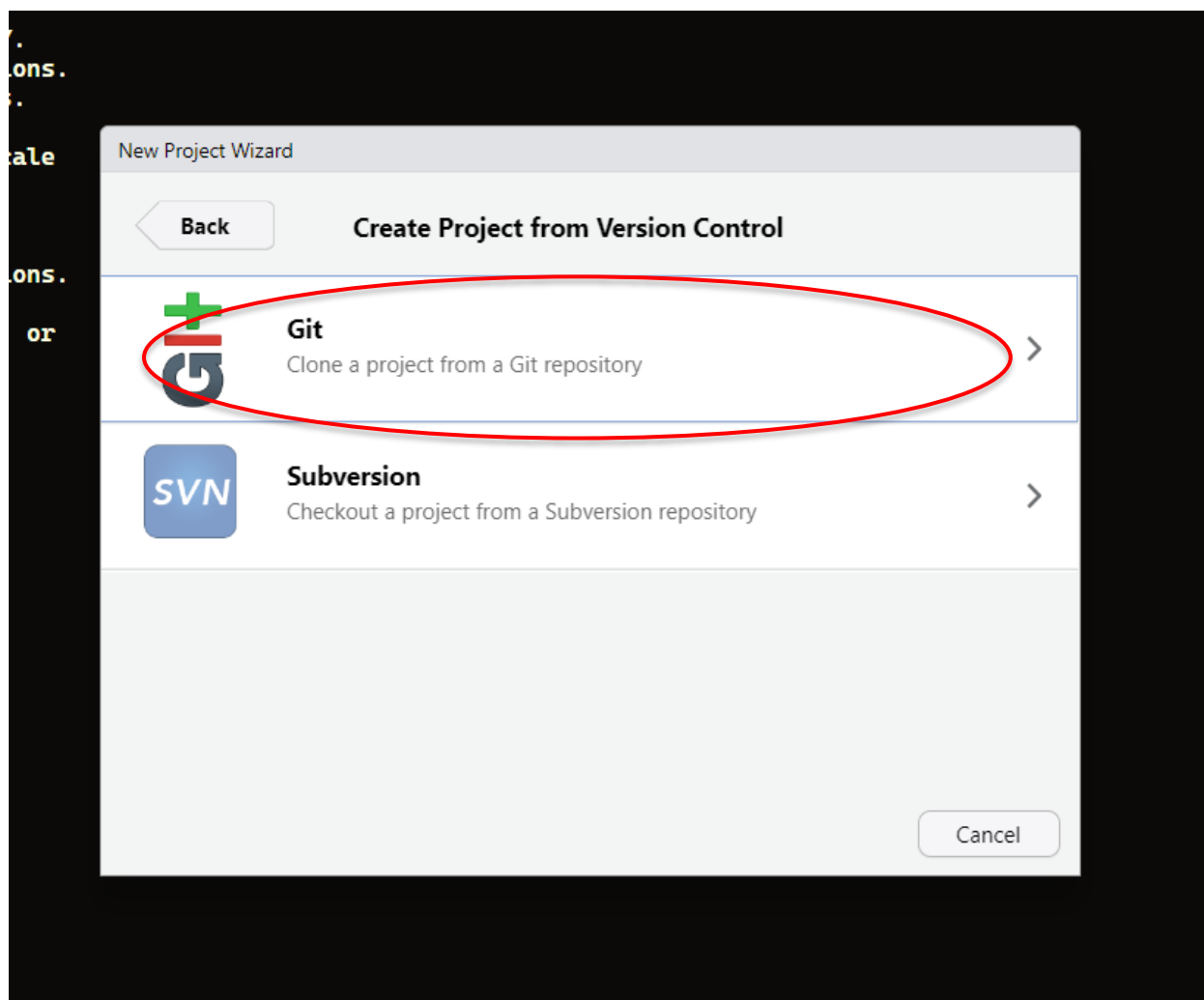


Figure 3- 7. Additional subsequent step to clone RBEROST repository in RStudio.

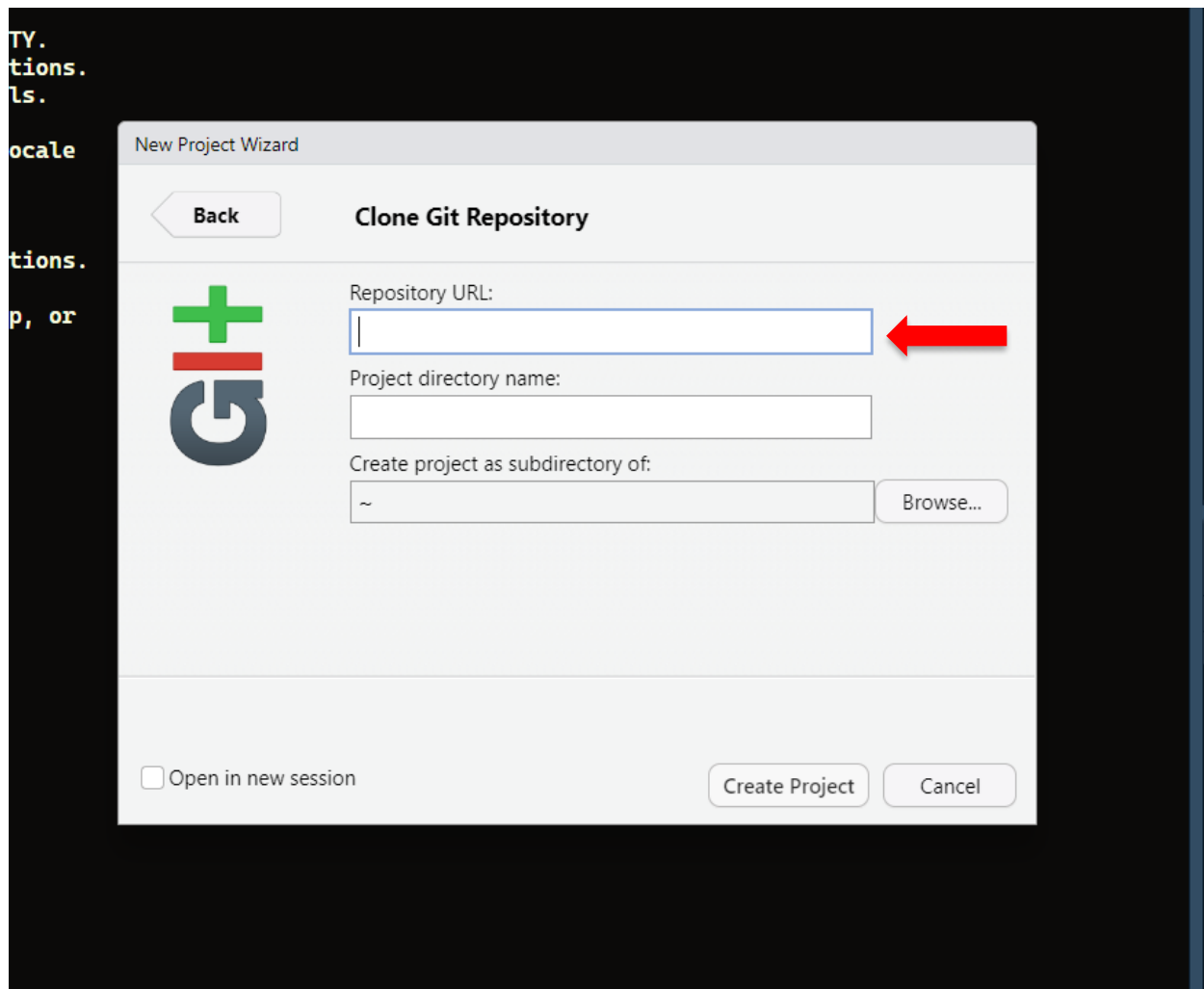


Figure 3- 8. Final step to clone RBEROST repository in RStudio.

3.1.4 Opening the project and running code

For the default pathways in RBEROST to work, you will need to open the Rproject via the **RBEROSTV1D.Rproj** file (Figure 3-9). Please note that the user will still need to manually input the working directory file path even when using the RProject. Double click the file to open the project in RStudio. Figure 3-9 shows the file in the unzipped folder. The project should open in RStudio, and the files can be seen on the bottom right. RBEROST can be executed through the **RunRBEROST-Northeast-Dynamic.Rmd** file, which is opened by clicking on it in the **Files** pane of RStudio (Figure 3-10).

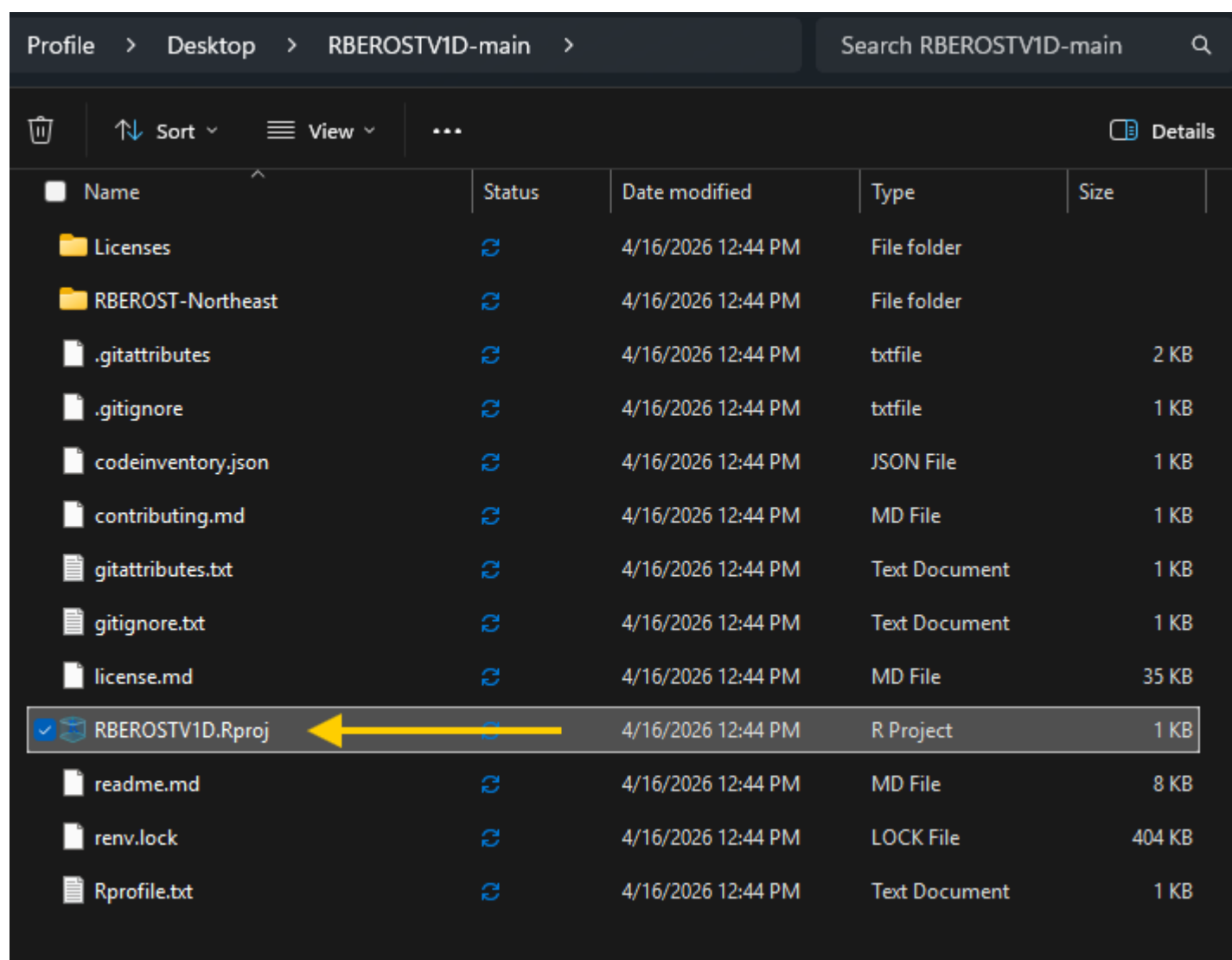


Figure 3- 9. RBEROST Rproject file.

RBEROST was developed in R version 4.3.1 and 4.3.2 and updated under R 4.4. RBEROST was developed with RStudio version 2025.05.1+ Build 513. The dynamic version of RBEROST includes enhanced folder structure with R scripts to preprocess the data and postprocess optimization results. All files are accessed through the **RunRBEROST-Northeast-Dynamic.Rmd** file, which serves as the primary interface for running the dynamic model. **RunRBEROST-Northeast-Dynamic.Rmd** will source the necessary scripts to run each part of RBEROST through three main stages: preprocessing, NEOS optimization and postprocessing.

Scripts used in the dynamic RBEROST are in the **./RBEROST-Northeast/R/** folder, where **./** refers to the file path of the unzipped Tier_1_Optimization-SSWR.5.3.2 folder. All scripts must retain their location and naming conventions to be called by the RBEROST preprocessing scripts. Key files include

1. **RunRBEROST-Northeast-Dynamic.Rmd** as the main execution file with three code chunks for the complete workflow,
2. **01_Optimization_Preprocessing_gateway-Northeast.R** which routes RBEROST to either preprocessing with or without uncertainty based on the IncludeUncertainty parameter,
3. **01_Optimization_Preprocessing-Northeast-Seasonal_V6_[PROD].R** which creates AMPL files without uncertainty for the dynamic seasonal model,
4. **01_Optimization_Preprocessing+Uncertainty-Northeast-Seasonal_[PROD].R** which creates AMPL files with uncertainty for dynamic analysis,
5. **01_Optimization_NEOSInteraction-Northeast.R** which handles submission of AMPL files to the NEOS server for optimization and
6. **02_Optimization_RunShiny-Northeast.R** which launches the Shiny application for results visualization.

The postprocessor sources multiple files including **Optimization_ServerFile-Northeast.R**, **Optimization_ServerFunctions_Postprocessor-Northeast.R**, **Optimization_UserInterfaceFile-Northeast.R** and **Optimization_UI_Postprocessor-Northeast.R** to build the Shiny application for results analysis. The dynamic version supports multi-year analysis configurable through the model_years parameter, seasonal modeling, uncertainty analysis with multiple scenarios controlled by n.scenarios, and batch processing including large-scale sensitivity analysis. Users can select from 33 available watersheds in the Long Island Sound region and configure planning horizons, interest rates and BMP comparison methods through the **RunRBEROST-Northeast-Dynamic.Rmd** interface.

The execution workflow consists of four main stages:

1. Users first configure R by installing and loading the required packages.
2. Second, users run the preprocessor by setting file paths, selecting watersheds, configuring uncertainty parameters and generating AMPL files. Note, running RBEROST with uncertainty will automatically produce AMPL scripts both with and without uncertainty, and users can choose to save or ignore the files without uncertainty based on their analysis needs.
3. Third, users submit optimization jobs to the NEOS server with user-specified email and comments.
4. Lastly, users launch an interactive RShiny application for results visualization and analysis, where .csv files can be downloaded with BMP implementation results.

RBEROST relies on several R packages to run. To reduce errors from incompatible package updates, RBEROST uses the `renv` package to generate a local library of packages with the same versions of the packages that were used during model development. RBEROST provides the necessary code to download and install the model version of each package in the **RunRBEROST-Northeast-Dynamic.Rmd** file (Figure 3-10).

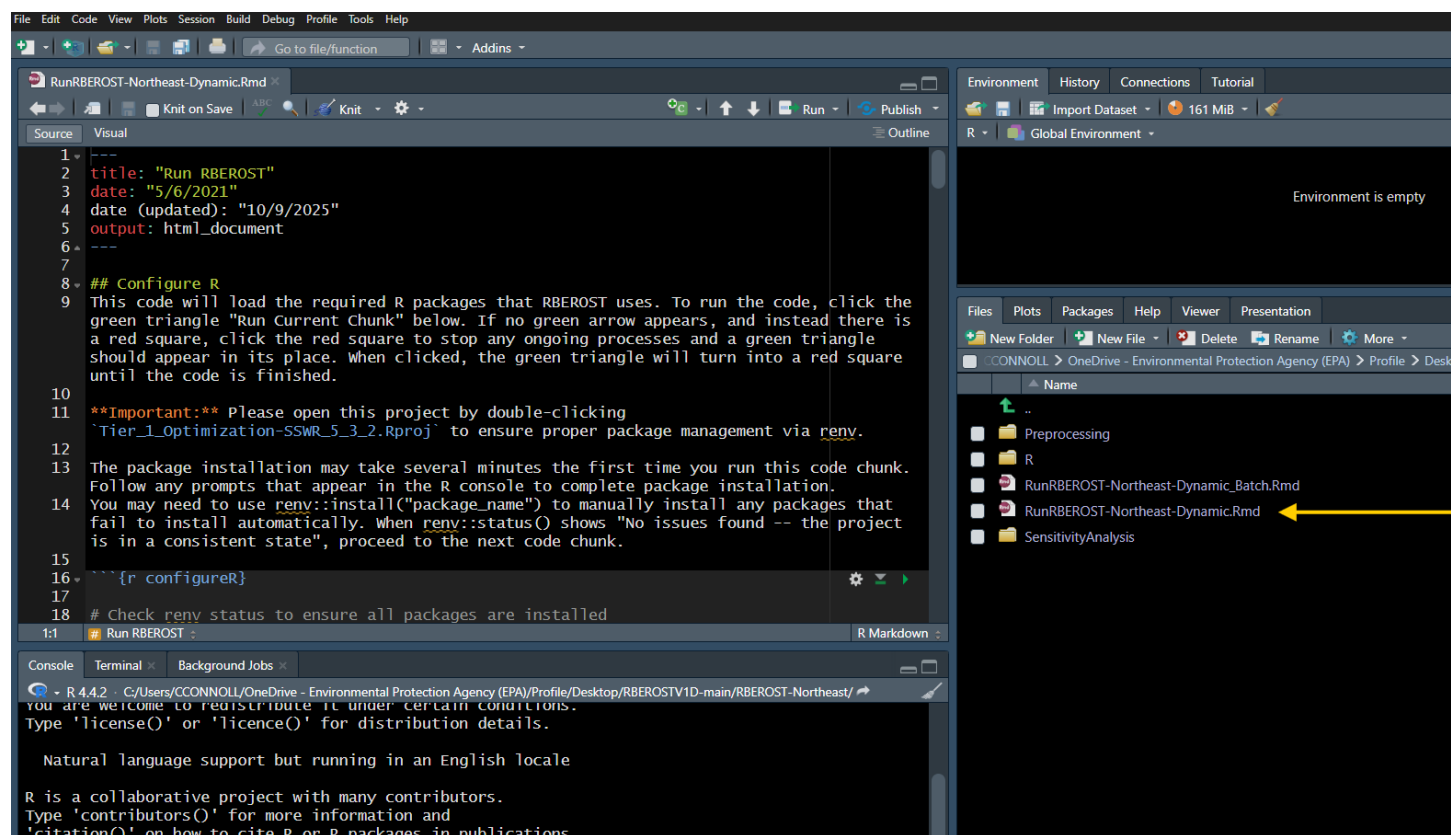


Figure 3- 10. Location of the file ‘RunRBEROST-Northeast-Dynamic.Rmd’ to open RBEROST through RStudio

The `renv` package generates a “`renv.lock`” file that keeps a record of all packages and package versions used in the RBEROST project. Users can use the `renv::status()` command to view the current state of package installation, `renv::restore()` to download and install packages from the `renv.lock` file and `renv::install()` to install packages individually if need be. Package installation can be an iterative process (Figure 3-12). During development we observed that `renv::restore()` does not always install every package on the first run. Users should follow `renv` instructions in the R console. In these cases the user may need to try multiple runs, or install the packages with `renv::install()` or `install.packages()` and then retry `renv::restore()`. Additionally, a yellow ribbon may appear at the top of the code window (Figure 3-11) which asks the user to install packages that are used in the code but not in the package library. Users can install packages by clicking “install” in the user interface and then re-running `renv::restore()`. This code will only need to be run once, the first time RBEROST is used, by clicking the green triangle on the first code chunk (indicated by the yellow arrow in Figure 3-12). While the code is running, the green triangle will change to a red square. The icon will change back to a green triangle when the code is finished. During package installation, you may be prompted with a question “Do you want to install from sources the package which needs compilation?” (Figure 3-13). Select “Yes.”

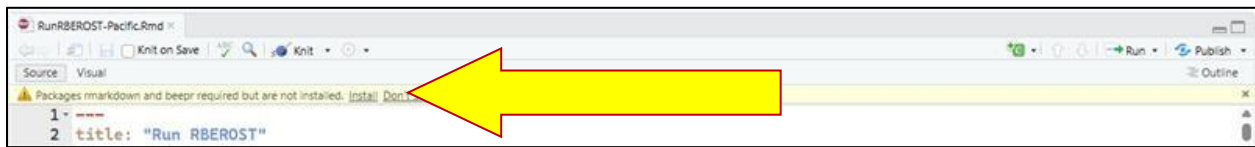


Figure 3- 11. Yellow install packages ribbon in Rstudio user interface.

Users may encounter an error when renv tries to install a program that needs compilation, e.g.,:

```
Error in system(paste(MAKE, p1(paste("-f", shQuote(makefiles))), "compilers"), :  
'make' not found
```

This means that R is trying to compile a package from source, but it cannot find the **make** build tool on your system. “make” is part of the standard development toolchain and is required for installing many R packages from source. If users encounter the above error the steps to fix it are as follows:

Install Rtools (for R 4.0+ use Rtools4.x):

1. Download from: <https://cran.r-project.org/bin/windows/Rtools/>
2. During installation, check "Add Rtools to PATH".
3. Restart R or RStudio.
4. Verify in R:

```
system("make --version")
```

You should see the version info.

```
make --version
```

Once packages are installed, users can proceed to the next code chunk where the relevant packages are loaded with `library()` as seen in Figure 3-14. If any packages are missing when the loading chunk is run, please return to the package installation chunk to install the missing packages.

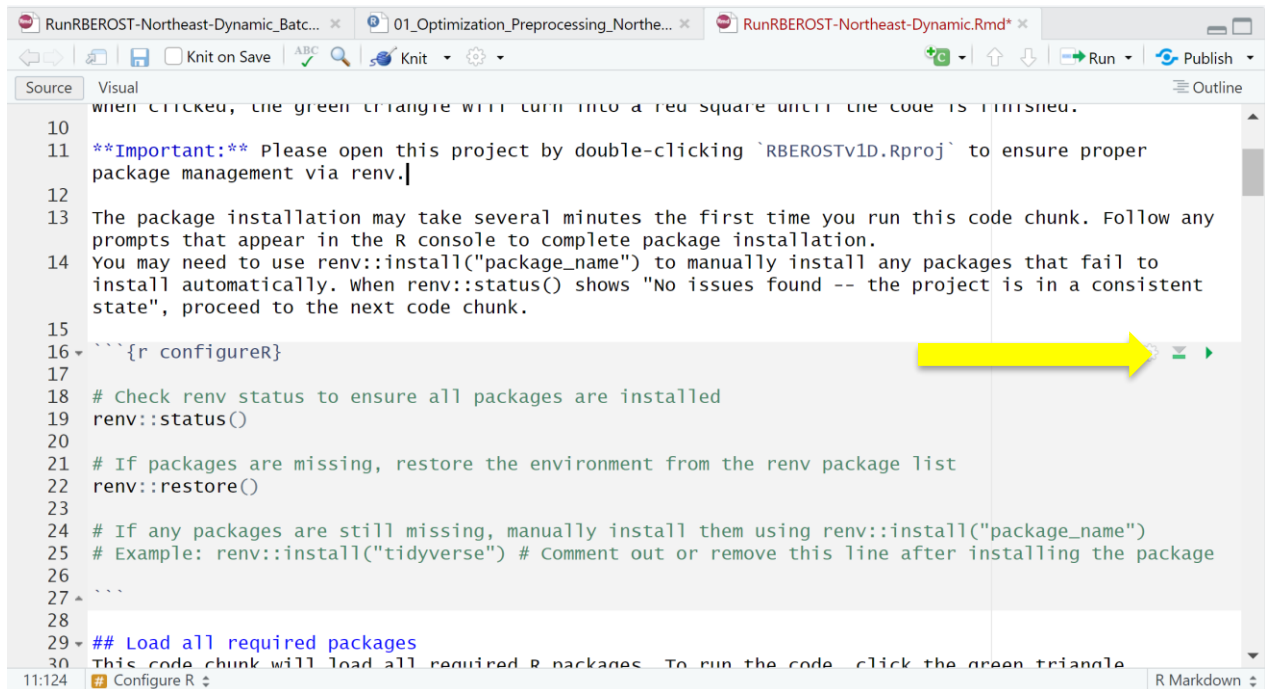


Figure 3- 12. Configuration code chunk where users will install R packages with renv by clicking on green caret.

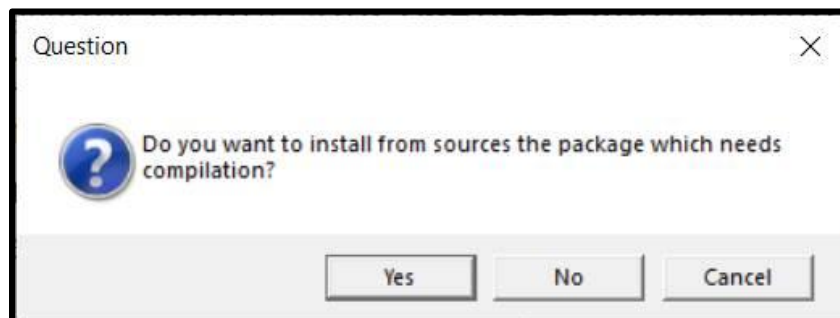


Figure 3- 13. A message users may encounter while installing packages.

```

31
32 {r loadpackages}
33 # Load required packages
34 library(tidyverse)
35 library(units)
36 library(dplyr)
37 library(measurements)
38 library(reshape2)
39 library(data.table)
40 library(stringr)
41 library(foreach)
42 library(shiny)
43 library(shinycssloaders)
44 library(tidygraph)
45 library(bit64)
46 library(beeper)
47 library(rneos)
48 library(doParallel)
49 library(magrittr)
50

```

1:1 Run RBEROST

Console Terminal Background Jobs

R 4.4.1 • C:/RBEROST/Tier_1_Optimization-SSWR.5.3.2/RBEROST-Pacific/

```

transpose
> library(stringr)
> library(foreach)
Warning: package 'foreach' was built under R version 4.4.3

```

Figure 3- 14. An image of what RStudio displays while code is running.

3.1.5 Optional code chunk to download lfs files

If the user chooses to install RBEROSTv1D from the remote repository directly or via RStudio, the user will need to download and unzip the lfs files into the Inputs folder. The next code chunk in [RunRBEROST-Northeast-Dynamic.Rmd](#) contains links to allow the user to download these files from a Zenodo archive and unzip them into the Inputs folder (Figure 3-15).

```

47 library(jquerylib)
48 library(foreign)
49
50
51 ## Download lfs files from Zenodo (required unless using git lfs to download large files)
52
53 When you download the repository from github, only placeholders for large files (>100MB) listed
54 in the .gitattributes file will be downloaded. If you are using git you can install lfs and
55 download these in git. Otherwise, you can download these files directly from a Zenodo archive
56 by cutting and pasting each of the following urls into a browser (or pointing to each of the urls
57 below and using<Ctrl-Click>) and saving each into the
58 .\RBEROSTV1D\RBEROST-Northeast\Preprocessing\Inputs folder then unzipping to that folder. Don't
59 create additional subdirectories under Inputs when unzipping any .zip files.
60
61 https://zenodo.org/record/20127975/files/ne_dynamic_sparrow_se_tn_final.zip?download=1
62 https://zenodo.org/record/20127975/files/Predict_NoBFlowN_wSeptic_final.zip?download=1
63 https://zenodo.org/record/20127975/files/Predict_P_final.zip?download=1
64 https://zenodo.org/record/20127975/files/ne_dynamic_sparrow_se_tp_final.zip?download=1
65 https://zenodo.org/record/20127975/files/ne_sparrow_model_input.zip?download=1
66 https://zenodo.org/record/20127975/files/RiparianEfficiencies_ICF25.zip?download=1"

```

Figure 3- 15. Optional code chunk for downloading large (lfs) files to Input folder to replace placeholders from github.

3.2 Preprocessing

This section details the steps required to develop the AMPL model files that are run through the NEOS server to solve the optimization. The user is required to update the Run Preprocessor section of the

code in **RunRBEROST-Northeast-Dynamic.Rmd** as well as the two .csv files **01_UserSpecs_BMPs.csv** and **01_UserSpecs_loadingtargets.csv**, which are in the **.Preprocessing/Inputs** folder (Figure 3-16). To run the preprocessing step, first edit the UserSpecs files and the available options within **RunRBEROST-Northeast-Dynamic.Rmd.-Northeast-Dynamic.Rmd**. Please update all the required input and output pathways with verified file paths. Incorrect file paths are a significant source of errors in RBEROST. A more detailed description of the various options available to the user in the file is included in section 4.2.1, Users may choose to alter various components of the UserSpec files, such as the **01_UserSpecs_BMP.csv** file, to better suit their objectives and additional data. Sections 3.2.1 and 3.2.2 as well as Appendix C – User Parameter Change Checklist contain additional information on how users may modify files to create their own optimization runs. The seasonal RBEROST model contains more variables that users must define prior to optimization runs compared to the original static RBEROST model. Costs, efficiencies, loadings and BMPs selected can each vary across time as well as geographic extent. Once input files and options have been set, click the green triangle in the upper right of the preprocessing code chunk. The green triangle will turn into a red square until the code is done running, at which point the red square will return to a green triangle. Users may encounter warning messages when running the preprocessor that displays warning about the creation of NA values. These warning messages do not reflect errors in the data and are normal occurrences that may happen when running the preprocessor. Users may also encounter warnings about deprecated functions or syntax. These warnings are typically due to package version differences from the development versions and do not typically impact the function of the model.

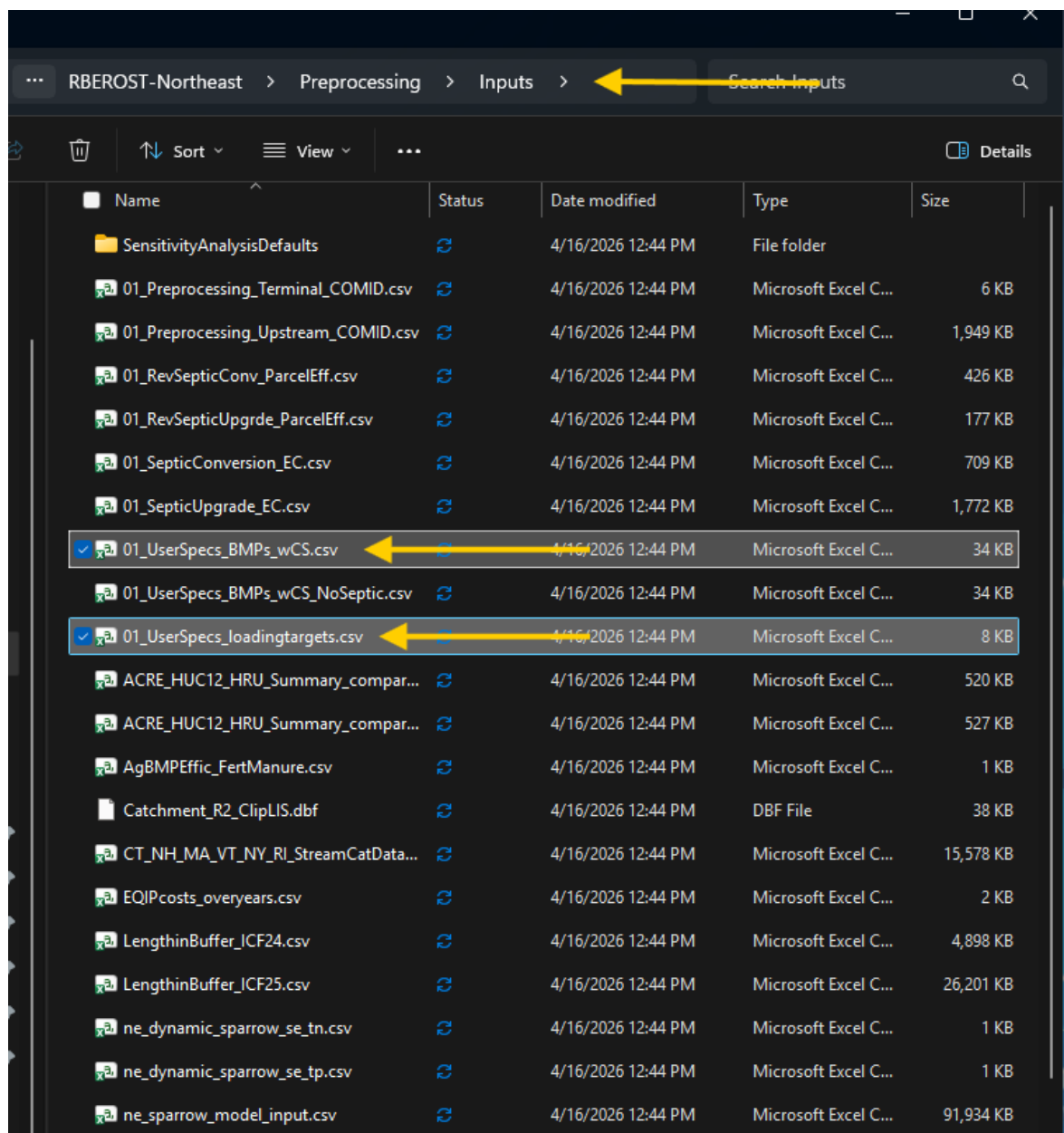


Figure 3- 16. File path and location of the two User Specification csv files used in RBEROST. User Specifications

3.2.1 User Specifications

3.2.1.1 User Settings within R Markdown files

There are many parameters that users can change in the **RunRBEROST-Northeast-Dynamic.Rmd** file (Figure 3-17). A summary of these variables is included in Table 3-4. The working directory must be specified. This is the base location for the user's local repository (path up to but not including RBEROST-Northeast subdirectory and including final "/"). These include **InPath**, which should point to the folder with the Preprocessing inputs, and **OutPath** which should point to the folder that the AMPL files will be written to. If RBEROST is opened from the **RBEROST.Rproj** file, and if file structure is left intact, these parameters do not need to be changed. However, if the user runs into issues with long path names (e.g., after installing the local repository within One Drive) then the user may choose to change the OutPath

so the final path for output files is shorter. The **user_specs_name** is the name of the csv file in the Inputs data folder that is provided for the user to select management actions to consider, and which contains BMP costs and efficiencies. Next, the **MODE** and **watershed_choices** variables determine which watersheds are modeled. **MODE** should be set to "Select". Only watersheds named in the **watershed_choices** vector will be modeled due to file size constraints. The data provided cover the geographic extent of 33 watersheds that drain to the Long Island Sound. A map of labeled watersheds included in the model can be found in Appendix A – Maps of Long Island Sound Model Spatial Extent. Furthermore, the **model_years** variable determines the number of years the model will be projected over. **Model_years** can accommodate runs of 1-20 years. SPARROW data does not exist beyond year 20 (2020).

Table 3-4. List of parameters that can be modified by user within RunRBEROST-Northeast-Dynamic.Rmd.

Parameter	Must change?	Default/ Example	Description
working_dir	Yes	"C:/RBEROSTV2D/"	Working directory for local repository (just above RBEROST-Northeast folder), must end in "/"
InPath	No	paste0(working_dir,"RBEROST-Northeast/Preprocessing/Inputs/")	Input path for data and codes.
OutPath	No	paste0(working_dir,"RBEROST-Northeast/Preprocessing/Outputs/")	Output path for data and codes.
user_specs_name	No	01_UserSpecs_BMPs_wCS.csv	Table of management actions with cost and efficiencies. Substitute 01_UserSpecs_BMPs_wCS_NoSeptic.csv to exclude septic BMPs.
MODE	No	Select	Set to "All" for entire model, or "Select" to run a subset of watersheds.
watershed_choices	Yes	c("Lower Connecticut River")	List of watersheds to run if MODE is set to "select"
model_years	No	c("03")	Continuous vector of years to model.
horizon	No	15	Cost annualization horizon.
interest_rate	No	0.03	Cost annualization interest rate.
actionable_load	No	TRUE	Set to TRUE to remove load types that cannot be reduced, or FALSE to model all relevant load types.
AgBMPcomparison	No	"No Practice"	Determines how Ag BMP efficiencies are determined in preprocessing.
IncludeUncertainty	Yes	No	Include uncertainty in model results if desired.
n.scenarios	No	3	Number of uncertainty scenarios.
scenariostepchange	No	0.01	Variation between each uncertainty scenario.
email	Yes	"example@email.com"	Email address where NEOS results should be sent in case of long run times (NOTE: may be blocked by firewall)
NEOSresults	No	paste0(working_dir,"RBEROST-Northeast/Preprocessing/Outputs")	
paramname	No	paste0(paste(watershed_choices, collapse = "_"), "_", Sys.Date())	Filename suffix for results

Other parameters include **horizon** which is the planning horizon used to annualize costs. The default value is 15 years. The variable **interest_rate** is the expected interest for the project, and the default value is 0.03, or 3 percent. The variable **actionable_load** determines whether the category of “other loads” is included in the model. Setting **actionable_load** to TRUE results in the exclusion of loads from other sources (“other sources”) from both the calculation of targets as well as the calculation of incremental storage and total load constraints. Note that setting this variable to FALSE, resulting in the inclusion of other loads, is much more likely to result in an infeasible result because other loads cannot be reduced and can often constitute a substantial portion of overall loads. Table 3- provides examples of total baseload breakdown by source category for two example watersheds in the first year. Note that the contribution of point sources and “Other Sources” can vary significantly by season. Spring snowmelt is a significant contributor to springtime loads in some watersheds. When utilizing RBEROST to model various scenarios, it should be noted that changing the **actionable_load** parameter changes the model implications significantly. Reducing nutrient loads for which BMPs are available below a given threshold does not necessarily imply meeting overall water quality goals within a water body. Users may try multiple runs and vary the **actionable_load** parameter and reduction target values to determine how much of the full load (including nonactionable loads) can be reduced.

Table 3-5. Estimated distribution of total watershed baseloads (fraction of total baseload) broken down by load source category.

Watershed (Year / Season)	Point Sources	Urban Sources	Ag Sources	Septic Sources	Other Sources
Niantic Bay (1/1)	0.0000	0.5307	0.2436	0.0001	0.2256
Niantic Bay (1/2)	0.0000	0.5537	0.1273	0.0003	0.3185
Niantic Bay (1/3)	0.0000	0.5662	0.0672	0.0005	0.3653
Niantic Bay (1/4)	0.0000	0.5402	0.1635	0.0002	0.2957
Connecticut River (1/1)	0.5165	0.0352	0.1949	0.0001	0.2533
Connecticut River (1/2)	0.1051	0.0135	0.0712	0.0001	0.8101
Connecticut River (1/3)	0.2530	0.0288	0.1127	0.0004	0.6050
Connecticut River (1/4)	0.1194	0.0447	0.2578	0.0002	0.5779

AgBMPcomparison can be set as “No Practice” or “Baseline” and determines how the efficiencies of agricultural BMPs will be determined using data from the USDA ACRE dataset. The ACRE data set was constructed by using SWAT and APEX to model the potential effectiveness of individual BMPs in reducing nutrient loads at the field scale for thousands of potential sites with different weather, soils, slope and crop type as compared to a background condition of either no pre-existing conservation practices or estimates of existing conservation practices. The latter estimates were based on a statistical survey of agricultural conservation practices using aerial photography (White et al. 2019). “No Practice” represents the efficiency of a practice in isolation compared to conventional agriculture, and “Baseline” represents the efficiency of adding to estimates of existing conservation practices and replacing them with a certain practice. **IncludeUncertainty** is a logical parameter and can be set as TRUE or FALSE. If TRUE, RBEROST will create AMPL scripts that include uncertainty calculations. If FALSE, it will not. Making AMPL scripts that include uncertainty may take several minutes. Variables **n.scenarios** and **scenariostepchange** only apply when running RBEROST with uncertainty. The variable **n.scenarios** sets the number of scenarios you would like to calculate for the uncertainty analysis, and **scenariostepchange** sets the relative difference between these scenarios. The different scenarios describe increasingly restrictive loading targets to be applied after the first scenario is solved with the user specifications from **01_UserSpecs_loadingtargets.csv** (Figure 3-16). Each successive scenario decreases the loading targets by the percentage given in **scenariostepchange**. This variable

scenariostepchange is equivalent to the additional margin of safety applied to each scenario. The default is to compute three scenarios with a 1 percent change in loading targets between scenarios. The lines of code to edit are marked with arrows in Figure 3-17.

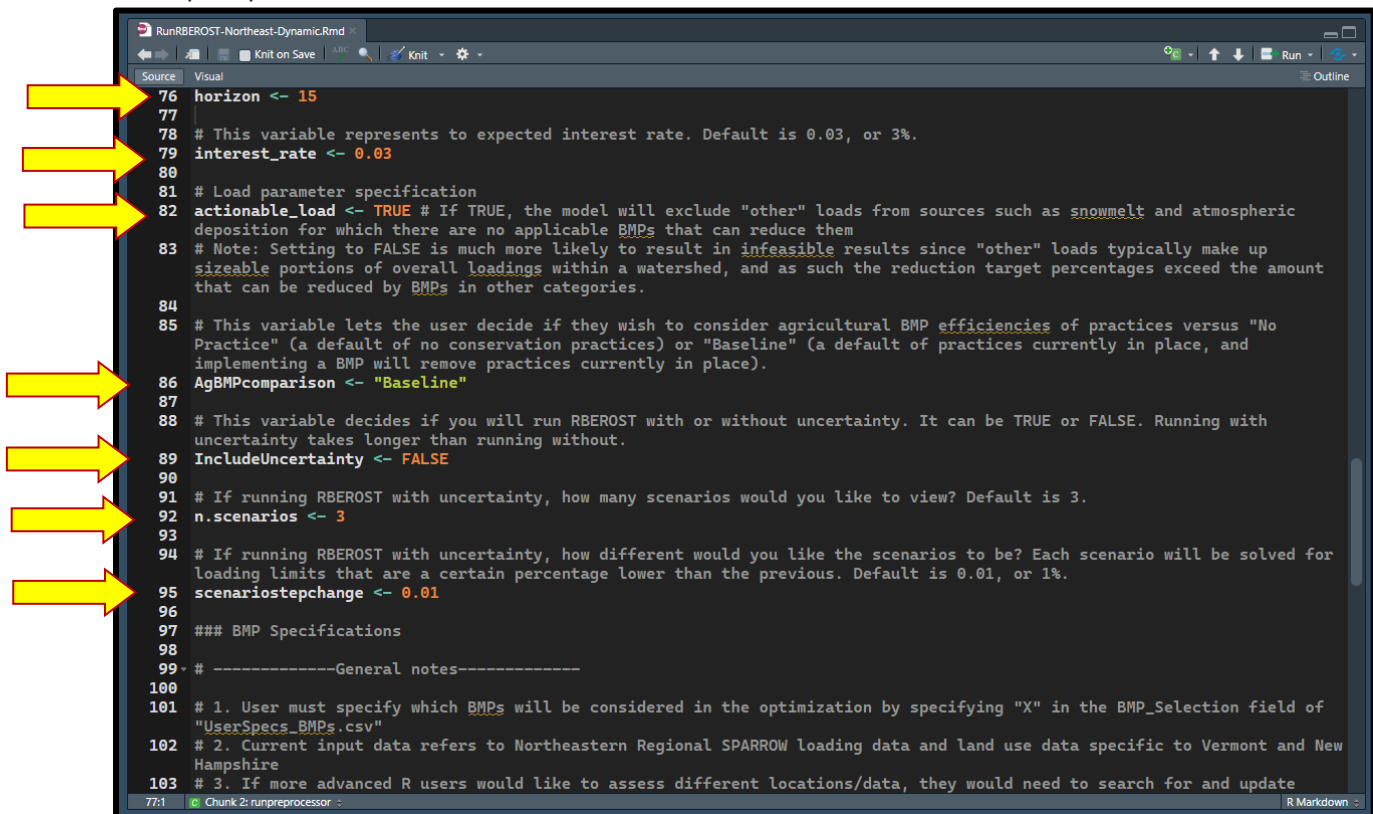
3.2.1.2 User Specifications for Management Action Inclusion, Costs and Efficiencies

Within the **01_UserSpecs_BMPs.csv** file, users also have the option to decide which management practices to be considered, as well as to modify estimated costs and efficiencies. (Figure 3-18). BMP options are specified by placing a capitalized “X” in the

BMP_Selection field of the **01_UserSpecs_BMPs.csv** file. Users can also choose a minimum and/or maximum implementation of each BMP as a fraction of total possible implementation.

01_UserSpecs_BMPs.csv includes all the BMPs available for model optimization per BMP category (agricultural, urban, point source and riparian buffer BMPs), state-specific capital and operations & maintenance costs to implement the BMP, units of the costs, user-specified design runoff depths for urban BMPs and buffer widths for riparian buffer BMPs. A generic agricultural BMP can be added by the user if desired if cost and efficiency data are available. When adding a BMP to the

01_UserSpecs_BMPs.csv do not use any special characters or spaces (e.g. /, [,], &, etc) to the BMP name column. Only use letters, numbers and underscores. Special characters and spaces produce errors in the post-processor.



```

76 horizon <- 15
77
78 # This variable represents to expected interest rate. Default is 0.03, or 3%.
79 interest_rate <- 0.03
80
81 # Load parameter specification
82 actionable_load <- TRUE # If TRUE, the model will exclude "other" loads from sources such as snowmelt and atmospheric
83 # Note: Setting to FALSE is much more likely to result in infeasible results since "other" loads typically make up
84 # sizeable portions of overall loadings within a watershed, and as such the reduction target percentages exceed the amount
85 # that can be reduced by BMPs in other categories.
86
87 # This variable lets the user decide if they wish to consider agricultural BMP efficiencies of practices versus "No
88 # Practice" (a default of no conservation practices) or "Baseline" (a default of practices currently in place, and
89 # implementing a BMP will remove practices currently in place).
90 AgBMPcomparison <- "Baseline"
91
92 # This variable decides if you will run RBEROST with or without uncertainty. It can be TRUE or FALSE. Running with
93 # uncertainty takes longer than running without.
94 IncludeUncertainty <- FALSE
95
96 # If running RBEROST with uncertainty, how many scenarios would you like to view? Default is 3.
97 n.scenarios <- 3
98
99 # If running RBEROST with uncertainty, how different would you like the scenarios to be? Each scenario will be solved for
100 # loading limits that are a certain percentage lower than the previous. Default is 0.01, or 1%.
101 scenariostepchange <- 0.01
102
103 ### BMP Specifications
104
105 # -----General notes-----
106
107 # 1. User must specify which BMPs will be considered in the optimization by specifying "X" in the BMP_Selection field of
108 # "UserSpecs_BMPs.csv"
109 # 2. Current input data refers to Northeastern Regional SPARROW loading data and land use data specific to Vermont and New
110 # Hampshire
111 # 3. If more advanced R users would like to assess different locations/data, they would need to search for and update

```

Figure 3- 17. A sample of lines of code for the user to edit in the preprocessing step.

BMP_Cat	BMP	BMP_Selection	frac_min	frac_max	capital_VT	capital_NH	capital_NY	capital_MA	capital_CT	capital_RI	operations	operations	operations	operations	operations
ag	Conservation	X	0	1	19.42	21.56	19.42	19.5	19.46	19.71	0	0	0	0	0
ag	Contour_Farming	X	0	1	8.97	9.9	9.24	9.24	9.1	9.2	0	0	0	0	0
ag	Filterstrip	X	0	1	0	0	0	0	0	0	143.7011	88.88516	144.1682	143.6738	143.773
ag	MIN_TILL	X	0	1	24.22	26.58	24.85	24.22	24.33	24.46	0	0	0	0	0
ag	Ponds	X	0	0.1	42253.2	10454.4	48738.8	32815.2	43027.6	43414.8	0	0	0	0	0
ag	Terrace_Waterway	X	0	1	2870.428	3097.814	2908.18	2983.495	3099.466	3090.754	0	0	0	0	0
ag	Terrace_Only	X	0	1	365.904	394.944	403.656	374.616	386.232	377.52	0	0	0	0	0
ag	Waterway_Only	X	0	1	2504.524	2702.87	2504.524	2608.879	2713.234	2713.234	0	0	0	0	0
ag	Fert_20	X	0	1	32.07	33.34	32.83	34.8	33.28	33.88	0	0	0	0	0
ag	Manure_Injection	X	0	1	50	50	50	50	50	50	0	0	0	0	0
ag	Cover_Crop	X	0	1	0	0	0	0	0	0	1127.793	1118.405	1207.414	1174.838	1147.075
point	CytecIndustrialP	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	SummitCorporationofAmeX	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	PharmaciaUpjohnComparX	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	KimberlyClarkCorporationX	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	ElectricBoatCorporationTfX	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	EastHamptonWPCFCRW	X	NA	NA	NA	NA	NA	NA	8514332 NA	NA	NA	NA	NA	NA	156927.2 NA
point	UniMetalSurfaceFinishingX	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	AnsoniaWPCFHRW	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	BranfordWPCF	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	BridgeportWestWPCF	X	NA	NA	NA	NA	NA	NA	3211661 NA	NA	NA	NA	NA	NA	55972.12 NA
point	NorthCaanWPCFHRW	X	NA	NA	NA	NA	NA	NA	38858006 NA	NA	NA	NA	NA	NA	264229.7 NA
point	CantonWPCFCRW	X	NA	NA	NA	NA	NA	NA	25056618 NA	NA	NA	NA	NA	NA	212114.5 NA
point	CheshireWPCF	X	NA	NA	NA	NA	NA	NA	0 NA	NA	NA	NA	NA	NA	0 NA
point	DanburyWPCFHRW	X	NA	NA	NA	NA	NA	NA	3211664 NA	NA	NA	NA	NA	NA	78928.67 NA
point	DerbyWPCFHRW	X	NA	NA	NA	NA	NA	NA	8624097 NA	NA	NA	NA	NA	NA	157391.5 NA

Figure 3- 18. A screenshot of the BMP UserSpecs file with example BMP selections.

Implementation of the majority of the BMPs requires both capital and operations & maintenance costs, except for several agricultural BMPs. All default costs are scaled to 2025\$ or 2026\$ (agricultural BMPs). The user can override these costs if they choose costs that are representative of costs per treated area for agricultural and riparian buffer BMPs, costs per treated volume for urban BMPs and costs per retrofit implementation for point source BMPs. Urban costs are specified assuming implementation in new developments rather than with retrofitting. RBEROST will adjust these costs internally based on the expected amount of retrofitting and complexity of installations in each catchment as judged by development intensity.

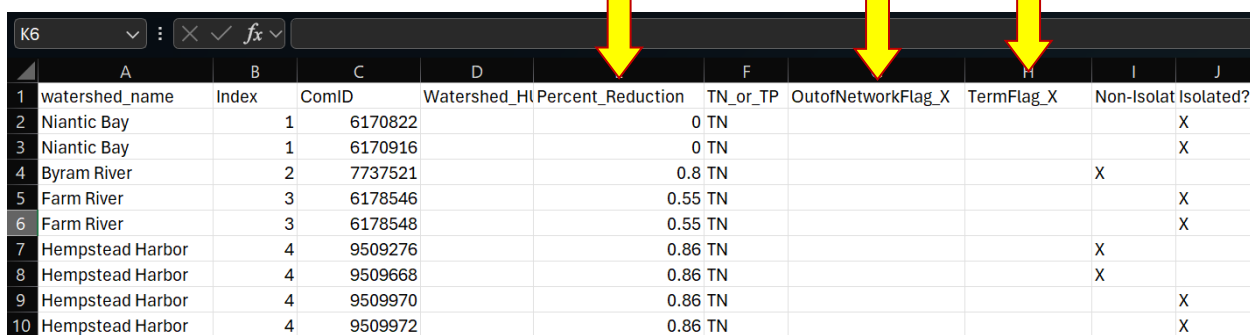
If cost data from additional states beyond the geographic extent of the included LIS watersheds must be included, these are added to the file **01_UserSpecs_BMPs.csv**. For each additional state, two columns must be added to this file, following the naming convention of “**capital_[two letter state code]**” and “**operations_[two letter state code]**”. The units for these costs must be the same across all states included, and units are indicated in the “**capital_units**” and “**operations_units**” fields. Possible units are “**ac**”, “**ft2**”, “**yd2**” and “**km2**” for agricultural and riparian buffer BMPs, “**ft3**” for urban BMPs and “**flat**” for point source BMPs. If optimizations that include uncertainty are performed for geographic ranges that extend to states beyond the ones in the case study, additional columns must also be added to the file **EQIPcosts_overyears.csv**. Column headings should follow the naming conventions of “**capital_[two letter state code]_[year]**” or “**operations_[two letter state code]_[year]**”, and cost units must be consistent across all columns.

Nutrient removal efficiency resulting from implementation of urban BMPs depends on the BMP design runoff depth. The **01_UserSpecs_BMPs.csv** file specifies default runoff depths for BMPs, but the user may specify alternate runoff depths in the **UserSpec_RD_in** field. The **Min_RD_in** and **Max_RD_in** fields specify the minimum and maximum runoff depths (i.e., upper and lower limit) that the user can specify (in inches) for the urban BMP **UserSpec_RD_in** field. RBEROST does not use the **Min_RD_in** and **Max_RD_in** fields for any calculations or data modifications, the fields are just for reference. Specification of riparian buffer width also affects the amount of nutrients removed by the buffers. Users can specify buffer widths in the same column, **UserSpec_RD_in**. A note is provided that, though presented to the user in the same column, the units for riparian buffer BMPs are different than for urban BMPs.

3.2.1.3 User Specifications for Load Reduction Goals

Users can modify loading targets (percent nutrient load reduction) by editing the **01_UserSpecs_loadingtargets.csv** (Figure 3-19) or by creating alternative versions of potential loading targets with adjusted file names and substituting these loading target file names in the Rmd file as described above. The loading targets included in the

01_UserSpecs_loadingtargets.csv are placeholders derived from the TetraTech (2022) summary as estimates of loads required to meet ecological endpoints in selected embayments. If the user is interested in determining what the maximum percent reduction is that can lead to a feasible optimization outcome across multiple watersheds, it may be more efficient to create multiple versions of the loading targets file with sequential changes in percent reductions. Target reductions are described as a percent reduction from current loading. Users can specify whether targets are for Total Nitrogen (TN) or Total Phosphorus (TP). If targets are not within the river network, they can be flagged with a capital X in **OutofNetworkFlag_X**. If the target is the terminal end of a watershed, it can be flagged with a capital X in **TermFlag_X**. **COMID** refers to the NHDPlus V2 catchment that contains the loading target. Once a user selects a percent reduction for a given watershed, the preprocessor will attempt to reduce the baseload for that watershed by the percent reduction given every season of every year chosen for the model. For example, if a user selects a 5% reduction for the Upper Middle Connecticut River watershed and a 2 year model, the model will seek to individually reduce the spring, summer, fall and winter baseloads by 5% for both years.



	A	B	C	D	E	F	G	H	I	J
	watershed_name	Index	ComID	Watershed_HI	Percent_Reduction	TN_or_TP	OutofNetworkFlag_X	TermFlag_X	Non-Isolat	Isolated?
1	Niantic Bay	1	6170822		0	TN				X
2	Niantic Bay	1	6170916		0	TN				X
3	Byram River	2	7737521		0.8	TN			X	
4	Farm River	3	6178546		0.55	TN				X
5	Farm River	3	6178548		0.55	TN				X
6	Hempstead Harbor	4	9509276		0.86	TN			X	
7	Hempstead Harbor	4	9509668		0.86	TN			X	
8	Hempstead Harbor	4	9509970		0.86	TN				X
9	Hempstead Harbor	4	9509972		0.86	TN				X

Figure 3- 19. A screenshot of the loading targets UserSpecs loading target file with the specifications.

3.2.2 Preprocess Data Inputs

The data preprocessing section of the code adjusts and formats the input data for AMPL model file development. Input data include baseline nutrient loads, land use information, watershed characteristics, BMP costs and BMP nutrient removal efficiencies. Users may choose to adjust preprocessing data inputs according to more recent information beyond the default information

included in the model. Note, however, that the inclusion of new BMPs would require revision of multiple preprocessing input files as well as code script modification. The provided data covers the geographic extent of the 33 watersheds that drain to the Long Island Sound or upstream lake TMDL watersheds in the Long Island Sound Basin only.

3.2.2.1 *Baseline Nutrient Loading and Land Use Data*

Recent regional SPARROW model outputs from the U.S. Geological Survey provide the basis for RBEROST baseline nutrient loading conditions (Moore et al., 2026). These models provide catchment-level nutrient loads and land use data by season and year. Catchments are specified using common identifiers (COMIDs) based on the National Hydrography Dataset (NHD) Plus Version 2 reach network (McKay et al., 2012). RBEROST also includes 2019 cropland and imperviousness land use data from the National Land Cover Database (Yang et al., 2018) provided at the NHDPlus V2 catchment level via StreamCat (Hill et al., 2015). For modeling that includes states outside of the region presented in the case study, StreamCat files for additional states can be added by changing the state abbreviations in the code such that the StreamCat API pulls different state data. (Search for the word “StreamCat” in the following files:

00_AccessStreamCatdata-Northeast.R,
01_Optimization_Preprocessing_Northeast_Seasonal_V6_PROD.R and
01_Optimization_Preprocessing_Uncertainty_Northeast_Seasonal_PROD.R.)

Data modifications entail classifying incremental baseline SPARROW loadings as point source, urban, agricultural, septic, or “other” loads. Incremental loads in SPARROW are stream loads that originate from the NHDPlus V2 catchment. This is differentiated from total loads in SPARROW which is the sum of incremental load for each catchment and delivered load from upstream. Other loads include loadings from sources that cannot be classified as point, urban, agricultural and septic (e.g., atmospheric deposition).

The optimization depends on agricultural and urban land area available for BMP implementation. Urban acreage data is calculated as the product of acreage available per catchment (from SPARROW; Schmadel et al., 2021) and the percent of developed land area, as defined as the sum of the share of land covered by four land use categories: Developed – Open space, Developed – Low intensity, Developed – Medium intensity, Developed – High intensity (from National Land Cover Data/StreamCat; Hill et al., 2015). Agricultural acreage is calculated as the product of incremental acreage available per catchment (from SPARROW; Ator, 2019) and the percent of area defined as cropland per catchment (from National Land Cover Data/StreamCat; Hill et al., 2015).

Urban BMP costs depend on the volume of water being treated. RBEROST relies on a water quality volume equation from the Vermont Stormwater Management Manual (Vermont Agency of Natural Resources, 2017). Water quality volume depends on a volumetric runoff coefficient, equal to $0.05 + 0.009 * \text{Imperviousness}$. The National Land Cover Dataset (NLCD) provides imperviousness data at the catchment level via StreamCat (Hill et al., 2015).

Riparian loading depends on the amount of impervious surface cover within the river corridor, as per Houle et al. (2019). These estimates are derived prior to modeling with RBEROST. They are provided in the [RiparianLoadings.csv](#) file. Riparian loading is a subset of urban, ag, point, or other loads and does not describe an additional source. Users who wish to use different methodologies to estimate riparian loads (e.g., including agricultural loads) may do so and overwrite the [RiparianLoadings.csv](#) file with a new file with the same naming conventions and format.

3.2.2.2 Watershed and Target Specifications

The `01_UserSpecs_loadingtargets.csv` file describes the locations of loading targets that will be included in RBEROST. The COMID identifiers are used to identify upstream contributing reaches to each target. These contributing reaches are included in the input baseline loading and land use datasets used in calculating inputs to each target waterbody.

The SPARROW regional models specify a delivery fraction (DEL_FRAC) that reflects the fraction of the incremental nutrient loads that are delivered to a flowline's terminal reach (in most cases, the ocean). RBEROST revises this delivery fraction to recognize each specified upstream loading target as the specified terminal reach. For instance, if the specified target COMID has a delivery fraction of 92 % in the SPARROW regional model dataset, indicating that 92 % of the incremental loads associated with the target reach are delivered to the terminal reach (ocean), the revised delivery fraction at this new target location is 100 %, indicating that 100 % of the incremental loads associated with the target are delivered to the corresponding target reach.

The delivery fractions for all reaches upstream of the target are revised based on the ratio of the revised target delivery fraction to the original terminal reach delivery fraction. For instance, if the SPARROW regional model data indicates that an upstream reach delivers 83 % of its incremental load to the terminal reach and that the target reach delivers 92 % of its incremental load to the terminal reach, then the revised delivery fraction for the upstream reach to the target reach is $(100\%/92\%)*(83\%) = 90\%$. Delivery fractions typically differ between TN and TP, and these calculations are performed separately for each nutrient target. RBEROST allows nested targets as well, in which case a stream reach that contributes to more than one loading target will be assigned more than one DEL_FRAC value, one for each target location.

3.2.2.3 BMP Costs and Efficiencies

BMP costs specified in the `01_UserSpecs_BMPs.csv` file are adjusted to per-acre units for agricultural BMPs, and per-square foot units for riparian buffer BMPs. Because point source BMPs are location-specific, the model code merges the wastewater treatment plant (WWTP) costs with each plant's reach location (COMID) prior to developing the AMPL model files. Default urban BMP capital costs represent new development costs. RBEROST will adjust costs for expected amounts of retrofits based on retrofit costs specified in Table 7-2 of the WMOST v3 Theoretical Documentation (Detenbeck et al., 2018). This adjustment is a weighted average of the new development cost (assigned to open and low development), retrofit costs (assigned to medium development) and difficult retrofit costs (assigned to high development). The development data originate from the NLCD (Yang et al., 2018) and are summarized by catchment (Hill et al., 2015). As in the WMOST model, default urban BMP operations and maintenance costs are assumed to be 5 % of capital costs, but this can be modified by the user if more accurate estimates are available.

RBEROST assumes a default planning horizon of 15 years and an interest rate for capital costs of 3 %. The user can adjust the horizon and interest_rate variables within the `RunRBEROST-Northeast-Dynamic.Rmd` file.

While some BMP efficiencies are BMP-specific, others are also location-specific. For agricultural BMPs with nutrient removal efficiencies that vary based on HUC12, the model code identifies the individual reaches that fall within each HUC12 and assigns the HUC12-specific efficiency to each reach, using either the "No Practice" or "Baseline" scenarios as specified by the user in the .Rmd file.. Point source BMP efficiencies are specific to each WWTP upgrade and urban BMP efficiencies are calculated based on the

user-specified runoff depth. As some urban BMP efficiencies are dependent on the infiltration rate of the soil underlying developed land, the urban BMP efficiency curves are selected based on the average infiltration rate. This information is provided in **NHD+infiltrationrates.csv** and is derived from gNATSGO soil layers (Soil Survey Staff, 2020b, 2020a).

Riparian buffer BMP efficiencies similarly depend on average infiltration rate, as well as the slope of the riparian buffer. The efficiency curves (as a function of buffer widths) of different riparian buffer BMPs for different nutrients at five representative buffer widths are provided in **RiparianEfficiencies.csv**. These efficiency curves have been assigned based on infiltration rate and slope of different buffer widths in each COMID. RBEROST will use the efficiency curve based on the closest match to the user-selected buffer width.

3.2.3 Write AMPL Model Files

Running the optimization model through the NEOS server requires the submission of three AMPL model files: the command file (.amp), the data file (.dat) and the model file (.mod). The command file contains instructions for the optimization server to solve the mathematical problem and defines what results should be displayed afterwards. The data file contains all the data used in the optimization, including values for each parameter across their geographic range. The model file defines the mathematical problem to be optimized and contains a list of constraints placed on the model during optimization. After running the preprocessing code, the AMPL model files will by default be written to the **.Preprocessing/Outputs** folder, unless otherwise specified in the **RunRBEROST-Northeast-Dynamic.Rmd** file (Figure 3-20). If the user chooses to run the model with uncertainty, these command, data and model files will be designated by “**_uncertainty**” in their name.

Several messages appear while running the RBEROST preprocessor (Figure 3-23). The first few indicate whether there are more COMIDs in the StreamCat datasets or in the SPARROW dataset of the contributing reaches to each target. The number of messages will match the number of targets. SPARROW COMIDs that are not in StreamCat, or StreamCat COMIDs that are not in SPARROW contribute to the `other_loads` parameters. There will also be messages related to the choice of actionable load components and how those have been processed for each target. The next messages will indicate when RBEROST has completed writing AMPL scripts. When running RBEROST with uncertainty analysis, AMPL scripts without uncertainty will be written first as part of the preprocessing. These AMPL files without uncertainty may be ignored or saved for further use.

Name	Status	Date modified	Type	Size
STcommand_dynamic_uncertainty.amp	✓	1/29/2026 10:32 AM	AMP File	7 KB
STcommand_seasonal.amp	✓	1/29/2026 10:31 AM	AMP File	1 KB
STdata_dynamic_uncertainty.dat	✓	1/29/2026 10:32 AM	DAT File	6,362 KB
STdata_seasonal.dat	✓	1/29/2026 10:31 AM	DAT File	3,382 KB
STmodel_dynamic_uncertainty.mod	✓	1/29/2026 10:32 AM	Movie Clip	72 KB
STmodel_seasonal.mod	✓	1/29/2026 10:31 AM	Movie Clip	69 KB

Figure 3- 20. An example of the outputs of running RBEROST preprocessor with and without uncertainty.

3.2.3.1 AMPL Command File

The command file specifies display characteristics of the NEOS optimization results. If the model is being run with uncertainty, the command file specifies the number of scenarios to run and alters the loading limits accordingly for each successive solve. The command file also calculates 200 ‘bootstraps,’ of cost and total loading based on the solved model’s suggested suite of BMPs. These bootstraps are 200 calculations of cost and total loading based on random sampling of values for all the necessary parameters in RBEROST. For most parameters, this is a normal distribution, though several parameters are described as uniform distributions. The mean value used for the resampling is the value used by RBEROST to solve the optimization problem, and the standard errors are additional parameters written to the **STdata_undertainty.dat** file.

3.2.3.2 AMPL Data File

The data file specifies the catchment- and BMP-specific data by season and year required for the NEOS optimization run, including:

- Baseline loadings, efficiency and cost data formatted by the R code;
- Total reduced loading values (loads_lim parameters) calculated based on the specified load reduction percentage;
- Total loading value for loads specified as “other loads” (other_loads parameters) that are not available for BMP loading reduction;
- The fraction of agricultural costs (agcost_frac) that reflect base payment versus actual agricultural BMP costs;
- The adjustment factor of urban costs (urban_cost_adjustment_coef); and
- Conversion factors for urban BMP treated water volume (acfttoft3) and precipitation (pcp).

These parameters are described in greater detail in Table 3-2. Additional parameters written when modeling with uncertainty are described in Table 3-3.

Depending on the geographic extent of optimization, the data file with uncertainty may become larger than the allowed upload size for NEOS. If this occurs, it is recommended to proceed using the AMPL scripts that do not contain the uncertainty analysis, or if multiple years are included, reduce the number of years in the analysis.

3.2.3.3 *AMPL Model File*

The model file specifies the following:

- COMIDs, BMPs, cost types and load types that are included in the optimization;
- The bases on which parameters vary (e.g., agricultural efficiencies vary by both reach and BMP);
- Constraints on parameters and variables (e.g., whether a variable represents binary conditions or a fraction less than or equal to 1);
- The total load reduction functions; and
- The cost minimization objective function.

The model file with uncertainty includes the same specifications for parameters that describe standard errors.

3.2.4 **Batch preprocessing**

Users can also process multiple watersheds sequentially from a list using the alternate R markdown file, `RunRBEROST-Northeast-Dynamic_Batch.Rmd`. Inputs to the .Rmd file are similar to those for the single watershed .Rmd file, with the exception that the user supplies a list of watersheds to process, and, if desired, a second list of watersheds to exclude from analysis. The batch .Rmd file creates a Batch Results folder under Outputs which contains an individual output folder for each watershed including an .amp, .mod, .dat, results .txt file and the `01_UserSpecs_loadingtargets_selected.csv` file required when running the postprocessor code chunk in the single watershed .Rmd file version.

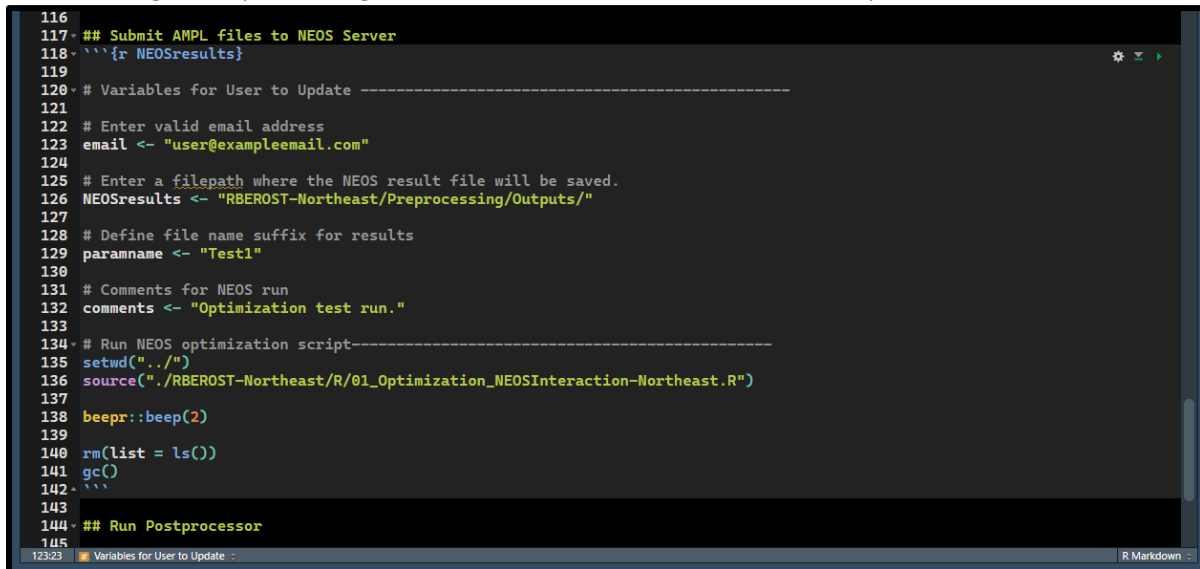
3.3 **NEOS Server**

RBEROST uses the CPLEX Optimizer available through NEOS to solve the optimization. CPLEX will allow XML calls from RBEROST to submit AMPL files to the server. Unfortunately, some local firewall settings will block these calls to and from NEOS because of the unusual port number used which is interpreted as a potential security risk. Users unable to take advantage of the automatic delivery of AMPL files to the NEOS server and receipt of an email link to results files can submit AMPL files to NEOS and retrieve results using the process outlined in Appendix B. Once preprocessing is complete and the data, model and command files are all present within the output folder, submission can be done directly from the RMD file within RStudio (Figure 3-21). Users do not need to move files to separate folders in order to submit a job request to NEOS. However, users may also upload files directly to the NEOS server manually through a web browser. More information on this process can be found in Appendix B – Manual NEOS Uploads Using Web Browser.

Generally, NEOS can solve models without uncertainty in less than five minutes, while models with uncertainty may take nearer to 20 minutes. Exact times may vary, perhaps greatly, depending on the complexity of the specified problem and the current demands on NEOS and on the CPLEX optimizer. Once the optimization has been solved, users will receive an email that contains a description of the model run according to the commands specified in command file, as well as the cost of the model run, BMP choice selections, and any errors or issues contained within the model run (Figure 3-22). Infeasible or failed results may be returned for various reasons. Failed results may stem from issues that

arise while setting up the model, such as missing input files, accidentally deleted code, missing code scripts, invalid input parameter selection, corrupted data, or excessive file size. These errors prevent the model from running, and users will see a very brief return message in these cases. Minor errors, such as presolve errors, may also arise and appear in the return email. However, these errors do not prevent the model from running and solving it based on other parameters. Watersheds without point sources often see this presolve error arise but can be optimized regardless.

Infeasible results may also result for other reasons. The NEOS CPLEX server can detect when the maximum selection would fail to meet the load reduction goal, returning an infeasible result accordingly. NEOS may return a “fixeps” version of the infeasible result in which the maximum BMP selection would fail to meet the reduction goals, but by an exceedingly slim margin. The “fixeps” equation signals that the model is *almost* able to solve the equation. An infeasible result may also simply occur when NEOS attempts many solutions but cannot find one that fits within the constraint equations. In such cases, it may be useful to revise the input parameter selection and slightly relax the model if possible. Doing so may be enough to allow for a feasible result on a subsequent model run.



```

116
117- ## Submit AMPL files to NEOS Server
118- ```{r NEOSresults}
119
120- # Variables for User to Update -----
121
122  # Enter valid email address
123  email <- "user@exampleemail.com"
124
125  # Enter a filepath where the NEOS result file will be saved.
126  NEOSresults <- "RBEROST-Northeast/Preprocessing/Outputs/"
127
128  # Define file name suffix for results
129  paramname <- "Test1"
130
131  # Comments for NEOS run
132  comments <- "Optimization test run."
133
134- # Run NEOS optimization script-----
135  setwd("../")
136  source("../RBEROST-Northeast/R/01_Optimization_NEOSInteraction-Northeast.R")
137
138  beep::beep(2)
139
140  rm(list = ls())
141  gc()
142- ```
143
144- ## Run Postprocessor
145

```

Figure 3- 21. Interface within the RMD file that allows users to submit model runs to NEOS without using the web interface.

Presolve eliminates 3727 constraints and 2364 variables.

Adjusted problem:

7716 variables:

3192 binary variables

4524 linear variables

4345 constraints, all linear; 21513 nonzeros

628 equality constraints

3717 inequality constraints

1 linear objective; 2157 nonzeros.

CPLEX 22.1.2: tech:threads = 4

CPLEX 22.1.2: optimal solution; objective 947097987.5

801 simplex iterations

solve_result_num = 0

solve_result = solved

cost.result = solved

cost = 947098000

set seasons := 1 2 3 4;

set years := 1;

point_dec [*] :=

;

Figure 3- 22. Example email text from a feasible result return from NEOS. Note that some presolve errors may occur but do not prevent optimization from occurring.

```

R 4.3.2 - C:/Tier_1_Optimization-SSWR5.3.2/
[1] "Reading in SPARROW TN data..."
=====
[1] "Reading in SPARROW TP data..."
=====
Note: For the 1st TN target, there are the same number of reaches in SPARROW and Streamcat
subsetting datasets.
All reaches available for BMP optimization
Note: For the 1st TP target, there are more reaches in the provided StreamCat
datasets than are included in SPARROW.
Only the reaches that are included in both datasets will be available
for BMP optimization. Loads from the remaining reaches will be included in
the 'other_loads' parameter.
[1] "Generating TN param load limit based only on actionable load components"
[1] "Generated TN param load limit based only on actionable load components"
[1] "Generating TP param load limit based only on actionable load components"
[1] "Generated TP param load limit based only on actionable load components"
[1] "Urban base costs are assumed to be the same across states. RBEROST is using values from VT."
[1] "Basing TN incremental equations only on actionable load components"
[1] "TN Incremental equations based only on actionable load components"
[1] "Basing TP incremental equations only on actionable load components"
[1] "Incremental TP equations based only on actionable load components"
[1] "RBEROST has finished writing AMPL scripts without uncertainty at 2025-07-28 15:46:11.334687"
Warning message:
package 'tidygraph' was built under R version 4.3.3
>

```

Figure 3- 23. Messages displayed by RBEROST preprocessor.

3.4 Postprocessing

The postprocessing R code combines optimization results from the NEOS server with the RBEROST input data to develop result summaries. The code uses R-Shiny, an R package that produces a summary of RBEROST either within the RStudio interface or in the default web browser. To run the postprocessing code, users can click the green triangle in the **Run Postprocessor** section of the **RunRBEROST-Northeast-Dynamic.Rmd** file (Figure 3-24). More than one result file can be viewed per session; however, the Shiny App will not automatically close if the viewing window is closed. To stop the Shiny App after closing the browser window, click the red square in the **Run Postprocessor** section. A green triangle will then appear that will allow the user to launch the application again.

If the RBEROST postprocessor is running within a browser, we recommend printing the results screen to a pdf file for your records.

```

143
144 ## Run Postprocessor
145
146 To run the postprocessor, click the green triangle below. A window will pop up either in RStudio or in your default web
browser. If no green arrow appears, and instead there is a red square, click the red square to stop any ongoing processes
and a green triangle should appear in its place. When clicked, the green triangle will turn into a red square until the
code is finished. Be aware that simply closing the browser window will not stop the Shiny app. You must also click the red
square below if you wish to relaunch the app.
147
148 ```{r runpostprocessor}
149 # Increase the max file size to accommodate dynamic SPARROW
150 options(shiny.maxRequestSize = 100000 * 1024^2)
151
152 source("../R/02_Optimization_RunShiny-Northeast.R", local = TRUE)
153 ```
154
77:1 | Chunk 2: runpostprocessor
R Markdown

```

Figure 3- 24. Run the RBEROST postprocessor.

3.4.1 Necessary Files

The postprocessor requires the user to upload at least five files and will display warning messages if files are not uploaded to the correct positions (Figure 3-25).

Choose Results File

Browse... NEOSresult_Lower Connecticut River_2026-05-06.txt

Upload complete

Choose the corresponding '01_UserSpecs_loadingtargets_selected.csv' file

Browse... 01_UserSpecs_loadingtargets_selected.csv

Upload complete

Choose WWTP File

Browse... WWTP_COMIDs.csv

Upload complete

Choose SPARROW Inputs File

Browse... ne_sparrow_model_input_short.csv

Upload complete

Choose State Cropland Streamcat Files

Browse... CT_NH_MA_VT_NY_RI_StreamCatData_2019.csv

Upload complete

Important: Selecting 'Preview Uploads' or 'View NEOS Results' before all uploads have completed may produce unexpected results. Previews can be found under the 'File Preview' tab, and generated reports can be found under the 'View Results' tab.

Preview Uploads View NEOS Results

Postprocessing Step

RBEROST Results Postprocessing Step

File Preview View Results

File Previews

NEOS Results File :

V1

Error at _cmdno 4 executing solve command
(file ampln, line 72440, offset 4667041):
error processing param point_effic_P:
Invalid subscript point_effic_P['NA'];point[] discarded.
or cplex_options...
Checking ampl.com for cplex_options...

User Inputs File (loading targets) :

watershed_name	Index	ComID	Watershed_HUC	Percent_Reduction	TN_or_TP	OutofNetworkFlag_X	TermFlag_X	Non-Isolated?	Isolated?	Number
Lower Connecticut River	35	7703046	NA	0.10	TN	NA	NA	X	NA	1

WWTP File :

State	Plant_Name	NPDES_ID	COMID
CT	HartfordWPCFCRW	CT0100251	7701054
CT	NewHavenEastWPCF	CT0100366	6178550
CT	BridgeportWestWPCF	CT0100056	7734269
CT	WaterburyWPCFHRW	CT0100625	7713302
CT	MattabassetWPCFCRW	CT0100307	7701178
CT	StamfordWPCFEN	CT0101087	7734319

SPARROW Inputs File :

ComID	TermFlag	IncAreaKm2	DivFrac	HUC_12	urban_km2	comid_oid	huc12	HUC_12_Rev	totalbanklength_ft
-90080001	0	0.00	0.52	NA	-99	NA	NA	NA	NA
-90078001	0	0.00	0.51	NA	-99	NA	NA	NA	NA
-90077001	0	0.00	0.20	NA	-99	NA	NA	NA	NA
-90071003	0	0.00	1.00	NA	-99	NA	NA	NA	NA
00075000	0	0.00	4.00	010	00	010	010	010	010

Figure 3- 26. Previewing files for RBEROST postprocessor.

The upload of StreamCat cropland files accepts all the state cropland files used in the optimization, which may be more than one file. Printing the file tab to pdf is the current recommended way of capturing metadata regarding which files are provided to the postprocessor. Required files include a result file from a NEOS optimization run, the list of WWTPs by their COMID locations, the SPARROW input file and the StreamCat files for all states included in the optimization. Default files have been stored in the .\RBEROSTV1D\RBEROST-Northeast\Preprocessing\Inputs\Rshiny Test Files – Northeast folder for ease in locating but see below for loading targets file. Note, as a part of the preprocessor script, a new “01_UserSpecs_loadingtargets_selected.csv” file is created, which only includes rows based on the users’ watershed choices and includes a new Number column identifier that is used for joining with the parsed NEOS results file. Each time the preprocessor is run, a new “01_UserSpecs_loadingtargets_selected.csv” file will be generated and saved over the previous version. Users should use this input file for the RShiny application.

3.4.2 Preview Files

The preview step is not necessary but may be helpful for users to ensure they have selected and uploaded the intended files. To preview files, click the [Preview Uploads](#) button and navigate to the [File Preview](#) tab (Figure 3-26).

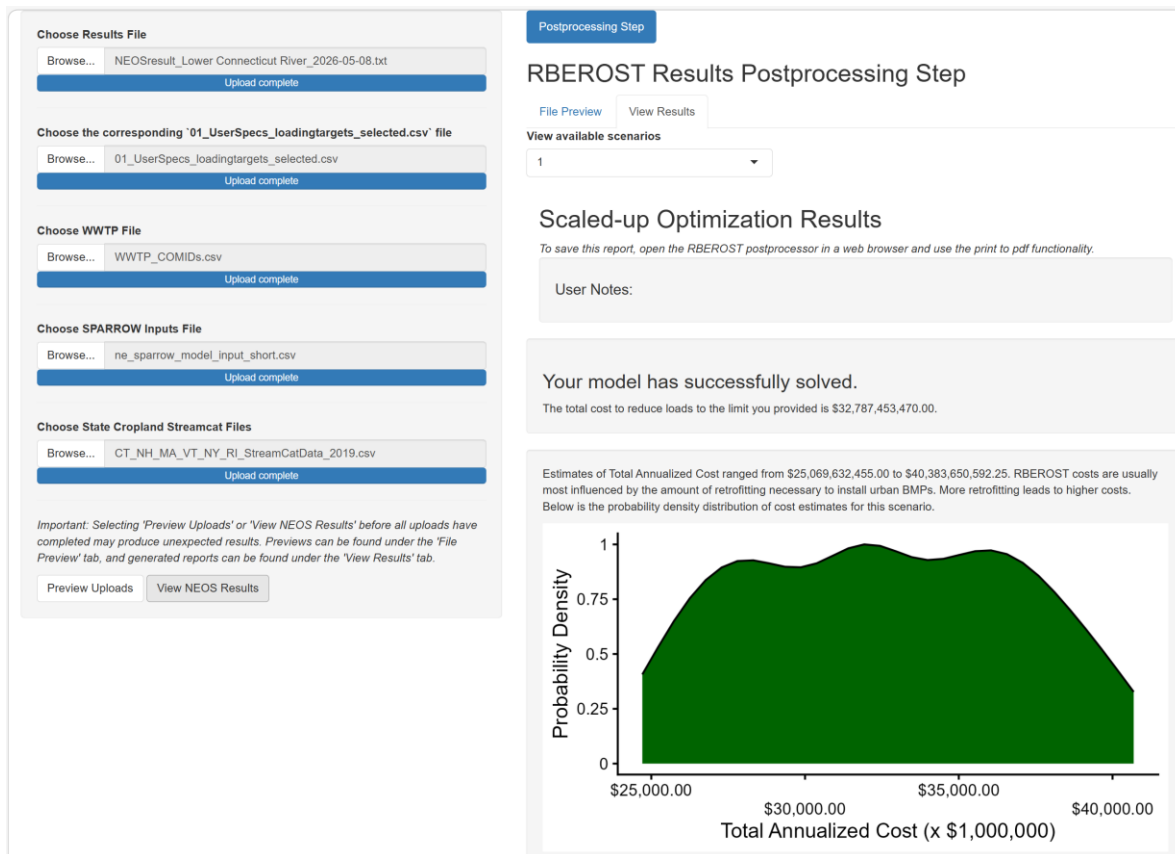


Figure 3- 27. Viewing RBEROST results in the RShiny postprocessor.

At any point, users can upload new files and preview them by selecting **Preview Uploads** again.

3.4.3 Display Results

To view results, select **View NEOS Results** and navigate to the **View Results** tab (Figure 3-27).

Estimates of Total Annualized Cost ranged from \$25,069,632,455.00 to \$40,383,650,592.25. RBEROST costs are usually most influenced by the amount of retrofitting necessary to install urban BMPs. More retrofitting leads to higher costs. Below is the probability density distribution of cost estimates for this scenario.

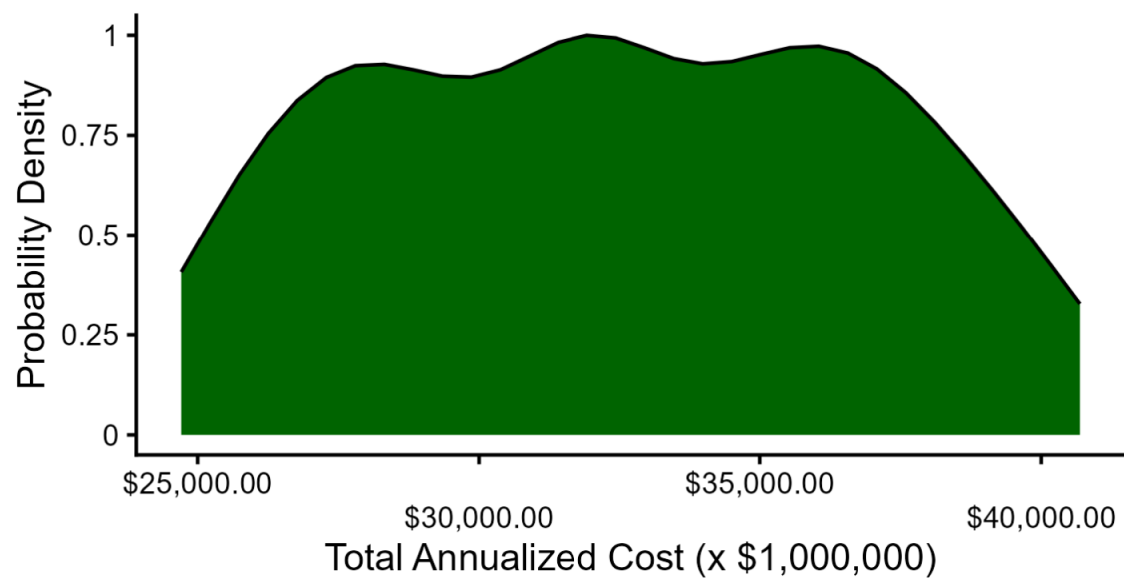


Figure 3- 28. Viewing results of uncertainty analysis in cost. The estimated range reflects the 95 % confidence interval of bootstrapped cost estimates. The probability density distribution of the bootstrapped cost estimates is shown in the figure.

Several panels will appear including an option to select the scenario the user wishes to view, the comments entered by the user to NEOS, a summary of the total cost calculated by RBEROST and a summary of BMP implementations by category. For RBEROST runs without uncertainty, only one scenario is available. For RBEROST runs with uncertainty, more than one scenario may be available, and only one will be displayed at a time. Additional panels displayed for models with uncertainty include a panel showing the expected probability density distribution of costs (Figure 3-28) and of loads (Figure 3-29) as well as a table summarizing the probability of meeting the series of targets (Figures 3-30, 3-31). If the model was unable to solve, a message will appear with the following explanation:

“RBEROST may fail to optimize models for a variety of reasons. Often, this is a result of loading targets that are too restrictive, or a result of not including enough BMPs for the model to use. It may be informative to run RBEROST again with less restrictive loading targets and/or including more BMPs in the optimization. Please check your user inputs and refer to the Model Sensitivity section of the documentation for more information.”

In addition, in the event of failure due to AMPL syntax, errors from CPLEX will be displayed. These errors can occasionally occur for a variety of reasons, including if the users have uploaded the AMPL files in the wrong order during job submission.

Choose Results File

Browse...
NEOSresult_Lower Connecticut River_

Upload complete

Choose the corresponding '01_UserSpecs_loadingtargets_selected.csv' file

Browse...
01_UserSpecs_loadingtargets_selecte

Upload complete

Choose WWTP File

Browse...
WWTP_COMIDs.csv

Upload complete

Choose SPARROW Inputs File

Browse...
ne_sparrow_model_input.csv

Upload complete

Choose State Cropland Streamcat Files

Browse...
CT_NH_MA_VT_NY_RI_StreamCatDa

Upload complete

Important: Selecting 'Preview Uploads' or 'View NEOS Results' before all uploads have completed may produce unexpected results. Previews can be found under the 'File Preview' tab, and generated reports can be found under the 'View Results' tab.

Preview Uploads

View NEOS Results

Figure 3- 25. Files necessary to run RBEROST post-processor.

Choose Results File

Browse... NEOSresult_Lower Connecticut River_2026-05-06.txt

Upload complete

Choose the corresponding '01_UserSpecs_loadingtargets_selected.csv' file

Browse... 01_UserSpecs_loadingtargets_selected.csv

Upload complete

Choose WWTP File

Browse... WWTP_COMIDs.csv

Upload complete

Choose SPARROW Inputs File

Browse... ne_sparrow_model_input_short.csv

Upload complete

Choose State Cropland Streamcat Files

Browse... CT_NH_MA_VT_NY_RI_StreamCatData_2019.csv

Upload complete

Important: Selecting 'Preview Uploads' or 'View NEOS Results' before all uploads have completed may produce unexpected results. Previews can be found under the 'File Preview' tab, and generated reports can be found under the 'View Results' tab.

Preview Uploads

Postprocessing Step

RBEROST Results Postprocessing Step

File Preview

View Results

File Previews

NEOS Results File :

V1

Error at _cmdno 4 executing solve command
(file ampln, line 72440, offset 4667041):
error processing param point_effic_P:
Invalid subscript point_effic_P['NA'];[point] discarded.
or cplex_options...
Checking ampl.com for cplex_options...

User Inputs File (loading targets) :

watershed_name	Index	ComID	Watershed_HUC	Percent_Reduction	TN_or_TP	OutofNetworkFlag_X	TermFlag_X	Non-Isolated?	Isolated?	Number
Lower Connecticut River	35	7703046	NA	0.10	TN	NA	NA	X	NA	1

WWTP File :

State	Plant_Name	NPDES_ID	COMID
CT	HartfordWPCFCRW	CT0100251	7701054
CT	NewHavenEastWPCF	CT0100366	6178550
CT	BridgeportWestWPCF	CT0100056	7734269
CT	WaterburyWPCFHRW	CT0100625	7713302
CT	MattabassetWPCFCRW	CT0100307	7701178
CT	StamfordWPCFEN	CT0101087	7734319

SPARROW Inputs File :

ComID	TermFlag	IncAreaKm2	DivFrac	HUC_12	urban_km2	comid_old	huc12	HUC_12_Rev	totalbanklength_ft
-90080001	0	0.00	0.52	NA	-99	NA	NA	NA	NA
-90078001	0	0.00	0.51	NA	-99	NA	NA	NA	NA
-90077001	0	0.00	0.20	NA	-99	NA	NA	NA	NA
-90071003	0	0.00	1.00	NA	-99	NA	NA	NA	NA
00076000	0	0.00	4.00	010	00	010	010	010	010

Figure 3- 26. Previewing files for RBEROST postprocessor.

Choose Results File

Browse... NEOSresult_Lower Connecticut River_2026-05-08.txt

Upload complete

Choose the corresponding '01_UserSpecs_loadingtargets_selected.csv' file

Browse... 01_UserSpecs_loadingtargets_selected.csv

Upload complete

Choose WWTP File

Browse... WWTP_COMIDs.csv

Upload complete

Choose SPARROW Inputs File

Browse... ne_sparrow_model_input_short.csv

Upload complete

Choose State Cropland Streamcat Files

Browse... CT_NH_MA_VT_NY_RI_StreamCatData_2019.csv

Upload complete

Important: Selecting 'Preview Uploads' or 'View NEOS Results' before all uploads have completed may produce unexpected results. Previews can be found under the 'File Preview' tab, and generated reports can be found under the 'View Results' tab.

Preview Uploads View NEOS Results

Postprocessing Step

RBEROST Results Postprocessing Step

View Results

View available scenarios

1

Scaled-up Optimization Results

To save this report, open the RBEROST postprocessor in a web browser and use the print to pdf functionality.

User Notes:

Your model has successfully solved.

The total cost to reduce loads to the limit you provided is \$32,787,453,470.00.

Estimates of Total Annualized Cost ranged from \$25,069,632,455.00 to \$40,383,650,592.25. RBEROST costs are usually most influenced by the amount of retrofitting necessary to install urban BMPs. More retrofitting leads to higher costs. Below is the probability density distribution of cost estimates for this scenario.

Figure 3- 27. Viewing RBEROST results in the RShiny postprocessor.

Estimates of Total Annualized Cost ranged from \$25,069,632,455.00 to \$40,383,650,592.25. RBEROST costs are usually most influenced by the amount of retrofitting necessary to install urban BMPs. More retrofitting leads to higher costs. Below is the probability density distribution of cost estimates for this scenario.

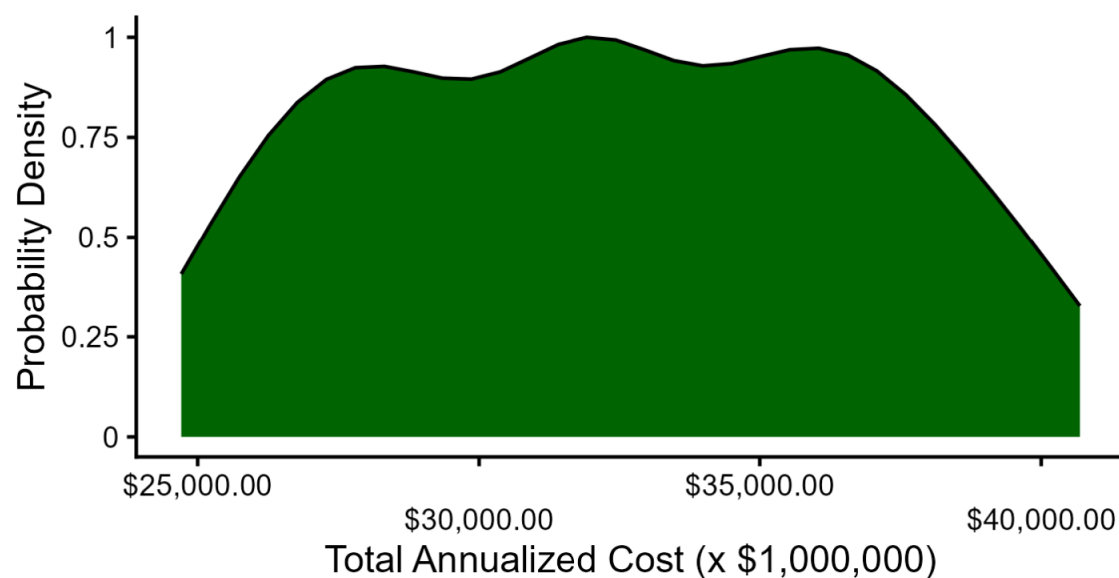


Figure 3- 28. Viewing results of uncertainty analysis in cost. The estimated range reflects the 95 % confidence interval of bootstrapped cost estimates. The probability density distribution of the bootstrapped cost estimates is shown in the figure.

Below are the probability density distributions of total annual load estimates at each of the loading targets specified by the user. The vertical dotted grey lines denote the specified target loads at each loading target. Probability densities to the left of the grey line are load estimates that meet the loading target, and probability densities to the right of the grey line are load estimates that exceed the loading target.

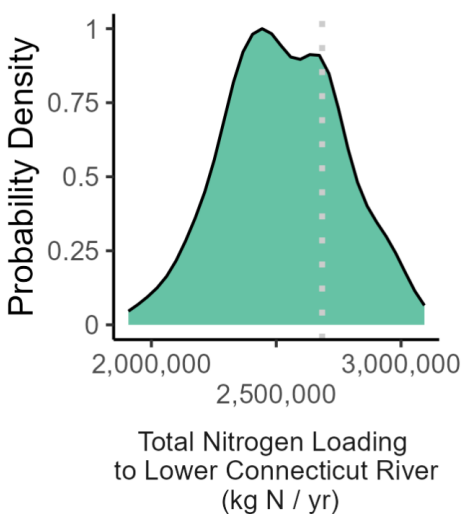


Figure 3- 29. Total annual expected loading to targets.

This suggested plan has an estimated minimum of 73% likelihood of meeting each of the specified loading targets. Likelihoods of meeting each specified loading target individually are listed in the table below. 'Likelihood' refers to the percentage of the probability density of estimated total annual load that falls below the user specified loading target.

Loading Target	Your Specified Target Load	95% CI of Estimated Annual Total Loading	Likelihood of Meeting Your Loading Target
Lower Connecticut River	10% reduction in TN; or 2,683,942.0 kg N/yr	2,061,410.4 - 2,971,983.7 kg N/yr	73%

Below are the probability density distributions of total annual load estimates at each of the loading targets specified by the user. The vertical dotted grey lines denote the specified target loads at each loading target. Probability densities to the left of the grey line are load estimates that meet the loading target, and probability densities to the right of the grey line are load estimates that exceed the loading target.

Figure 3- 30. Scenario of optimization postprocessor results when running RBEROST with uncertainty.

3.4.4 Download Detailed Results

COMID-specific BMP implementation decisions can be downloaded from the RShiny App for any single scenario (Figure 3-31). Point-source decisions are not available for download and are displayed within the RShiny interface. Additional contextual information is included in the downloadable csv files including the HUC12 code and state for each COMID, the size of the catchment and the total area of agricultural land for ag BMPs, or urban land for urban BMPs, or the total river reach length for riparian buffer BMPs.

The model did not implement any WWTP retrofits.

The model chose to implement the total area of the following agricultural BMPs:

Cover crops (ac)	Filterstrip (ac)	Terrace Waterway (ac)
12,698.5	9,750.3	2,480.5

[Download Agricultural BMPs by COMID](#)

The model chose to implement the total area of the following urban BMPs.

Enhanced Biofiltration w ISR (ac)	Gravel Wetland (ac)	Porous Pavement w underdrain (ac)
88,312.4	1.3	7,986.7

[Download Urban BMPs by COMID](#)

The model chose to implement the total length of the following riparian buffer BMPs.

Forested Buffer (ft of bank)	Grassed Buffer (ft of bank)
24,489.6	3,191,289.4

[Download Riparian Buffer BMPs by COMID](#)

The model chose to implement the total number of parcels for the following septic BMPs.

class 1 upgrade (parcels)	class 2 upgrade (parcels)	class 3 upgrade (parcels)	class 4 upgrade (parcels)	class 5 upgrade (parcels)	class 6 upgrade (parcels)	class 7 upgrade (parcels)	class 8 upgrade (parcels)
80	413	742	3,874	1	39	68	407

[Download Septic BMPs by COMID](#)

Figure 3- 31. Downloading RBEROST results.

3.5 Uncertainty Parameters

Estimates of parameter uncertainties are needed to support both sensitivity analyses and evaluation of uncertainty associated with load and cost estimates. Most of these values were derived from the literature (Table 3-3). Uncertainty in some parameters are not included as parameters in the model but rather are hard coded into the AMPL scripts. These include uncertainty around point_costs, point_effic_N and point_effic_P and agcost_frac. Uncertainty around point costs were estimated based on the predicted contractor markup, and assuming that additional engineering services or detailed designs would have a similar markup (JJ Environmental, 2015). Point efficiencies assumed a coefficient of variation of 20 % (JJ Environmental, 2015). Other loads for nitrogen and phosphorus were derived from SPARROW models (Schmadel et al., 2021). The ratio of actual agricultural costs to base payments (agcost_frac) was modeled as a uniform distribution between 1 and 1.67 (i.e., base payment costs ranging from 60 %-100 % of actual costs).

3.6 N Loads for Long Island Sound North and South Shores

The dynamic SPARROW model provides estimates of all source loads for North Shore watersheds draining to the Long Island Sound except for septic N loads, for which separate estimates had to be derived so that the effect of management actions could be applied to the appropriate sources. The original output of the SPARROW model includes estimates of loads from urban lands which includes both stormwater and septic sources. Septic N loads for coastal catchments draining into the north shore of the Long Island Sound were calculated based on the Vaudrey N loading model (Vaudrey et al. 2017), which captures reasonable groundwater travel distances and times to the Long Island Sound. For the North Shore of the Long Island Sound, we only compiled data on septic systems within the groundwater model domain of Barclay and Mullaney, (2021). Barclay and Mullaney's model outputs indicate that most of the groundwater travel times to either inland or coastal waters in the nearshore model domain are less than one year, and most groundwater is discharging into inland surface waters rather than to the coast.

The Vaudrey model for the watersheds draining into the south shore of Long Island Sound estimates annual septic N loading parsed out into loading zones draining into individual embayments (e.g., FAR, NEAR, BUFF zones) of the Long Island Sound. Annual septic N loads were estimated for each NHDPlus catchment by 1) calculating the fraction of catchment area that overlaps with each of the aforementioned Vaudrey loading zone boundaries, 2) multiplying the catchment fractional coverage by the septic N load for the respective loading zone(s) and 3) summing the fractional-zone N load(s) for the NHDPlus catchment for each catchment. Septic N loads were adjusted for the population on septic based on estimates of the number of septic systems per NHDPlus catchment. The calculated NHDPlus septic N load was then subtracted from the USGS SPARROW developed land N load.

The SPARROW model was designed to predominantly capture loads from surface waters, not groundwater, so the domain was restricted to North Shore watersheds. For catchments entering the south shore of the Long Island Sound, annual N loads were separately estimated for a subset of SPARROW parameters (atmospheric deposition, urban non-septic, agricultural fertilizer, developed fertilizer) as well as for septic sources using:

1. USGS-modeled nonpoint source annual N loading rates covering Long Island (every 500 sq-m) (Monti 2024),
2. annual NLCD class areas (USGS 2024a) calculated for the Vaudrey N loading zones (BUFF, NEAR, FAR) and
3. attenuation factors affecting N retention and transport within the Vaudrey loading zones for each embayment.

Using the centroids of the modeled 500 m × 500 m areas, average annual N loading rates were estimated within the boundaries of catchments and embayments of the south shore of Long Island Sound for the following nonpoint sources: atmospheric deposition, residential fertilizer, residential fertilizer + pet waste, agriculture + livestock waste and septic systems. N source types were combined in some cases to match NLCD classes that make up the SPARROW output. Annual NLCD raster files were downloaded for the area covering Long Island, and NLCD areas were calculated for each Vaudrey loading zone within each catchment entering the south shore embayments. Some NLCD classes were separated by turf and impervious cover by using percentages obtained from the USGS annual NLCD user guide (USGS 2024b). Attenuation factors were obtained from the Vaudrey N loading model (excel workbook) for each NLCD class by south shore loading zone and embayment. In addition, modeled low- and high-range attenuation factors were obtained. Source annual N loads were estimated by multiplying

corresponding attenuation factors (mean, low and high range), NLCD class areas and N type loading rates for each Vaudrey loading zone and embayment. Annual N loads were then estimated for each NHDPlus catchment using the proportion of overlap with the Vaudrey N loading zones, providing mean, low-range and high-range values that correspond to the SPARROW parameters.

The remaining SPARROW parameters were estimated using other individual methods. Annual N loads from snowfall were derived from the SPARROW input for catchments draining into north shore of Long Island Sound, where an average rate was calculated across NHDPlus catchments for each year and season (although snowmelt only occurs in season = 2) for all North Shore watersheds with an area $\leq$ to the largest catchment draining into the Long Island Sound south shore. This served as a reasonable average rate that can be applied to similar sized catchments draining into the south shore. The average rate was applied to the areas of the NHDPlus catchments draining the south shore to estimate a mean N load from snowfall. Municipal (point-source) N loads were obtained from the EPA Hypoxia Task Force Nutrient Model (US EPA 2025) using SPDES/NPDES permit numbers for WWTPs in New York State reported in (New York State Department of Environmental Conservation 2025). Locations of WWTPs were intersected with catchments draining into the south shore of Long Island Sound. Annual WWTP N loads for 2007-2020 were obtained from the Hypoxia Task Force Nutrient Model (US EPA 2025). Annual N loads were then linearly interpolated back to year 2000 for each NHDPlus catchment. N load from storage was calculated using Long Island Sound data reported by (Jahn and Walter 2025), where the fraction of total N load due to storage was calculated for each year in the dataset by the following equation: $(\text{“Instantaneous loads with wells”} - \text{“Loads based on steady state flow”}) / \text{“Instantaneous loads with wells”}$. The fraction difference was then multiplied by the total N load (i.e., sum of all SPARROW N type loads by season) to derive the storage component. For non-zero loads, annual SPARROW N loads were then subtracted by the N load from storage. The remaining SPARROW N load parameters were set to zero: biological nitrogen fixation and agricultural manure. In the case of agricultural manure, this N load type is already accounted for.

4 Model Sensitivity

4.1 Sensitivity Analysis Overview

RBEROST results are sensitive to provided data and user inputs. Model parameters are not known with absolute accuracy but have some associated uncertainty associated with them. The same approach that is used for solving optimizations with uncertainty can be used to inform a sensitivity analysis, (i.e., determining how “sensitive” cost and nutrient loading estimates are to variation in parameter inputs). RBEROST has built-in capacity to estimate uncertainty around the optimized cost and total loading. This allows the tool to estimate likely ranges of the total cost as well as the exceedance probability of the nutrient targets (or the inverse—the probability of success) for a given combination of BMPs. The range of results are created through a bootstrapping approach, which recalculates the total cost of the optimized lowest-cost result and total loading at each loading target multiple times. Sensitivity analysis is accomplished by sampling each parameter one at a time from estimated parameter distributions (See Table 4-3 and Section 4.5 for details).

Sensitivity analyses were carried out for three New England watersheds. The analysis tested parameter-BMP combinations across dynamic RBEROST models for Pequonnock River (heavily urbanized), Mystic Harbor (mixed urban-agricultural) and Bantam Lake (agricultural with moderate urban development) to determine which parameters most strongly influence optimization results. Tolerances are defined as the relative amount of variation in the estimate of each parameter that produces RBEROST results with costs that are within 10% of the reference cost, and that produces results with relative amounts of spending on each BMP that are within 10% of the amount spent on that BMP in the reference result. The reference result is the RBEROST solution to a problem with all BMPs included and no limits on implementation (sometimes referred to as the ‘unconstrained’ model. Parameters are considered sensitive if a change of less than 10% produces results that are more than 10% different from the reference. Parameters are considered highly sensitive if a change of less than 1% produces loading or cost results that are more than 10% different from the reference.

4.2 Sensitivity Analysis Summary

Local sensitivity analysis of three contrasting New England watersheds reveals that parameter sensitivity patterns depend critically on watershed land use characteristics and source loading distributions. Findings demonstrate that the most sensitive parameters shift dramatically based on the relative importance of urban versus agricultural nutrient sources within each watershed.

Across all three watersheds, the nitrogen loading target at the terminal reach consistently emerges as the most critical factor affecting RBEROST model success, with tolerance ranges between 0.996 and 1.01 representing < 1% variation that produces > 10 % changes in model outputs. However, the specific BMP parameters that exhibit high sensitivity vary substantially by watershed type. The heavily urbanized Pequonnock River watershed shows high sensitivity to urban baseline loads (tolerance 0.99-1.01), Enhanced Biofiltration with Internal Storage Reservoir implementation potential (tolerance 1.0) and Gravel Wetland parameters (tolerances ranging from 0.98-1.05). Agricultural parameters in Pequonnock River are generally not sensitive, with tolerances spanning 0.01 to 100 for most agricultural BMP costs and efficiencies.

Mystic Harbor exhibits a mixed sensitivity pattern reflecting its combination of urban development and agricultural land use. Agricultural baseline loads show moderate sensitivity with a tolerance range of

0.98 to 1.02, while urban baseline loads show similar moderate sensitivity at 0.95 to 1.04. Filterstrip nitrogen efficiency emerges as highly sensitive (tolerance 0.995-1.01), as does the maximum implementation fraction for filter strips (tolerance 1.0), indicating that agricultural practices play a more important role in this mixed-use watershed than in the heavily urbanized Pequonnock River. Urban parameters including Enhanced Biofiltration with Internal Storage Reservoir and Gravel Wetland costs, efficiencies and implementation potential all show high sensitivity, with Enhanced Biofiltration costs only tolerant to changes between 1.0 and 1.01, representing highly sensitive responses to even minimal cost increases.

Bantam Lake displays a distinctly agricultural sensitivity profile, reflecting its greater proportion of agricultural land compared to the other two watersheds. Agricultural baseline nitrogen loads show high sensitivity with a tolerance range 0.996 to 1.004, while agricultural baseline phosphorus loads demonstrate similarly high sensitivity at 0.995 to 1.01. Filter strip nitrogen efficiency shows high sensitivity (tolerance 0.99-1.02), as does filter strip phosphorus efficiency (tolerance 0.97-1.01). Both Minimum Tillage and Manure Injection nitrogen efficiencies show moderate sensitivity (tolerance 0.95-1.1 and 0.95-1.05 respectively), while filter strip capital costs are sensitive in both directions (tolerance 0.9-1.1). Urban parameters show much lower sensitivity in Bantam Lake than in the other two watersheds, with most urban BMP parameters displaying "not sensitive" classifications and tolerance ranges spanning 0.01 to 100.

4.3 Model Success and Loading Target Sensitivity

User-specified nitrogen loading targets at the terminal reach represent the single most critical parameter determining whether RBEROST can successfully solve the optimization problem across all three watersheds examined in this analysis. Global sensitivity analysis involving 1,000 bootstrap iterations for Pequonnock River demonstrates this relationship through clear binary success patterns as a function of terminal reach target loads. Each iteration varied parameter estimates sampled from distributions describing likely parameter values, with most data-based parameters sampled from normal distributions based on uncertainty analysis and user-specified parameters drawn from uniform distributions encompassing their allowed ranges.

The sensitivity testing framework employed consistent parameter variation approaches across all three watersheds. Maximum implementation fractions were sampled between 70–100 % to reflect realistic upper bounds on BMP adoption. Loading target reductions were sampled between 2–20 % to span the range of potential management goals. Minimum implementations were held constant at 0 % because minimum implementation requirements impose additional constraints on the optimization problem that more frequently prevent the optimizer from finding feasible solutions. This design choice reflects the practical consideration that requiring minimum implementations of multiple BMPs simultaneously substantially increases the likelihood of model infeasibility.

The three watersheds examined show consistently high sensitivity to loading targets but with subtle differences in their tolerance ranges. Pequonnock River loading limits show high sensitivity with tolerance 0.99 to 1.01, meaning that parameter changes of < 1 % in either direction produce loading output changes > 10 %. Mystic Harbor displays similar high sensitivity with tolerance range 0.99 to 1.01 for nitrogen loading limits. Bantam Lake shows the tightest tolerance of the three watersheds, with nitrogen loading limits tolerant to changes only between 0.996 and 1.004 and phosphorus loading limits between 0.995 and 1.002, reflecting the additional constraint of managing both nitrogen and phosphorus simultaneously in this lake-draining watershed (Figure 4-1).

Success rates varied across the three watersheds, reflecting differences in the availability and cost-effectiveness of applicable BMPs given each watershed's land use composition. In the heavily urbanized Pequonnock River watershed, the model successfully solved 635 of 1,000 scenarios, with failures typically occurring when nitrogen reduction targets exceeded the combined capacity of available urban stormwater BMPs. Optimization failures across all watersheds occurred when loading targets were overly restrictive relative to BMP capabilities, when maximum implementation restrictions constrained the solution space excessively, or when combined parameter uncertainty created infeasible constraint combinations.

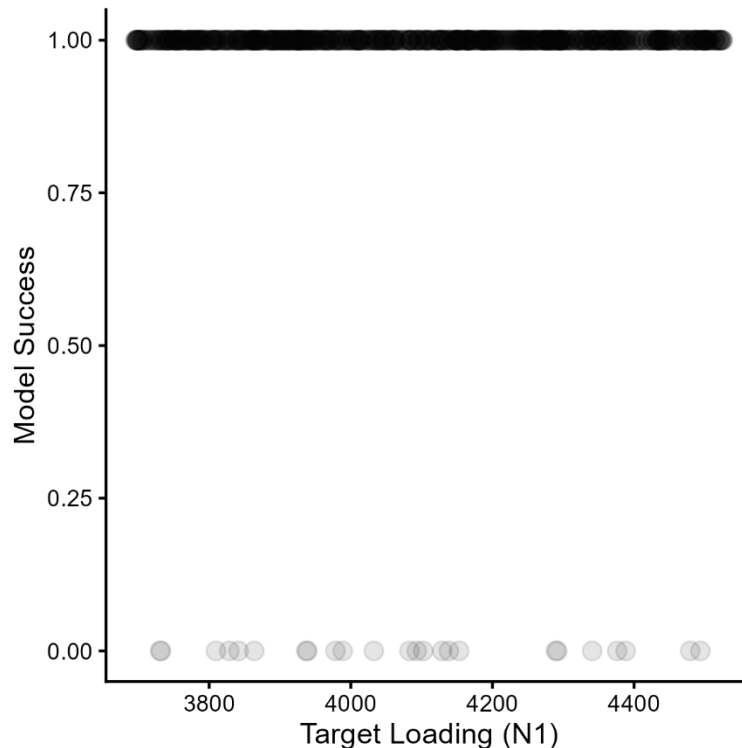


Figure 4- 1. Model success compared to nitrogen loading target for Bantam Lake

When users encounter model infeasibility, several approaches can restore feasibility. Relaxing maximum BMP implementation restrictions provides the optimizer with greater flexibility to allocate BMPs across the watershed. Including additional BMP options expands the solution space by providing alternative pathways to achieve loading reductions. Reducing or eliminating intermediate loading targets at non-terminal reaches decreases the number of constraints that must be simultaneously satisfied. Re-evaluating the realism of loading reduction goals in light of available BMP capacity ensures that targets remain achievable given the physical and economic constraints of the watershed. Occasionally optimization of scenarios fails even when target loading is less restrictive, a phenomenon that occurs as a function of an overly constrained problem and may be alleviated by relaxing maximum restrictions on BMPs, by including more BMPs, or by relaxing intermediate loading targets.

4.4 Cost Sensitivity to Loading Targets

RBEROST predictions of annualized total cost increase exponentially as nitrogen loading targets become more restrictive, a relationship that holds consistently across all three watersheds examined in this

analysis (Figure 4-2). This exponential cost-loading relationship reflects the fundamental economics of pollution control, where initial nutrient reductions are achieved cost-effectively through implementation of high-efficiency BMPs at the most suitable sites, while approaching progressively stricter targets requires implementing more expensive practices or deploying BMPs at less optimal locations.

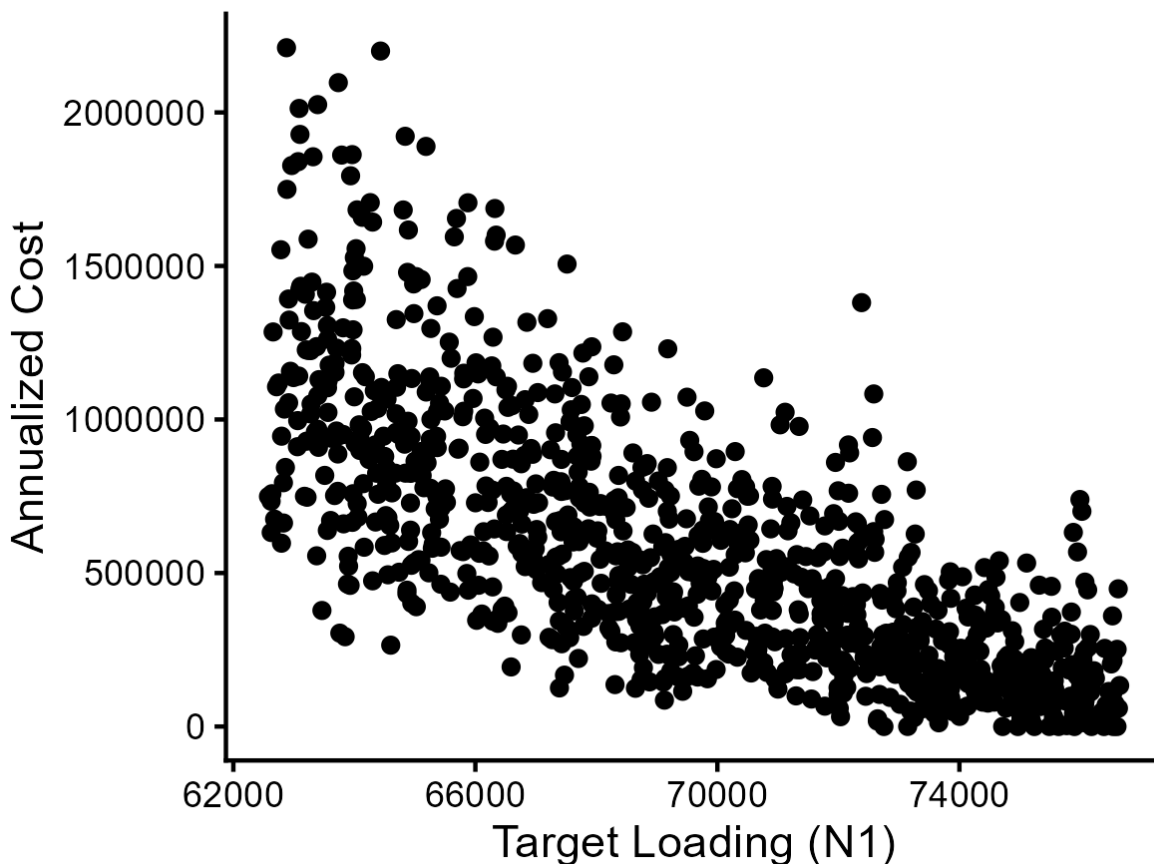


Figure 4- 2. Annualized cost compared to nitrogen loading target for the Pequonnock River.

Analysis of 635 feasible solutions from the Pequonnock River watershed reveals distinct phases in the cost-loading relationship. Costs rise modestly for loading reductions up to ~ 10 %, as the model selects the most cost-effective BMPs first. Exponential cost escalation begins > 10–15 % reduction targets, as the optimizer exhausts the supply of highly cost-effective options and must turn to progressively more expensive alternatives. The highest variability in costs occurs at moderate reduction levels between 5–15 %, where parameter uncertainty substantially affects which specific BMPs are selected. At more stringent reduction targets > 15 %, cost variability decreases as the model approaches full implementation of all available BMP options, leaving less room for substitution between alternatives. The cost-loading relationship for the heavily urbanized Pequonnock River differs substantially from agricultural watersheds due to several structural factors. The predominance of urban stormwater sources requires expensive infrastructure-based BMPs rather than lower-cost agricultural management practices. Limited agricultural land area restricts opportunities for cost-effective agricultural practice adoption that might provide less expensive loading reductions. Higher baseline capital costs for urban stormwater management infrastructure increase the absolute cost of achieving any given level of reduction. Reduced cost-efficiency at higher implementation levels reflects site limitations that prevent universal application of the most cost-effective urban BMPs.

4.5 BMP Selection Patterns Across Watershed Types

4.5.1 Point Source BMPs

Point source BMP implementation patterns differ markedly across the three watersheds, reflecting differences in existing wastewater treatment infrastructure and the relative contribution of point sources to total watershed loading. The Pequonnock River watershed analysis shows no point source BMP implementation across tested scenarios (Figure 4-3), indicating that existing wastewater treatment plants already achieve high removal efficiencies and that the cost-effectiveness of additional point source upgrades is lower than urban BMPs (Figure 4-3). Point source loads contribute minimally to the Pequonnock River optimization problem, and further point source improvements are not required to meet the tested loading targets.

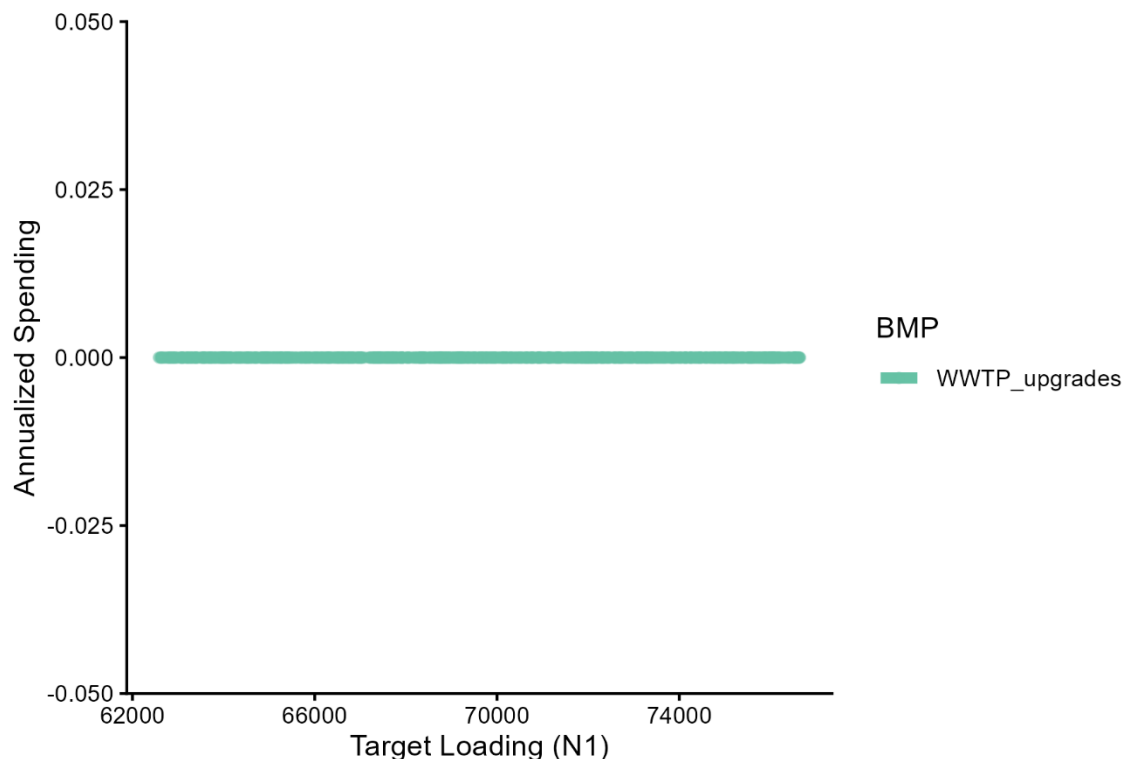


Figure 4- 3. Point source BMP spending by nitrogen target for the Pequonnock River.

Sensitivity analysis confirms that point source parameters exhibit "not sensitive" classifications across all three watersheds. In Pequonnock River, point source costs and efficiencies show tolerances spanning 0.01 to 100, meaning that even 100-fold changes in these parameters would not substantially alter optimization results. Mystic Harbor shows identical insensitivity to point source parameters with tolerance ranges of 0.01 to 100 for both capital costs, operations costs and nitrogen removal efficiency. Bantam Lake similarly displays no sensitivity to point source costs or efficiency, with one exception being that phosphorus point source efficiency is listed as "not applicable," suggesting that point sources contribute negligible phosphorus loads in this watershed.

The consistent insensitivity to point source parameters across all three watersheds suggests that existing wastewater treatment infrastructure in these New England watersheds generally achieves sufficiently high removal rates that further improvements are not cost-competitive with other BMP options. This

finding contrasts with some other regions where point source upgrades represent the most cost-effective approach to achieving nutrient loading reductions. Users applying RBEROST to watersheds with older or less efficient wastewater treatment infrastructure may observe different sensitivity patterns for point source parameters.

4.5.2 Agricultural BMPs

Agricultural BMP selection and sensitivity patterns vary dramatically across the three watersheds, directly reflecting the proportion of agricultural land use in each system. These patterns provide perhaps the clearest illustration of how watershed characteristics drive parameter sensitivity.

Pequonnock River, with minimal agricultural land area, shows virtually no agricultural BMP spending across scenarios and exhibits universal insensitivity to agricultural parameters. All agricultural costs, both capital and operations, show tolerance ranges of 0.01 to 100 for all practices including Conservation, Contour Farming, Fertilizer Reduction, Filter Strips, Minimum Tillage, Manure Injection, Ponds and various terrace practices. Agricultural nitrogen removal efficiencies similarly show "not sensitive" classifications with tolerance ranges of 0.01 to 100 for most practices. Maximum implementation fractions for agricultural BMPs show tolerances of 0.01 to 1, again indicating no sensitivity. Agricultural baseline nitrogen loads in Pequonnock River are not sensitive with tolerance 0.01 to 5, while agricultural land area is not sensitive with tolerance 0.01 to 100. This comprehensive insensitivity reflects the simple fact that agricultural sources represent such a small fraction of total watershed loading that even dramatic changes in agricultural BMP costs or efficiencies cannot significantly alter the optimal solution.

Mystic Harbor presents a mixed agricultural sensitivity profile that reflects its intermediate position between heavily urbanized and predominantly agricultural watersheds (Figure 4-4). Filterstrip parameters emerge as particularly sensitive in this mixed-use watershed. Filterstrip nitrogen efficiency shows high sensitivity with a tolerance range of 0.995 to 1.01, meaning that < one % changes in this parameter produce > 10 % changes in model outputs. The maximum implementation fraction for filter strips shows high sensitivity as well, with tolerance of 1.0 in the decreased direction, indicating that any reduction in the maximum allowable filterstrip implementation substantially alters optimization results. Filterstrip capital costs show moderate sensitivity with tolerance 0.9 to 1.1, meaning that 10–90 % changes in costs are required to produce 10 % output changes. Other agricultural practices in Mystic Harbor generally show lower sensitivity, with most parameters classified as "not sensitive." Agricultural baseline nitrogen loads show moderate sensitivity with tolerance 0.98 to 1.02, indicating that this source category contributes meaningfully but not dominantly to total watershed loading.

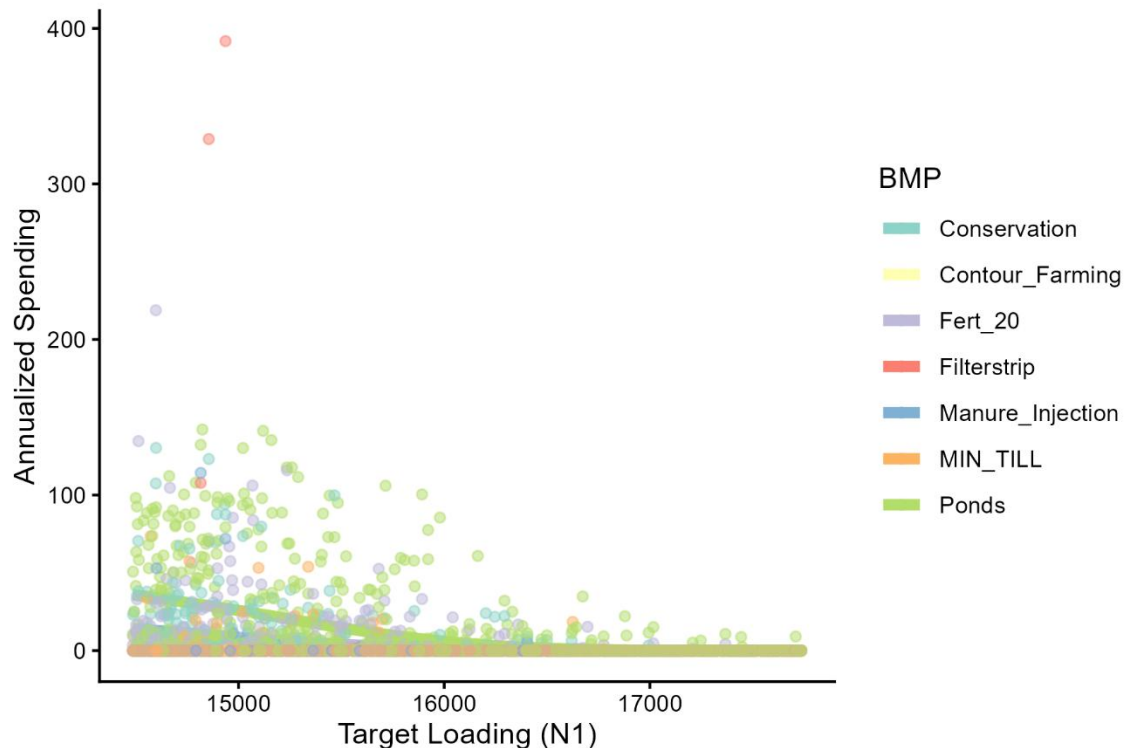


Figure 4- 4. Agricultural BMP spending by nitrogen target for the Mystic Harbor watershed.

Bantam Lake displays the most comprehensive agricultural sensitivity profile of the three watersheds, consistent with its greater proportion of agricultural land use. Agricultural baseline nitrogen loads show high sensitivity with tolerance range 0.996 to 1.004, the tightest tolerance observed for any baseline loading parameter across the three watersheds. Agricultural baseline phosphorus loads similarly show high sensitivity at 0.995 to 1.01, reflecting the dual-nutrient management requirements for this lake system. Filterstrip nitrogen efficiency shows high sensitivity (tolerance 0.99-1.02), as does filterstrip phosphorus efficiency (tolerance 0.97-1.01), with the phosphorus efficiency showing particularly high sensitivity to increases (tolerance 1.01). These tight tolerances indicate that filter strips represent a frequently selected, cost-effective practice in this watershed, and that precise characterization of their nutrient removal performance is critical for accurate optimization results.

Beyond filter strips, Bantam Lake shows moderate sensitivity to several other agricultural parameters. Minimum Tillage nitrogen efficiency shows sensitivity with tolerance 0.95 to 1.1, indicating that 5–10 % changes can substantially affect results. Manure Injection nitrogen efficiency demonstrates sensitivity at 0.95 to 1.05, while Manure Injection phosphorus efficiency shows even higher sensitivity at 0.97 to 1.01. Filterstrip capital costs are sensitive in both directions with tolerance 0.9 to 1.1, as are Minimum Tillage capital costs (tolerance 0.7-1.1), Manure Injection capital costs (tolerance 0.9-1.1) and Pond capital costs (tolerance 0.95-1.1). Agricultural land area shows sensitivity with tolerance 0.95 to 1.05, and the agricultural cost fraction adjustment factor shows similar sensitivity at 0.95 to 1.05. Maximum implementation fractions show moderate sensitivity for Manure Injection (tolerance 0.9) and Ponds (tolerance 0.95).

The agricultural sensitivity patterns across these three watersheds provide clear guidance for model application. In heavily urbanized watersheds like Pequonnock River, users can employ approximate agricultural parameter estimates without substantially affecting results. In mixed watersheds like Mystic

Harbor, careful characterization of a few key agricultural practices, particularly filter strips, becomes important. In agricultural watersheds like Bantam Lake, comprehensive and accurate agricultural data collection is essential, with particular attention to frequently selected practices like filter strips, tillage practices and manure management.

4.5.3 Urban BMPs

Urban BMP selection exhibits clear patterns but with substantial differences across the three watersheds that reflect their varying degrees of urbanization and the relative importance of urban sources to total watershed loading. The sensitivity results provide critical insights into which urban BMP parameters require careful characterization for accurate optimization.

Pequonnock River, being heavily urbanized, shows strong sensitivity to several specific urban BMP parameters (Figure 4-5). Enhanced Biofiltration with Internal Storage Reservoir emerges as a critical practice with high sensitivity across multiple parameters. The maximum implementation fraction for Enhanced Biofiltration with Internal Storage Reservoir shows high sensitivity (tolerance 1.0) in the decreased direction, meaning that any reduction below the reference value substantially alters results. Implementation potential shows similar high sensitivity with a tolerance of 1.0, indicating that site suitability assessment is critical for this practice. Nitrogen removal efficiency shows high sensitivity with tolerance 0.999 to 1.03, demonstrating that performance characterization must be precise. Design depth shows sensitivity with tolerance 0.8 to 1.02, being more sensitive to increases than decreases. Costs show sensitivity with tolerance 0.8 to 1.02, also more sensitive to increases.

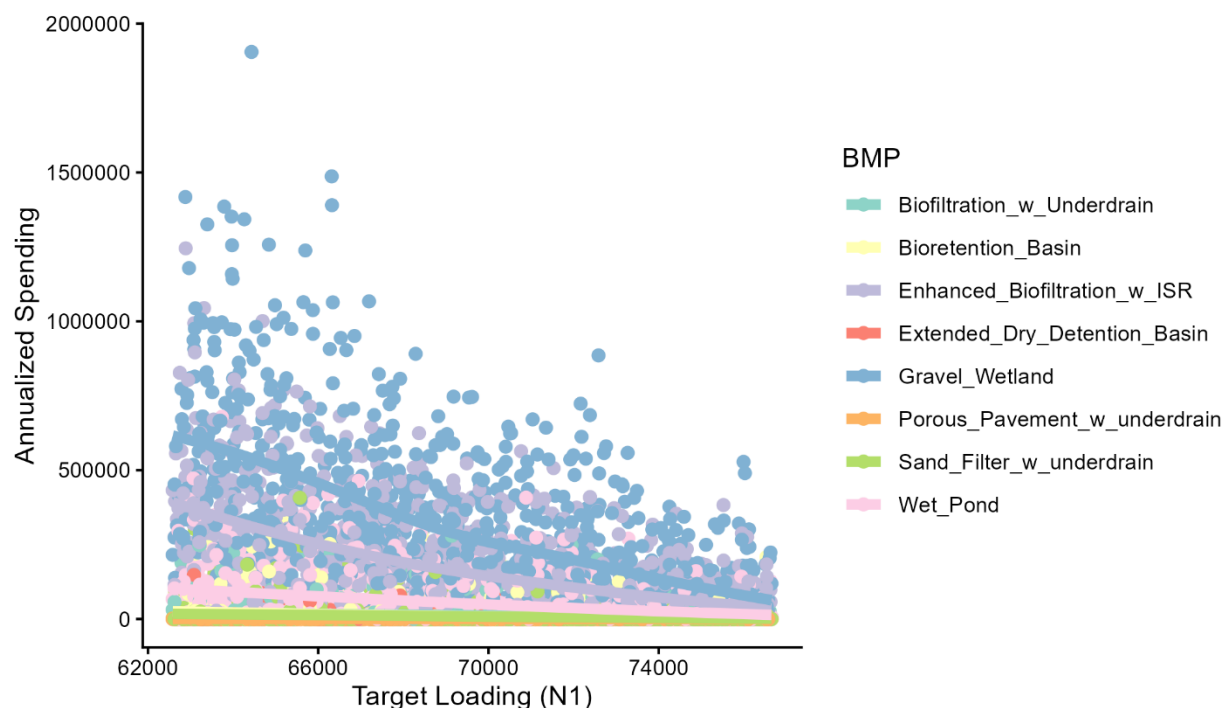


Figure 4- 5. Urban BMP spending by nitrogen target for the Pequonnock River

Gravel Wetland parameters in Pequonnock River also exhibit high sensitivity, though the tolerance ranges differ slightly from Enhanced Biofiltration with Internal Storage Reservoir. Maximum implementation fraction shows high sensitivity with a tolerance of 0.995 in the decreased direction.

Implementation potential shows high sensitivity at a tolerance of 0.995. Nitrogen removal efficiency shows sensitivity to decreased values (tolerance 0.98) and high sensitivity to increased values (tolerance 1.005), indicating that overestimating the efficiency has larger consequences than underestimating it. Design depth shows sensitivity in both directions with tolerance 0.98 to 1.05. Costs similarly show sensitivity in both directions at 0.98 to 1.05.

Porous Pavement with underdrain shows moderate sensitivity for certain parameters in Pequonnock River. Design depth shows sensitivity to decreased values only with a tolerance of 0.95, as do costs with a tolerance of 0.95. However, nitrogen removal efficiency shows high sensitivity to increased values with a tolerance of 1.001, indicating that even minimal overestimation of efficiency substantially affects results.

Other urban BMPs in Pequonnock River generally show lower sensitivity. Biofiltration with Underdrain, Bioretention Basin, Infiltration practices, Sand Filter with underdrain, Wet Pond, Extended Dry Detention Basin and Grass Swale with detention show "not sensitive" classifications for most parameters with tolerances ranging from 0.5 to 100. Urban baseline nitrogen loads in Pequonnock River show high sensitivity with a tolerance of 0.99 to 1.01, indicating that accurate urban load characterization is critical. Urban land area shows sensitivity with a tolerance of 0.95 to 1.05, as does urban runoff coefficient at 0.95 to 1.05. The urban cost adjustment coefficient shows sensitivity at 0.95 to 1.05 and general urban capital costs show sensitivity with a tolerance of 0.9 to 1.1.

Mystic Harbor displays similarly high sensitivity to Enhanced Biofiltration with Internal Storage Reservoir and Gravel Wetland, but with some differences in specific tolerance ranges that reflect the mixed land use characteristics of this watershed. Enhanced Biofiltration with Internal Storage Reservoir costs show high sensitivity in both directions with a tolerance range of 1.0 to 1.01, representing even tighter tolerances than observed in Pequonnock River. Nitrogen efficiency shows high sensitivity in both directions with a tolerance of 1.0 to 1.01 as well. Design depth shows high sensitivity in both directions at 1.0 to 1.01. The maximum implementation fraction shows moderate sensitivity with a tolerance of 0.98 in the decreased direction. Implementation potential shows sensitivity at 0.95.

Gravel Wetland in Mystic Harbor shows high sensitivity across multiple parameters. Costs show high sensitivity in both directions with a tolerance of 1.0 to 1.01. Nitrogen efficiency shows high sensitivity in both directions at 1.0 to 1.01. Design depth shows high sensitivity in both directions at 1.0 to 1.01. Maximum implementation fraction shows high sensitivity with tolerance 1.0 in the decreased direction. Implementation potential shows high sensitivity at tolerance 1.0.

Urban baseline nitrogen loads in Mystic Harbor show moderate sensitivity with a tolerance of 0.95 to 1.04, a slightly wider range than Pequonnock River but still indicating meaningful sensitivity. Urban land area shows sensitivity at 0.95 to 1.1, as does urban runoff coefficient at 0.95 to 1.1. The urban cost adjustment coefficient shows sensitivity at 0.95 to 1.1. Urban capital costs show sensitivity at 0.9 to 1.1. Bantam Lake, with its lower proportion of urban development, shows markedly lower sensitivity to urban parameters compared to the other two watersheds. Urban baseline nitrogen loads show only moderate sensitivity with a tolerance of 0.95 to 1.05, substantially wider than the 0.99 to 1.01 tolerance observed in Pequonnock River. All urban design depth parameters show "not sensitive" classifications with tolerances ranging from 0.5 to 100 for all urban BMP types. Maximum implementation fractions for all urban BMPs are not sensitive with tolerances of 0.01 to 1. All urban BMP implementation potentials are not sensitive with tolerances of 0.01 to 100. Urban costs for all practices are not sensitive with tolerances of 0.5 to 100. Urban runoff coefficient is not sensitive with tolerance 0.01 to 100. The urban

cost adjustment coefficient is also not sensitive at 0.01 to 100.

This contrast in urban parameter sensitivity across the three watersheds provides clear operational guidance. In heavily urbanized watersheds like Pequonnock River, users must carefully characterize Enhanced Biofiltration with Internal Storage Reservoir and Gravel Wetland parameters through site surveys, local performance monitoring and detailed cost estimates. In mixed watersheds like Mystic Harbor, similar careful characterization remains important for these key urban practices. In predominantly agricultural watersheds like Bantam Lake, urban parameter estimates can be less precise without substantially affecting optimization results, though reasonable estimates should still be employed.

The choice of which urban BMPs to implement depends on the exact estimates of efficiency and cost in both Pequonnock River and Mystic Harbor. To test if this affects a certain model scenario, users may wish to run RBEROST several times, changing the costs in the BMP specification file slightly for each run. If RBEROST subsequently chooses different urban BMPs, additional modeling for urban BMPs may be advantageous. Generally, users will benefit by having the most accurate estimates of cost, efficiency, suitability of sites for each BMP and the expected difficulty of retrofitting existing urban infrastructure with urban BMPs.

4.5.4 Riparian BMPs

Riparian buffer implementation shows minimal spending and universally low sensitivity across all three watersheds, indicating that riparian restoration is not a cost-effective approach to nutrient load reduction in these particular systems (Figure 4-6). This finding holds regardless of watershed land use composition, suggesting that factors other than land use proportions determine the cost-effectiveness of riparian buffers.

Pequonnock River shows no sensitivity to any riparian parameters, with tolerances of 0.01 to 100 for both Forested Buffer and Grassed Buffer costs, capital and operations. Riparian nitrogen removal efficiencies are not sensitive with a tolerance of 0.01 to 100. Maximum implementation fractions are not sensitive with tolerances 0.01 to 1. Riparian baseline nitrogen loads are not sensitive at 0.01 to 100. Mystic Harbor displays identical insensitivity to riparian parameters. All costs show a tolerance of 0.01 to 100, all removal efficiencies show a tolerance of 0.01 to 100, all maximum fractions show a tolerance of 0.01 to 1, and riparian baseline loads show a tolerance of 0.01 to 100.

Bantam Lake similarly shows comprehensive insensitivity to riparian parameters across both nitrogen and phosphorus metrics. All costs, efficiencies, maximum implementation fractions and baseline loads show tolerances of 0.01 to 100, indicating "not sensitive" classifications.

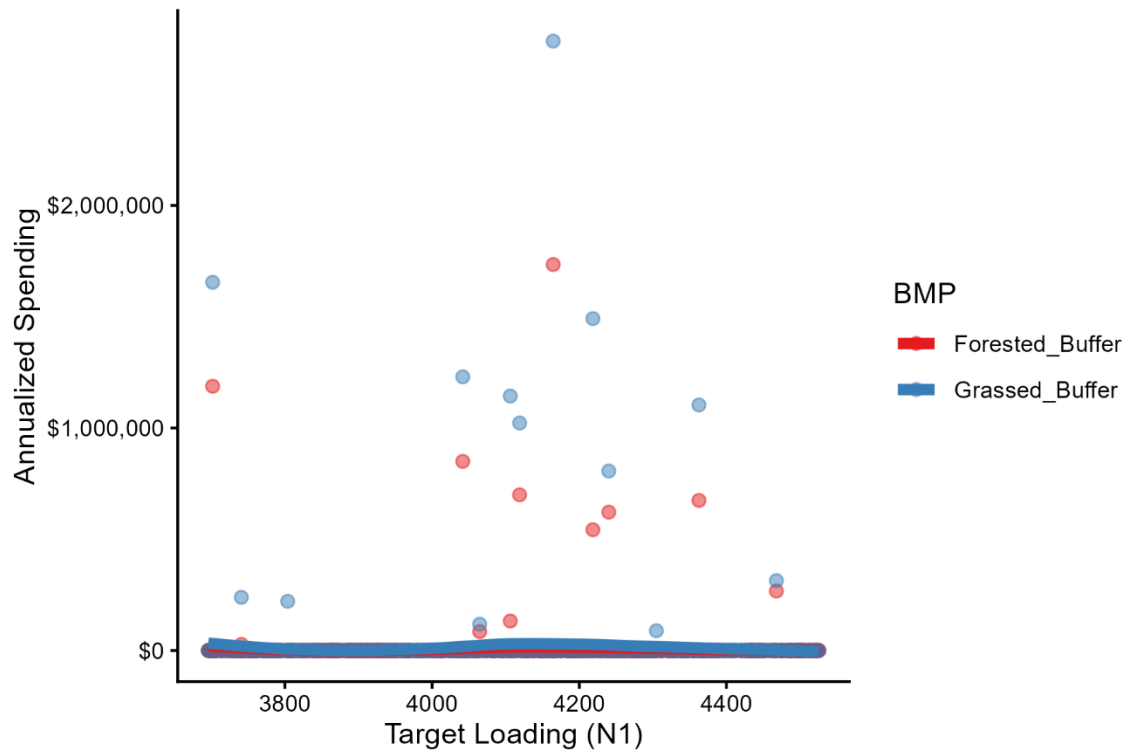


Figure 4- 6. Riparian buffer BMP spending by nitrogen target for Bantam Lake.

This consistent low sensitivity to riparian buffer parameters likely reflects several factors. Limited riparian restoration opportunities in these developed landscapes restrict the total area available for buffer establishment. Higher cost-effectiveness of other BMP types, particularly agricultural practices in Bantam Lake and urban stormwater BMPs in Pequonnock River and Mystic Harbor, means that the optimizer selects those alternatives preferentially. Riparian loads representing a small fraction of total watershed loading reduces the potential impact of riparian restoration. Physical constraints, including existing development in riparian areas, limit buffer establishment opportunities.

Table 4-1: RBEROST tolerance to variation in User Specifications.

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
ag_frac_max	Conservation	not sensitive	not applicable
ag_frac_max	Contour_Farming	not sensitive	not applicable
ag_frac_max	Fert_20	not sensitive	not applicable
ag_frac_max	Filterstrip	not sensitive	not applicable
ag_frac_max	MIN_TILL	not sensitive	not applicable
ag_frac_max	Manure_Injection	sensitive	not applicable
ag_frac_max	Ponds	sensitive	not applicable
ag_frac_max	Terrace_Only	not sensitive	not applicable
ag_frac_max	Terrace_Waterway	not sensitive	not applicable
ag_frac_max	Waterway_Only	not sensitive	not applicable
ag_frac_min	Conservation	not applicable	not applicable
ag_frac_min	Contour_Farming	not applicable	not applicable
ag_frac_min	Fert_20	not applicable	not applicable
ag_frac_min	Filterstrip	not applicable	not applicable
ag_frac_min	MIN_TILL	not applicable	not applicable
ag_frac_min	Manure_Injection	not applicable	not applicable
ag_frac_min	Ponds	not applicable	not applicable
ag_frac_min	Terrace_Only	not applicable	not applicable
ag_frac_min	Terrace_Waterway	not applicable	not applicable
ag_frac_min	Waterway_Only	not applicable	not applicable
loads_lim_N1	limits	highly sensitive	highly sensitive
loads_lim_P1	limits	highly sensitive	highly sensitive
ripbuf_frac_max	Forested_Buffer	not sensitive	not applicable
ripbuf_frac_max	Grassed_Buffer	not sensitive	not applicable
ripbuf_frac_min	Forested_Buffer	not applicable	not applicable
ripbuf_frac_min	Grassed_Buffer	not applicable	not applicable
urban_design_depth	Biofiltration_w_Underdrain	not sensitive	not sensitive
urban_design_depth	Bioretention_Basin	not sensitive	not sensitive
urban_design_depth	Enhanced_Biofiltration_w_ISR	not sensitive	not sensitive
urban_design_depth	Extended_Dry_Detention_Basin	not sensitive	not sensitive
urban_design_depth	Grass_Swale_w_detention	not sensitive	not sensitive
urban_design_depth	Gravel_Wetland	not sensitive	not sensitive
urban_design_depth	Infiltration_Basin	not sensitive	not sensitive
urban_design_depth	Infiltration_Chamber	not sensitive	not sensitive
urban_design_depth	Infiltration_Trench	not sensitive	not sensitive
urban_design_depth	Porous_Pavement_w_subsurface_infiltration	not sensitive	not sensitive
urban_design_depth	Porous_Pavement_w_underdrain	not sensitive	not sensitive
urban_design_depth	Sand_Filter_w_underdrain	not sensitive	not sensitive
urban_design_depth	Wet_Pond	not sensitive	not sensitive
urban_frac_max	Biofiltration_w_Underdrain	not sensitive	not applicable
urban_frac_max	Bioretention_Basin	not sensitive	not applicable

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
urban_frac_max	Enhanced_Biofiltration_w_ISR	not sensitive	not applicable
urban_frac_max	Extended_Dry_Detention_Basin	not sensitive	not applicable
urban_frac_max	Grass_Swale_w_detention	not sensitive	not applicable
urban_frac_max	Gravel_Wetland	not sensitive	not applicable
urban_frac_max	Infiltration_Basin	not sensitive	not applicable
urban_frac_max	Infiltration_Chamber	not sensitive	not applicable
urban_frac_max	Infiltration_Trench	not sensitive	not applicable
urban_frac_max	Porous_Pavement_w_subsurface_infiltration	not sensitive	not applicable
urban_frac_max	Porous_Pavement_w_underdrain	not sensitive	not applicable
urban_frac_max	Sand_Filter_w_underdrain	not sensitive	not applicable
urban_frac_max	Wet_Pond	not sensitive	not applicable
urban_frac_min	Biofiltration_w_Underdrain	not applicable	not applicable
urban_frac_min	Bioretention_Basin	not applicable	not applicable
urban_frac_min	Enhanced_Biofiltration_w_ISR	not applicable	not applicable
urban_frac_min	Extended_Dry_Detention_Basin	not applicable	not applicable
urban_frac_min	Grass_Swale_w_detention	not applicable	not applicable
urban_frac_min	Gravel_Wetland	not applicable	not applicable
urban_frac_min	Infiltration_Basin	not applicable	not applicable
urban_frac_min	Infiltration_Chamber	not applicable	not applicable
urban_frac_min	Infiltration_Trench	not applicable	not applicable
urban_frac_min	Porous_Pavement_w_subsurface_infiltration	not applicable	not applicable
urban_frac_min	Porous_Pavement_w_underdrain	not applicable	not applicable
urban_frac_min	Sand_Filter_w_underdrain	not applicable	not applicable
urban_frac_min	Wet_Pond	not applicable	not applicable

Table 4-2: RBEROST tolerance to variation in parameters in RBEROST datasets.

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
ag_costs_capital	Conservation	not sensitive	not sensitive
ag_costs_capital	Contour_Farming	not sensitive	not sensitive
ag_costs_capital	Fert_20	not sensitive	not sensitive
ag_costs_capital	Filterstrip	sensitive	sensitive
ag_costs_capital	MIN_TILL	not sensitive	sensitive
ag_costs_capital	Manure_Injection	sensitive	sensitive
ag_costs_capital	Ponds	sensitive	sensitive
ag_costs_capital	Terrace_Only	not sensitive	not sensitive
ag_costs_capital	Terrace_Waterway	not sensitive	not sensitive
ag_costs_capital	Waterway_Only	not sensitive	not sensitive
ag_costs_operations	Conservation	not sensitive	not sensitive
ag_costs_operations	Contour_Farming	not sensitive	not sensitive
ag_costs_operations	Fert_20	not sensitive	not sensitive
ag_costs_operations	Filterstrip	not sensitive	not sensitive
ag_costs_operations	MIN_TILL	not sensitive	not sensitive
ag_costs_operations	Manure_Injection	not sensitive	not sensitive
ag_costs_operations	Ponds	not sensitive	not sensitive
ag_costs_operations	Terrace_Only	not sensitive	not sensitive
ag_costs_operations	Terrace_Waterway	not sensitive	not sensitive
ag_costs_operations	Waterway_Only	not sensitive	not sensitive
ag_effic_N	Conservation	not sensitive	not sensitive
ag_effic_N	Contour_Farming	not sensitive	not sensitive
ag_effic_N	Fert_20	not sensitive	not sensitive
ag_effic_N	Filterstrip	highly sensitive	sensitive
ag_effic_N	MIN_TILL	sensitive	sensitive
ag_effic_N	Manure_Injection	sensitive	sensitive

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
ag_effic_N	Ponds	not sensitive	not sensitive
ag_effic_N	Terrace_Only	not sensitive	not sensitive
ag_effic_N	Terrace_Waterway	not sensitive	sensitive
ag_effic_N	Waterway_Only	not sensitive	not sensitive
ag_effic_P	Conservation	not sensitive	not sensitive
ag_effic_P	Contour_Farming	not sensitive	not sensitive
ag_effic_P	Fert_20	not sensitive	not sensitive
ag_effic_P	Filterstrip	sensitive	highly sensitive
ag_effic_P	MIN_TILL	not sensitive	not sensitive
ag_effic_P	Manure_Injection	sensitive	highly sensitive
ag_effic_P	Ponds	not sensitive	not sensitive
ag_effic_P	Terrace_Only	not sensitive	not sensitive
ag_effic_P	Terrace_Waterway	not sensitive	sensitive
ag_effic_P	Waterway_Only	not sensitive	not sensitive
area	ag	sensitive	sensitive
area	urban	not sensitive	not sensitive
baseloads_N1	ag	highly sensitive	highly sensitive
baseloads_N1	other	not sensitive	not sensitive
baseloads_N1	point	not sensitive	not sensitive
baseloads_N1	septic	not sensitive	not sensitive
baseloads_N1	storage	not sensitive	not sensitive
baseloads_N1	urban	sensitive	sensitive
baseloads_P1	ag	highly sensitive	highly sensitive
baseloads_P1	other	not sensitive	not sensitive
baseloads_P1	point	not sensitive	not sensitive
baseloads_P1	storage	not sensitive	not sensitive
baseloads_P1	urban	highly sensitive	highly sensitive

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
point_costs	capital	not sensitive	not sensitive
point_costs	operations	not sensitive	not sensitive
point_efficient_N	point	not sensitive	not sensitive
point_efficient_P	point	not applicable	not applicable
riparianload_N1	riparian	not sensitive	not sensitive
riparianload_P1	riparian	not sensitive	not sensitive
riparianremoval_N1	Forested_Buffer	not sensitive	not sensitive
riparianremoval_N1	Grassed_Buffer	not sensitive	not sensitive
riparianremoval_P1	Forested_Buffer	not sensitive	not sensitive
riparianremoval_P1	Grassed_Buffer	not sensitive	not sensitive
ripbuf_costs_capital	Forested_Buffer	not sensitive	not sensitive
ripbuf_costs_capital	Grassed_Buffer	not sensitive	not sensitive
ripbuf_costs_operations	Forested_Buffer	not sensitive	not sensitive
ripbuf_costs_operations	Grassed_Buffer	not sensitive	not sensitive
runoff_coeff_urban	urban	not sensitive	not sensitive
septic_bmp_implement	none	not sensitive	not sensitive
septic_costs_capital	none	not sensitive	not sensitive
septic_costs_operati	none	not sensitive	not sensitive
septic_efficient_N	none	not sensitive	not sensitive
unbuffered_banklength	Forested_Buffer	not sensitive	not sensitive
unbuffered_banklength	Grassed_Buffer	not sensitive	not sensitive
urban_bmp_implementationpotential	Biofiltration_w_Underdrain	not sensitive	not sensitive
urban_bmp_implementationpotential	Bioretention_Basin	not sensitive	not sensitive
urban_bmp_implementationpotential	Enhanced_Biofiltration_w_ISR	not sensitive	not sensitive
urban_bmp_implementationpotential	Extended_Dry_Detention_Basin	not sensitive	not sensitive
urban_bmp_implementationpotential	Grass_Swale_w_detention	not sensitive	not sensitive
urban_bmp_implementationpotential	Gravel_Wetland	not sensitive	not sensitive

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
urban_bmp_implementationpotential	Infiltration_Basin	not sensitive	not sensitive
urban_bmp_implementationpotential	Infiltration_Chamber	not sensitive	not sensitive
urban_bmp_implementationpotential	Infiltration_Trench	not sensitive	not sensitive
urban_bmp_implementationpotential	Porous_Pavement_w_subsurface_infiltration	not sensitive	not sensitive
urban_bmp_implementationpotential	Porous_Pavement_w_underdrain	not sensitive	not sensitive
urban_bmp_implementationpotential	Sand_Filter_w_underdrain	not sensitive	not sensitive
urban_bmp_implementationpotential	Wet_Pond	not sensitive	not sensitive
urban_cost_adjustment_coef	NA	not sensitive	not sensitive
urban_costs	Biofiltration_w_Underdrain	not sensitive	not sensitive
urban_costs	Bioretention_Basin	not sensitive	not sensitive
urban_costs	Enhanced_Biofiltration_w_ISR	not sensitive	not sensitive
urban_costs	Extended_Dry_Detention_Basin	not sensitive	not sensitive
urban_costs	Grass_Swale_w_detention	not sensitive	not sensitive
urban_costs	Gravel_Wetland	not sensitive	not sensitive
urban_costs	Infiltration_Basin	not sensitive	not sensitive
urban_costs	Infiltration_Chamber	not sensitive	not sensitive
urban_costs	Infiltration_Trench	not sensitive	not sensitive
urban_costs	Porous_Pavement_w_subsurface_infiltration	not sensitive	not sensitive
urban_costs	Porous_Pavement_w_underdrain	not sensitive	not sensitive
urban_costs	Sand_Filter_w_underdrain	not sensitive	not sensitive
urban_costs	Wet_Pond	not sensitive	not sensitive
urban_costs	capital	not sensitive	not sensitive
urban_costs	operations	not sensitive	not sensitive
urban_effic_N	Biofiltration_w_Underdrain	not sensitive	not sensitive
urban_effic_N	Bioretention_Basin	not sensitive	not sensitive
urban_effic_N	Enhanced_Biofiltration_w_ISR	not sensitive	not sensitive
urban_effic_N	Extended_Dry_Detention_Basin	not sensitive	not sensitive

Parameter	Set member (BMP or other)	Tolerance to decreased values	Tolerance to increased values
urban_effic_N	Grass_Swale_w_detention	not sensitive	not sensitive
urban_effic_N	Gravel_Wetland	not sensitive	not sensitive
urban_effic_N	Infiltration_Basin	not sensitive	not sensitive
urban_effic_N	Infiltration_Chamber	not sensitive	not sensitive
urban_effic_N	Infiltration_Trench	not sensitive	not sensitive
urban_effic_N	Porous_Pavement_w_subsurface_infiltration	not sensitive	not sensitive
urban_effic_N	Porous_Pavement_w_underdrain	not sensitive	not sensitive
urban_effic_N	Sand_Filter_w_underdrain	not sensitive	not sensitive
urban_effic_N	Wet_Pond	not sensitive	not sensitive
urban_effic_P	Biofiltration_w_Underdrain	not sensitive	not sensitive
urban_effic_P	Bioretention_Basin	not sensitive	not sensitive
urban_effic_P	Enhanced_Biofiltration_w_ISR	not sensitive	not sensitive
urban_effic_P	Extended_Dry_Detention_Basin	not sensitive	not sensitive
urban_effic_P	Grass_Swale_w_detention	not sensitive	not sensitive
urban_effic_P	Gravel_Wetland	not sensitive	not sensitive
urban_effic_P	Infiltration_Basin	not sensitive	not sensitive
urban_effic_P	Infiltration_Chamber	not sensitive	not sensitive
urban_effic_P	Infiltration_Trench	not sensitive	not sensitive
urban_effic_P	Porous_Pavement_w_subsurface_infiltration	not sensitive	not sensitive
urban_effic_P	Porous_Pavement_w_underdrain	not sensitive	not sensitive
urban_effic_P	Sand_Filter_w_underdrain	not sensitive	not sensitive
urban_effic_P	Wet_Pond	not sensitive	not sensitive

5 Data Dictionary

Table 5-1 lists the input files that are read into the preprocessing and postprocessing R code, describes the contents of the inputs and specific data fields used in the model, units of data and data sources.

Table 5-1 lists the output files that are created by the postprocessing R code, the data units and the sources of associated data.

Table 5-1 : Data Dictionary of Input Files.

Input File Name	Input File Description	Data Units (If Applicable)	Source(s)
01_UserSpecs_BMPs_wCS.csv	User specification file for selection of BMPs to be included in optimization model, BMP capital and operations & maintenance costs and urban BMP runoff depth specification. File includes default capital and operations & maintenance costs as well as default runoff depths. Minimum and maximum runoff depth fields (Min_RD_in and Max_RD_in) indicate the range between which the user can specify urban BMP runoff depths. Min_RD_in and Max_RD_in fields are not used in scaled-up optimization model code. Separate columns must be included for the capital and operations & maintenance costs for each state included in the geographic range of optimization. Column titles must follow the naming convention “capital_[two letter state code]” and “operations_[two letter state code]”.	Per-area and per-volume BMP cost units described in file. Runoff depth in inches. Buffer width in feet.	BMP cost data sources described in Table 3-1. Minimum and maximum runoff depths for Urban BMPs from New Hampshire MS4 permit BMP performance curves (New Hampshire Department of Environmental Services, 2020b). Minimum and maximum buffer widths from Houle et al. (2019).
01_UserSpecs_loadingtargets.csv	User specification file for providing loading targets to include in RBEROST. Fields include Waterbody name, NHD+ COMID, Percent Reduction, choice of TN or TP, flags for out of network and flags for terminal reaches. If the user has loading targets on a mass basis, they should input the percent reduction from current loadings that the mass target represents.	Percent	EPA
01_Preprocessing_Terminal_COMID.csv	File providing the list of terminal COMIDs assigned to each watershed contained within the LIS model spatial extent and 01_UserSpecs_loadingtargets.csv.	Not applicable	EPA
01_Preprocessing_Upstream_COMID.csv	File providing the list of terminal and upstream COMIDs for each watershed within the LIS model spatial extent. This file contains the entirety of the COMIDs in the model.	Not applicable	EPA

Input File Name	Input File Description	Data Units (If Applicable)	Source(s)
ACRE_HUC12_HRU_Summary_compareBaseline.csv	Agricultural BMP efficiencies summarized by HUC12 (or if unavailable, HUC10 and/or HUC8) based on ACRE database. "HUC" fields represent the hydrologic unit code (HUC12/HUC10/HUC8). "Scenario" field represents BMP scenarios, modeled by White et al. (2019) for the ACRE database. "MeanTN_Effic" field reflects average nitrogen removal efficiency per scenario, with respect to "Baseline" conditions.	Percent	Original ACRE database from White et al. (2019). Data summarized at HUC-levels by RBEROST associated R scripts.
ACRE_HUC12_HRU_Summary_compareNoPractice.csv	Agricultural BMP efficiencies summarized by HUC12 (or if unavailable, HUC10 and/or HUC8) based on ACRE database. "HUC" fields represent the hydrologic unit code (HUC12/HUC10/HUC8). "Scenario" field represents BMP scenarios, modeled by White et al. (2019) for the ACRE database. "MeanTN_Effic" field reflects average nitrogen removal efficiency per scenario, with respect to "No Practice" conditions.	Percent	Original ACRE database from White et al. (2019). Data summarized at HUC-levels by RBEROST associated R scripts.
AgBMPEffic_FertManure.csv	Agricultural BMP nitrogen removal efficiencies for 20 percent fertilizer reduction and manure injection (do not vary by location). "Category" field reflects the agricultural BMP category within RBEROST, "BMP" field reflects the best management practice considered and "N_Efficiency" field reflects the nitrogen removal efficiency.	Percent	EPA, NRCS

Input File Name	Input File Description	Data Units (If Applicable)	Source(s)
EQIPcosts_overyears.csv	EQIP costs for agricultural and riparian buffer BMPs for years 2018-2021. “BMP_Category” field reflects whether a BMP is implemented on agricultural (row crop) land or in riparian buffers. “BMP” field reflects the best management practice considered, “capital_units” reflect the cost units for capital costs as cost / unit area, “operations_units” reflect the cost units for operation costs as cost / unit area. The remaining fields describe the costs from EQIP payment schedules follow the naming convention “[capital or operations costs]_[state]_[year]”. At least one year of data must be provided for each state included in the geographic range of optimization.	Dollars per unit area	U.S. Department of Agriculture Staff. 2021
LengthinBuffer_ICF25.csv	The amount of each stream reach that is in riparian buffers of certain widths, as of 2016. Fields include “comid,” “totalbanklength_ft,” which is two times the stream reach, “totalbanklength_ft_se” describing the standard error associated with the total bank length and fields describing the length of stream bank in buffer and the standard error around those values. Naming conventions for the rest of the fields are [Riparian buffer BMP]_["buffer" for estimates, "buffer_se" for standard errors]_[minimum buffer width to be considered "in buffer"]_ft.	Feet	Stream reaches from McKay et al. 2012, landcover from Yang et al. 2018. Summaries provided following R scripts and ArcPython model builder scripts associated with RBEROST.
ne_sparrow_model_input.csv	Northeastern Regional SPARROW Model input dataset. Fields used in scaled-up optimization model include “HUC_12,” “comid,” “IncAreaKM2” (incremental area per catchment) and “urban_km2” (urban area per catchment).	Incremental and urban areas in km2	Ator (2019)
NHD+infiltrationrates.csv	The average infiltration rate in each COMID. Fields include “comid,” “infiltrationrate_inperhr” (the average infiltration rate in inches per hour) and “infiltrationrate_inperhr_dist” (a resampling of infiltration rates used for RBEROST uncertainty analysis. Values are a character string of 10 samples that are parsed within RBEROST)	inches per hour, list of values in inches per hour	Hydrologic soil group values from Soil Survey Staff (2020a, 2020b). Infiltration rates for hydrologic soil groups from USDA NRCS 2009

Input File Name	Input File Description	Data Units (If Applicable)	Source(s)
RiparianEfficiencies_ICF24.csv	Performance curves describing nutrient removal efficiency in riparian buffers. Fields include “comid,” and fields with character entries of functions that RBEROST will parse to calculate efficiency. All functions are functions of buffer width and the coefficients depend on slope and hydrologic soil group. Naming convention follows [N or P, designating which nutrient removal efficiency is being described]_[type of riparian buffer BMP]_[approximate width of buffer]. Though all equations are a function of buffer width, the slope and hydrologic soil group may differ based on the buffer width which may change the coefficients. Columns that describe uncertainty are designated with “_uncertainty” at the end. Uncertainty columns contain a character string of 10 different equations produced by sampling the slope and hydrologic soil groups of each buffer width. These character strings are parsed within RBEROST.	Percent	Methodology: Houle et al. 2019. River reach shape files: McKay et al. 2012. Hydrologic soil group: gNATSGO (Soil Survey Staff, 2020a; Soil Survey Staff, 2020b;). Slope data: Verdin, 2017
RiparianLoadings_ICF25.csv	Nutrient loading from riparian zones. Fields include “comid,” “N_riparian_kgyr” “P_riparian_kgyr,” “N_riparian_kgyr_se,” and “P_riparian_kgyr_se,” where N or P designates the nutrient being described and _se designates standard errors.	kg-N/yr; kg-P/yr; kg-N/yr; kg-P/yr	Methodology for calculating loading: Houle et al. 2019. Land cover data: Jin et al. 2019, Yang et al. 2018. River reach shape files: McKay et al. 2012

Input File Name	Input File Description	Data Units (If Applicable)	Source(s)
UrbanBMPPPerformanceCurves.csv	Performance curves that describe the nutrient removal efficiency of urban BMPs. Fields include: “BMP” (the best management practice under consideration), “Pollutant” (designates if the performance curve relates to N or P), “InfiltrationRate_inperhr” (the infiltration rate that the performance curve assumes. BMPs that do not depend on infiltration rate have “NA.”), “Best.Fit.Curve” (a character string that lists the form of the performance curve), and “Coef.1,” “Coef.2” and “Coef.3.” The “Coef” columns list the coefficients that correspond with the function in “Best.Fit.Curve.” Standard error around these coefficients are given in “Coef.1_se,” “Coef.2_se” and “Coef.3_se.”	y in “Best.Fit.Curve” are in units percent, where x is the rating depth of the BMP in inches.	New Hampshire Department of Environmental Service (2020b)
WWTP_BaselineRemoval_Finergrain.csv	An extension of WWTP_COMIDs_BslnRemoval.csv that includes loading for each year available in New Hampshire Department of Environmental Services 2020a. RBEROST uses the “load_lbdy_2014/2015/2016/2017/2018” columns to calculate the standard error of change over time.	kg/yr or lb/day as designated in column headings	New Hampshire Department of Environmental Services (2020a)
WWTP_COMIDs_BslnRemoval.csv	Percentage change in WWTP effluent loadings from 2014-2018 based on Newport Wastewater Treatment Facility NPDES Permit No. NH0100200. Data used to scale point source loading estimates from the Northeastern Total Nitrogen SPARROW Model (representing 2012 loadings) to current (as of 2018) loadings. “Rem_2014_2018_load_ch” field represents percent loading change from 2014-2018 per WWTP.	Percent	New Hampshire Department of Environmental Services (2020a)
WWTP_COMIDs.csv	Crosswalk between the WWTPs included in the scaled-up optimization model example application and their locations along the NHDPlus Version 2 flow network.	Not applicable	McKay et al. (2012), Ator (2019), EPA

Input File Name	Input File Description	Data Units (If Applicable)	Source(s)
WWTP_RemovalEffici_Upd.csv	Estimated nitrogen removal efficiencies based on BioWin modeling. “Category” field reflects the point source BMP category within the scaled-up optimization model code, “BMP” field reflects the WWTP considering implementation of low-cost retrofits, and “N_Efficiency” field reflects the nitrogen removal efficiency.	Percent	JJ Environmental (2015)
Unclassified_Septic_Parcel_LIS.xlsx	Counts of unclassified (Class 0) parcels within each COMID, used to calculate septic load per parcel in preprocessing code.	Count	EPA REGRID parcel database (2025b)
01_RevSepticUpgrde_ParcelEff.csv	Counts of parcels available for each class of upgrades in each COMID, as well as the efficiency per parcel.	Count and fractions	EPA REGRID parcel database (2025b), Como et al. 2015, Parcel databases by town in CT and RI
01_RevSepticConv_ParcelEff.csv	Counts of parcels available for each class of sewer conversions in each COMID, and efficiency per parcel.	Count and fractions	EPA REGRID parcel database (2025b)
PredictNoBFlowN_WSeptic.csv	SPARROW model outputs that provide per COMID nitrogen loadings broken down by source categories such as urban runoff, agricultural manure application, atmospheric deposition, septic system and many more.	kg-N/yr	Schamdel et al (2021), EPA
ne_dynamic_sparrow_se_tn.csv	The standard errors for SPARROW model nitrogen loadings. These form the basis of uncertainty bounds.	kg-N/yr	Schamdel et al (2021)
Predict_P.csv	SPARROW model outputs that provide per COMID phosphorous loadings broken down by source categories such as urban runoff, agricultural manure application, atmospheric deposition and many more.	kg-P/yr	Schamdel et al (2021)
ne_dynamic_sparrow_se_tp.csv	The standard errors for SPARROW model phosphorous loadings. These form the basis of uncertainty bounds.	kg-P/yr	Schamdel et al (2021)
CT_NH_MA_VT_NY_RI_StreamCatData_2019.csv	Contains the percentage of land use cover percentages for each model COMID by state. Each COMID has percentage values for percent open water, percent shrub, percent grassland, percent low density urban and a variety of other categories.	Percentage	EPA

Table 5-2. Data Dictionary of Output Files.

Input File Name	Input File Description	Data Units (If Applicable)
AgBMP_by_ComID_Scenario1.csv (or other scenarios, if not renamed by user)	Agricultural (row crop) BMP implementation by COMID. Fields include “COMID,” “HUC_12” (12 digit hydrologic unit code), “IncAreaKm” (the area of the catchment defined by COMID), “ag_km2” (the area within that COMID defined as ag), “PctCrop2011Cat” (the percent of NLCD 2011 pixels categorized as row crop in the COMID as summarized by StreamCat) and “StateAbbrev” (VT or NH). Any additional fields follow the naming convention of [BMP name]_[either FraclImplement (the fraction of ag area treated with this BMP) or areaac (the total acreage of area treated with this BMP)]	_FraclImplement fields as fractions, _areaac fields as acres
Urban_by_ComID_Scenario1.csv (or other scenarios, if not renamed by user)	Urban BMP implementation by COMID. Fields include “COMID,” “HUC_12” (12 digit hydrologic unit code), “IncAreaKm” (the area of the catchment defined by COMID), “urban_km2” (the area within that COMID defined as urban) and “StateAbbrev” (VT or NH). Any additional fields follow the naming convention of [BMP name]_[either FraclImplement (the fraction of urban area treated with this BMP) or areaac (the total acreage of area treated with this BMP)]	_FraclImplement fields as fractions, _areaac fields as acres
RipBufBMP_by_ComID_Scenario1.csv (or other scenarios, if not renamed by user)	Riparian buffer BMP implementation by COMID. Fields include “COMID,” “HUC_12” (12 digit hydrologic unit code), “IncAreaKm” (the area of the catchment defined by COMID), “totalbanklength_ft” (the distance of bank within the COMID, or 2 times “riverreachlength_ft”), “StateAbbrev” (VT or NH) and “riverreachlength_ft” (the length of river). Additional fields follow the naming convention of [BMP name]_LengthImplemented_ft.	Length, in ft

SepticBMP_by_ComID_Scenario1.csv (or other scenarios, if not renamed by user)	Septic system BMP implementation by COMID. Fields include "COMID," "HUC_12" (12 digit hydrologic unit code), "IncAreaKm2" (the area of the catchment defined by COMID) and "StateAbbrev" (state abbreviation). Any additional fields follow the naming convention of [BMP name]_ParcelsImplement (the number of septic parcels treated with this BMP). BMP names include class_[1-9]_upgrade for septic system upgrades by efficiency class and Sewer_convert for sewer conversion projects.	_ParcelsImplement fields as number of parcels (integer values)
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Legend

Embayment Watersheds

- Byram River, CT and NY
- Connecticut River, CT
- Farm River, CT
- Hempstead Harbor, NY
- Housatonic River, CT
- Huntington Bay, NY
- Huntington Harbor, NY
- Little Narragansett Bay, CT
- Lloyd Harbor, NY
- Mamaroneck River, NY
- Manhasset Bay, NY
- Mount Sinai Harbor, NY
- Mystic Harbor, CT
- New Haven Harbor, CT
- Niantic Bay, CT
- Nissequogue River, NY
- Northport-Centerport Harbor Complex, NY
- Norwalk Harbor, CT
- Oyster Bay / Cold Spring Harbor Complex, NY
- Pawcatuck River, CT and RI
- Pequonnock River, CT
- Port Jefferson Harbor, NY
- Saugatuck Estuary, CT
- Southport Harbor / Sasco Brook, CT
- Stonington Harbor, CT
- Stony Brook Harbor, NY
- Thames River, CT

0 70 140 280 Kilometers

Esri, TomTom, Garmin, SafeGraph, FAO, METI/NASA, USGS, EPA, NPS, USFWS

Figure A- 1. Map of watersheds available for the Long Island Sound RBEROST Model.

Appendix B – Manual NEOS Uploads Using Web Browser

In the event that the API submission through RStudio cannot be utilized, model runs can be submitted to NEOS using a web browser interface. Users will need to create an account with NEOS if they have not already done so and ensure they are logged in. Once a new job has been created for CPLEX optimization, a user may upload the data, model and command files. Additional comments and settings may be chosen as well. Once a run has been processed, outputs can be returned by email or can be viewed on the NEOS results webpage. Once results are accessed through the NEOS web interface, a user can copy and save them as a text file, allowing the files to be run through the postprocessor as they would be if a model run was submitted through the RStudio API interface.

Users will click on the following link and upload the AMPL model, data and commands files to the Web Submission Form: <https://neos-server.org/neos/solvers/lp:CPLEX/AMPL.html> (Figure B-1). Before submitting, the user will specify the e-mail address to receive an update when the model run is complete. Users are also highly encouraged to include a descriptive note of the model in the comments section (Figure B-2). Please refer to the WMOST User Guide (*Detenbeck et al.*, 2018) for additional information on how to run optimization models on the NEOS server. All AMPL files must be of the same type -- either with uncertainty, or without uncertainty. Mixing scripts of the two types will produce errors and the model will not solve. If you are running RBEROST with the uncertainty analysis, you only need to submit the AMPL files titled with "uncertainty". The other files that were created, which describe the same problem without the uncertainty analysis, can be ignored, saved for reference, or submitted separately.

Once the optimization has solved, users will receive an email from the server with a link, a job number and a password. In some cases, receipt of emails may be hindered by local firewall settings so code has been modified to report NEOS jobnames and passwords for each submitted job so that users can later retrieve details if desired. Users can retrieve their results from the server with this information (Figure B-3). Users save the NEOS server optimization results by selecting all text within the results window (CTRL+A), pasting it into a text file (via text editor such as Notepad) and saving it as a text file (.txt) to a location that makes sense for them (Figure B-4). Example output files are provided in .

\Postprocessor\Input. If NEOS was unable to optimize the model, the returned result may be very short, simply stating that optimization failed. This result file may still be submitted to the RBEROST postprocessor.

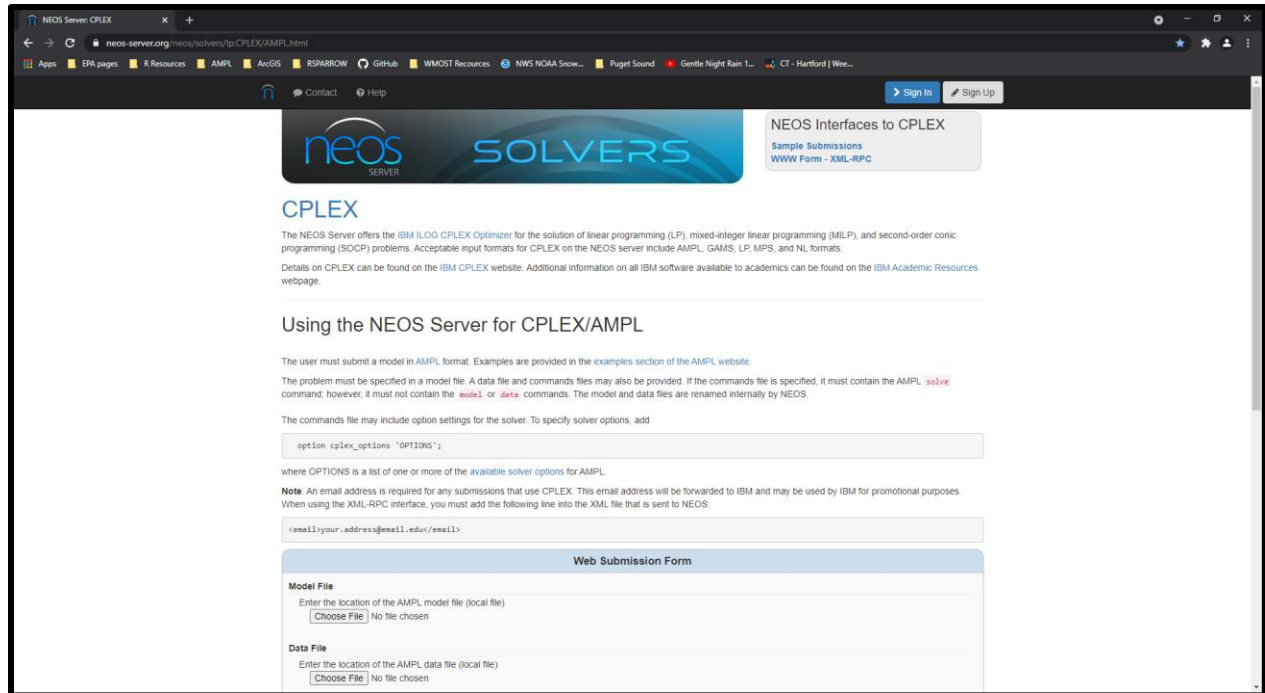


Figure B- 1. The CPLEX Optimizer home page hosted by NEOS.

MPL.html

GIS RSPARROW GitHub WMOST Resources NWS NOAA Snow... Puget Sound Gentle Night Rain 1... CT - Hartford | Wee...

Contact Help Sign In Sign U

When using the XML-RPC interface, you must add the following line into the XML file that is sent to NEOS:

<email>your.address@email.edu</email>

Web Submission Form

Model File
Enter the location of the AMPL model file (local file)
Choose File STmodel.mod

Data File
Enter the location of the AMPL data file (local file)
Choose File STdata.dat

Commands File
Enter the location of the AMPL commands file (local file)
Choose File STcommand.amp

Comments
RBEROST demonstration
Upper CT river, 3 loading targets.
Porous Pavement w/ subsurface infiltration excluded, 10%
cap on ag ponds
No uncertainty

Additional Settings
☐ Dry run: generate job XML instead of submitting it to NEOS
☐ Short Priority: submit to higher priority queue with maximum CPU time of 5 minutes
E-Mail address: chamberlin.catherine@epa.gov

Please do not click the 'Submit to NEOS' button more than once.

Submit to NEOS Clear this Form

By submitting a job, you have accepted the [Terms of Use](#)

Figure B- 2. Example of a user input submission to CPLEX using the web interface to run RBEROST.

NEOS Server: Job Administration

neos-server.org/neos/admin.html

Apps EPA pages R Resources AMPL ArcGIS RSPARROW GitHub WMOST Resources NWS NOAA Show... Puget Sound Gentle Night Rain 1... CT - Hartford | Wee...

Contact Help Sign In Sign Up

neos OPTIMIZATION
SERVER

Enter the job number and the password of the job you wish to kill/view.
You can leave these blank if viewing the queue.

Enter the job number: 10459757

Enter the password for job: aZmYhHfRi

☐ View Job Queue

☒ View Job Results

☐ Kill or Dequeue Job

Submit

We want to keep our services as available and free as possible. Please consider making a [contribution](#) to help us keep the optimizations flowing.

[Terms of Use](#) · [Acknowledgements](#) · [Questions and Comments](#)

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Figure B- 3. Results can also be retrieved through the NEOS Job Administration page.

The screenshot shows the NEOS Server Results page for Job #10459757. The browser address bar shows the URL: `neos-server.org/neos/cgi-bin/nph-neos-solver.cgi`. The page header includes the NEOS SERVER logo and the word "RESULTS".

On the left side, there is a sidebar with the following information:

- NEOS Server Version
- Job# : 10459757
- Password : aZmYhHRi
- User :
- Solver : lp:CPLEX
- Start : 2021-05-
- End : 2021-05-
- Host : prod-sub
- Disclaimer:
- This information is implied warranty. I of any kind concern information for any
- Announcements:
- We will require an on the NEOS Server
- Please see: <https://neos-guid>

The main content area displays a table of results. The table has columns for Job ID, Solver, and various numerical values. The data is as follows:

Job ID	Solver	Value 1	Value 2	Value 3	Value 4	Value 5	Value 6	Value 7
'9332520'	0	0.9	0	0.1	0	0	0	0
'9332526'	0	0.9	0	0.1	0	0	0	0
'9332528'	0	0.9	0	0.1	0	0	0	0
'9332532'	0	0.9	0	0.1	0	0	0	0
'9332534'	0	0.9	0	0.1	0	0	0	0
'9332536'	0.9	0	0	0.1	0	0	0	0
'9332538'	0.9	0	0	0.1	0	0	0	0
'9332542'	0.9	0	0	0.1	0	0	0	0
'9332546'	0.9	0	0	0.1	0	0	0	0
'9332548'	0.9	0	0	0.1	0	0	0	0
'9332550'	0.9	0	0	0.1	0	0	0	0
'9332552'	0.9	0	0	0.1	0	0	0	0
'9332558'	0	0.9	0	0.1	0	0	0	0
'9332560'	0	0.9	0	0.1	0	0	0	0
'9332564'	0	0.9	0	0.1	0	0	0	0
'9332566'	0.9	0	0	0.1	0	0	0	0
'9332568'	0	0.9	0	0.1	0	0	0	0
'9332570'	0.9	0	0	0.1	0	0	0	0
'9332582'	0.9	0	0	0.1	0	0	0	0
'9332584'	0.9	0	0	0.1	0	0	0	0
'9336412'	0	1	0	0	0	0	0	0
'9336426'	0	0.9	0	0.1	0	0	0	0
'9336430'	0	0.9	0	0.1	0	0	0	0
'9336432'	0	0	0	0.1	0.9	0	0	0
'9336434'	0	1	0	0	0	0	0	0
'9336440'	0	0.9	0	0.1	0	0	0	0
'9336444'	0	0.9	0	0.1	0	0	0	0
'9336446'	0	1	0	0	0	0	0	0

Below the table, there are several lines of text:

- Job 10459757 has finished
- You are using the solver
- Comments
- RBEROST demonstration
- Upper CT river, 3 load
- Porous Pavement w/ sub
- With uncertainty
- Checking ampl.mod for
- Checking ampl.com for
- Executing AMPL.
- processing data.

A Notepad window titled "Untitled - Notepad" is open over the table, showing the same data. The NEOS Server Home link is visible at the bottom right.

Figure B- 4. Saving NEOS results as a text file through the web interface.

Appendix C – User Parameter Change Checklist

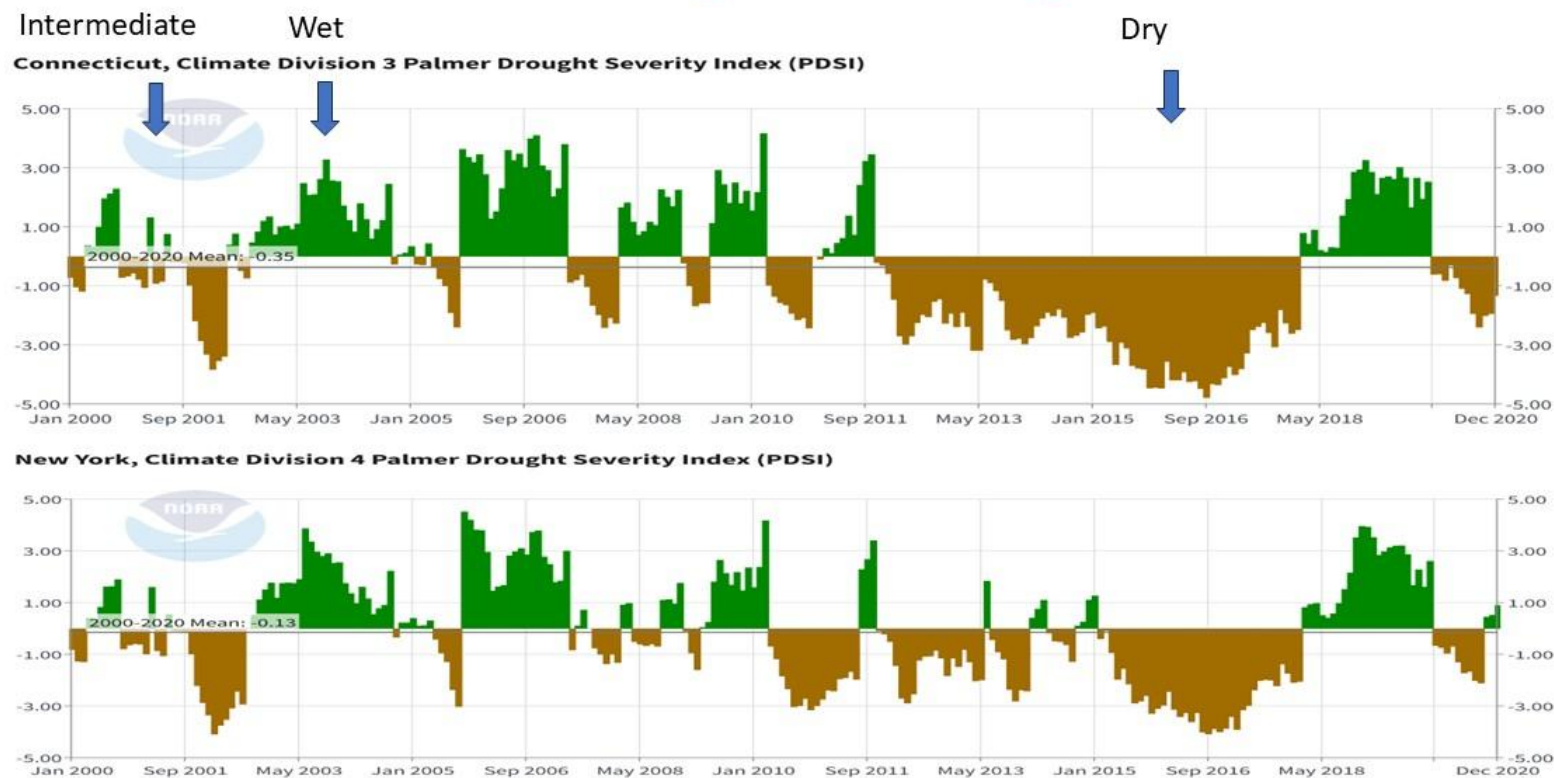
Table C- 1: Checklist of User-defined parameters in the LIS RBEROST RMD file.

Parameter	Description	Adjusted?
InPath	Input path for data and codes. Should not need to be updated if package downloaded together.	
OutPath	Output path for data and codes. Should not need to be updated if package downloaded together.	
MODE	Set to “All” for entire model, or “Select” to run a subset of watersheds.	
watershed_choices	List of watersheds to run if MODE is set to “select”	
model_years	Continuous vector of years to model.	
horizon	Cost annualization horizon.	
interest_rate	Cost annualization interest rate.	
actionable_load	Set to TRUE to remove load types that cannot be reduced, or FALSE to model all relevant load types.	
AgBMPcomparison	Determines how Ag BMP efficiencies are determined in preprocessing.	
IncludeUncertainty	Include uncertainty in model.	
n.scenarios	Number of uncertainty scenarios.	
scenariostepchange	Variation between each uncertainty scenario.	

Appendix D – Large Watershed Optimizations

Optimizations without uncertainty were carried out using the dynamic Long Island Sound RBEROST for the largest watersheds in the LIS Basin: Connecticut River (Lower Connecticut R, Lower Middle Connecticut R, Upper Middle Connecticut R, Upper Connecticut R), Housatonic R, Thames River, New Haven Harbor and Pawcatuck R. To cover the range of interannual variability in temperature and precipitation over the course of the RBEROST time period (2000 – 2020), optimizations were carried out for each of three years representing average Palmer Drought Severity Index (PDSI, 2001), wettest year (highest PDSI, 2003) and driest year (lowest PDSI, 2016) (Figure D-1). Initial targets selected for optimizations were identified from TetraTech (2022), and then reduced systematically to determine largest achievable loading targets if the initial target was infeasible. The actionable_load option was set to “TRUE” to focus on reducing source loads that are subject to management actions included in RBEROST (i.e., excluding spring snowmelt, baseflow, atmospheric deposition and storage). The AgBMPcomparison variable was set to “No Practice” such that ACRE database comparisons of SWAT model runs with and without agricultural conservation practices of interest were compared to estimate percent load reductions (Figure D-2).

Palmer Drought Severity Index



Office of Applied Science and Environmental Solutions

Figure D- 1. Monthly Palmer Drought Severity Index from 2000 to 2020 for coastal Connecticut (top) and coastal New York (bottom). Retrieved from: <https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/regional/time-series/101/pdsi/1/0/2000-2020>

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140
141 # Select years to include in the model (e.g., c("00","01","02"))
142 # Note: years are available from 00 to 20, and numbers less than ten must have preceeding zeros.
143 model_years <- c("01") #
144
145 # This variable represents the planning horizon in terms of years. Default is 15.
146 horizon <- 15
147
148 # This variable represents to expected interest rate. Default is 0.03, or 3%.
149 interest_rate <- 0.03
150
151 # Load parameter specification
152 actionable_load <- TRUE # If TRUE, the model will exclude "other" loads from sources such as snowmelt
    and atmospheric deposition for which there are no applicable BMPs that can reduce them
153 # Note: Setting to FALSE is much more likely to result in infeasible results since "other" loads
    typically make up sizeable portions of overall loadings within a watershed, and as such the reduction
    target percentages exceed the amount that can be reduced by BMPs in other categories.
154
155 # This variable lets the user decide if they wish to consider agricultural BMP efficiencies of
    practices versus "No Practice" (a default of no conservation practices) or "Baseline" (a default of
    practices currently in place, and implementing a BMP will remove practices currently in place).
156 AgBMPcomparison <- "No Practice"
157
158 # This variable decides if you will run RBEROST with or without uncertainty. It can be TRUE or FALSE.
    Running with uncertainty takes longer than running without.
159 IncludeUncertainty <- FALSE
160
161 # If running RBEROST with uncertainty, how many scenarios would you like to view? Default is 3.
162 n.scenarios <- 3
163
164 # If running RBEROST with uncertainty, how different would you like the scenarios to be? Each
    scenario will be solved for loading limits that are a certain percentage lower than the previous.
    Default is 0.01, or 1%.
165 scenariostepchange <- 0.01
166
167 ### BMP Specifications
168
169 # -----General notes-----
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Figure D- 2. RBEROST screenshot showing options selected for large watershed optimizations.

Initial and achievable targets for each of the optimization runs are summarized in Table D-1. The portion of the target percent N load reduction that could be met varied across watersheds as well as across average, wet and dry years. Ability to meet a fixed percent N load reduction was usually lowest in wet years but not necessarily greatest in dry years because the distribution of loads across point sources (with more expensive load reductions) versus nonpoint sources (with less expensive load reductions) varied across wet and dry years. For the Connecticut River, the initial target percent N load reduction of 33 percent was feasible in the Upper Connecticut River and Upper Middle Connecticut River segment, but not achievable in the lower two segments. In the wettest years, only 9-10 percent N load reductions could be achieved in the lower two segments. Ability to achieve load reductions was seasonally constrained by the dominance of “non-actionable” load reductions such as spring snowmelt, groundwater, atmospheric deposition and storage (legacy nutrient) contributions (See Table 3-5 for example.)

Table D-1. Results of optimizations for least-cost solutions to meeting nitrogen loading targets for largest tributaries to Long Island Sound, including evaluations for average, wet and dry years (based on Palmer Drought Severity Index). Actionable sources do not include atmospheric deposition, baseflow (groundwater), spring snowmelt, or legacy nutrients.

Watershed	TetraTech target load percent N reduction	Year	Wet/Dry/ Average	Feasible solution for "actionable sources"?	Largest target % reduction for given year with feasible solution	Cost
Housatonic R	64	2003	W	N	22	18,697,349,830
Housatonic R	64	2001	A	N	42	155,805,812
Housatonic R	64	2016	D	N	57	30,291,130,370
New Haven Harbor	34	2003	W	N	9	8,769,480,742
New Haven Harbor	34	2001	A	N	28	8,283,754,193
New Haven Harbor	34	2016	D	N	15	5,336,939,889
Thames R	52	2003	W	N	22	6,133,993,147
Thames R	52	2001	A	Y	52	19,275,692
Thames R	52	2016	D	N	51	92,130,068,540
Upper CT R	33	2003	W	Y	33	33,468
Upper CT R	33	2001	A	Y	33	100,5865
Upper CT R	33	2016	D	Y	33	83,074
Upper Middle CT R	33	2003	W	Y	33	762,286
Upper Middle CT R	33	2001	A	Y	33	197,551
Upper Middle CT R	33	2016	D	Y	33	4,173,845
Lower Middle CT R	33	2003	W	N	9	92,278
Lower Middle CT R	33	2001	A	N	28	89,176,296,160
Lower Middle CT R	33	2016	D	N	8	1,519,330,852
Lower CT R	33	2003	W	N	10	33,128,215,120
Lower CT R	33	2001	A	N	14	17,597,4181
Lower CT R	33	2016	D	N	25	17,616,231,670
Pawcatuck R	61	2003	W	N	21	8,262,209,375
Pawcatuck R	61	2001	A	N	25	8,229,707
Pawcatuck R	61	2016	D	N	11	1,059,373,416

The frequency of different least-cost management options selected also varied across watersheds (Figures D-3, D-4), and, to a lesser extent, across years. Among agricultural conservation practices, implementation of filter strips, cover crops and terracing with grassed waterways were common elements of solutions, while minimum tillage and nutrient management plans were also selected for the upper portions of the Connecticut River. Among urban stormwater BMPs, implementation of enhanced biofiltration with internal storage reservoir (ISR; Houle et al. 2017), gravel wetlands and infiltration basins (Upper and Upper Middle Connecticut River only) were common solutions. Permeable pavement with subsurface infiltration was a common solution across watersheds but implemented to a lesser extent because an upper constraint was set for implementation. (Permeable pavement is more practical for parking lots than for heavy use roadways.) Neither grassed nor forested buffers were included in least cost solutions. For the large watersheds evaluated, septic upgrades were included in some but not all years for all of the lower river segments except for the Pawcatuck, in part because we only considered septic upgrades within a one-year groundwater travel-time distance from the Long Island Sound shoreline. Septic solutions were more prevalent in the dry year, possibly because septic inputs comprised a larger component of loadings during these periods. Neither wastewater treatment plant upgrades nor septic to sewer conversions were included in any of the least-cost solutions evaluated.

Files listing BMP implementation locations can be joined to catchment shapefiles to show the distribution of least-cost BMP implementations. Shown below are the distributions of agricultural and urban BMPs, riparian buffer restorations and septic upgrades for the Lower Connecticut River for a 2003 (wet year) optimization (Figures D-7 to D-9). Only a 10% nitrogen load reduction could be achieved after least-cost solutions were implemented on most of the catchments in the watershed segment. Further details for each of the optimization runs can be found in the Preprocessing/Outputs folder of the RBEROST repository, including the .pdf screenshots of the graphical user interface showing result summaries and .csv files listing location of least cost solutions by catchment COMID, which can be joined to a catchment shapefile.

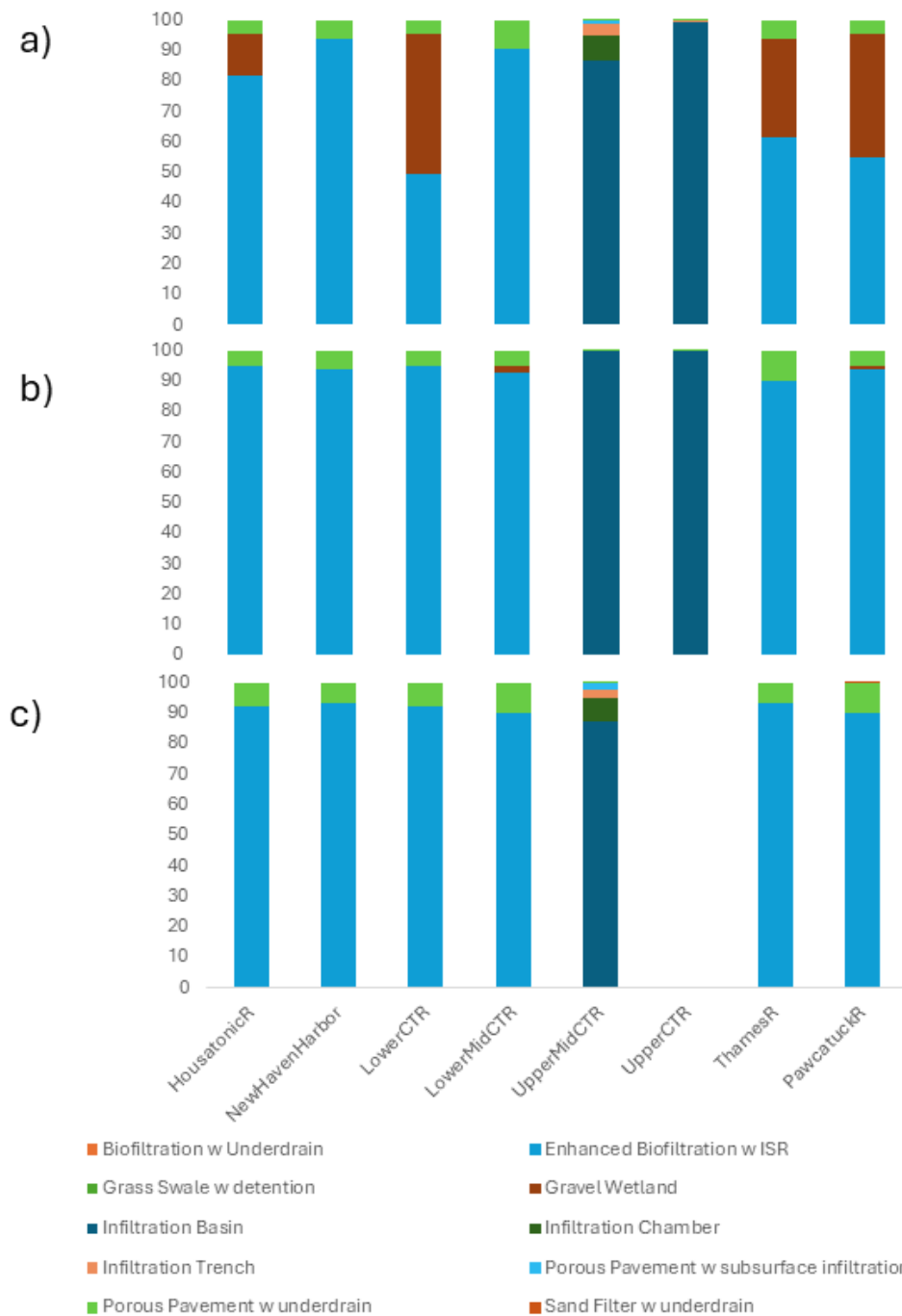


Figure D- 3. Percent of optimized urban area treated by stormwater BMP for a) average, b) dry and wet precipitation years for large watersheds in Long Island Sound Basin to meet largest N reduction possible less than or equal to proposed target.

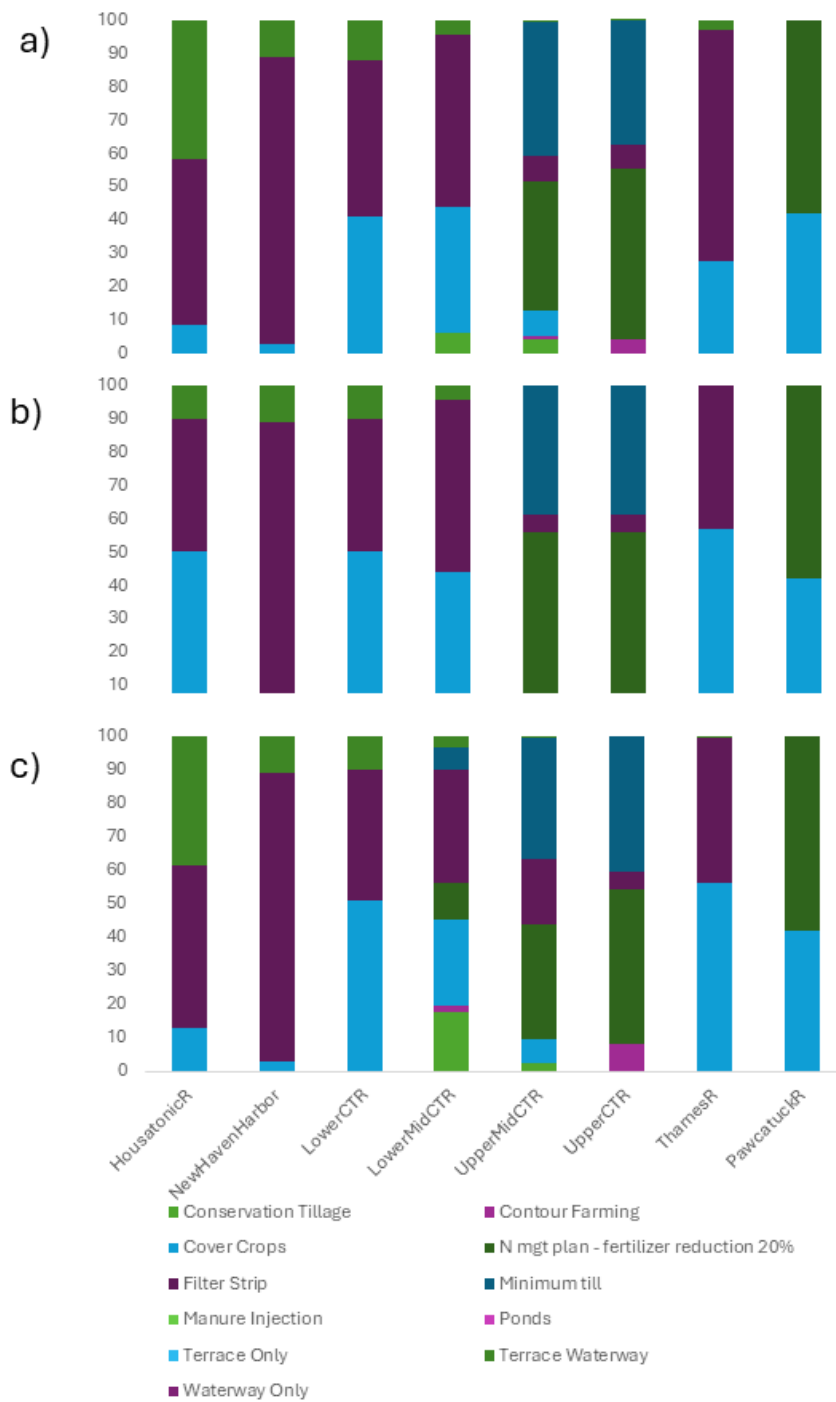


Figure D- 4. Percent of optimized treated agricultural area by agricultural conservation practice for a) average, b) dry and wet precipitation years for large watersheds in Long Island Sound Basin to meet largest N reduction possible less than or equal to proposed target.

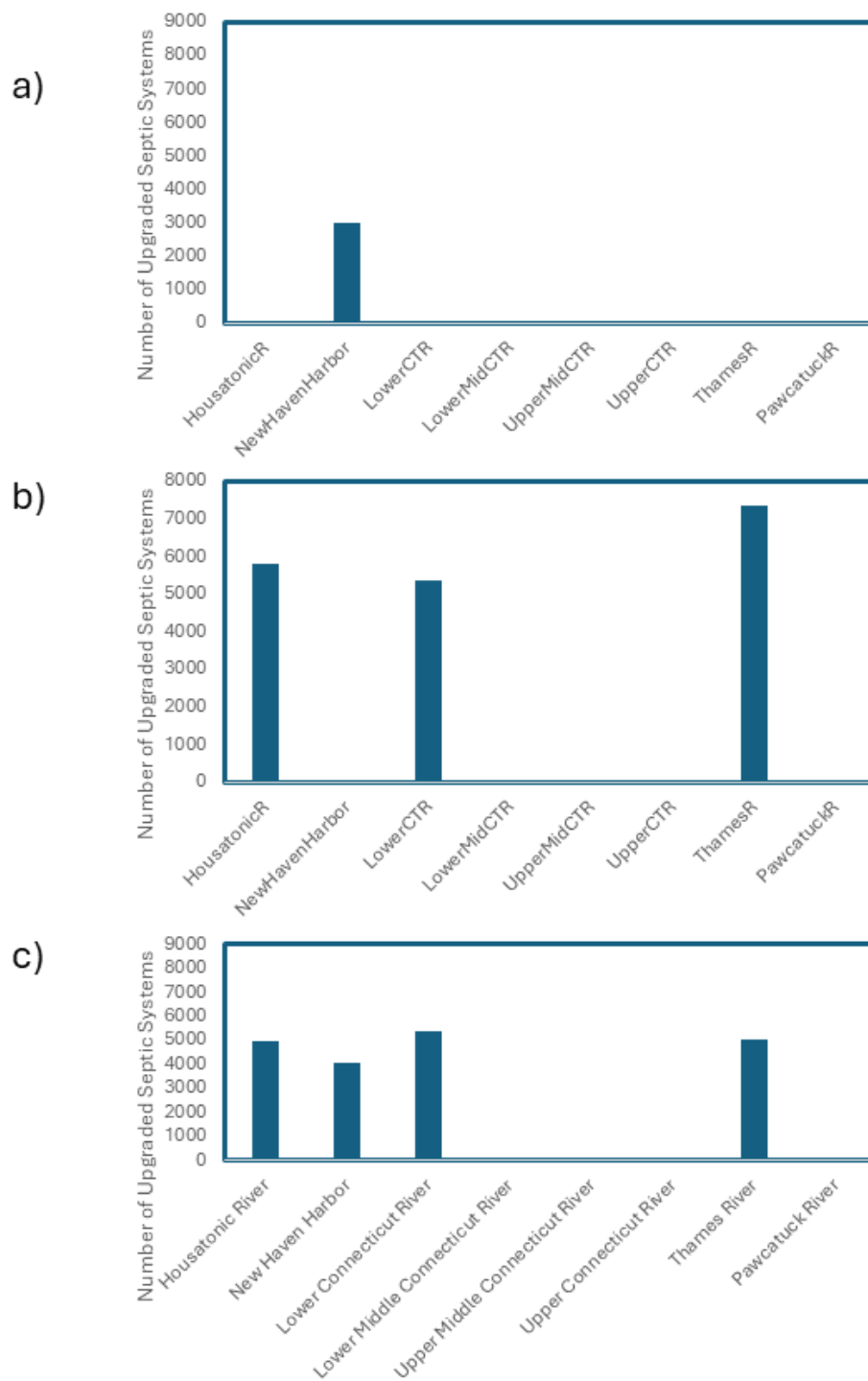


Figure D- 5. Optimized number of septic systems to be upgraded for a) average, b) dry and c) wet precipitation years for large watersheds in Long Island Sound Basin to meet largest N reduction possible less than or equal to proposed target.

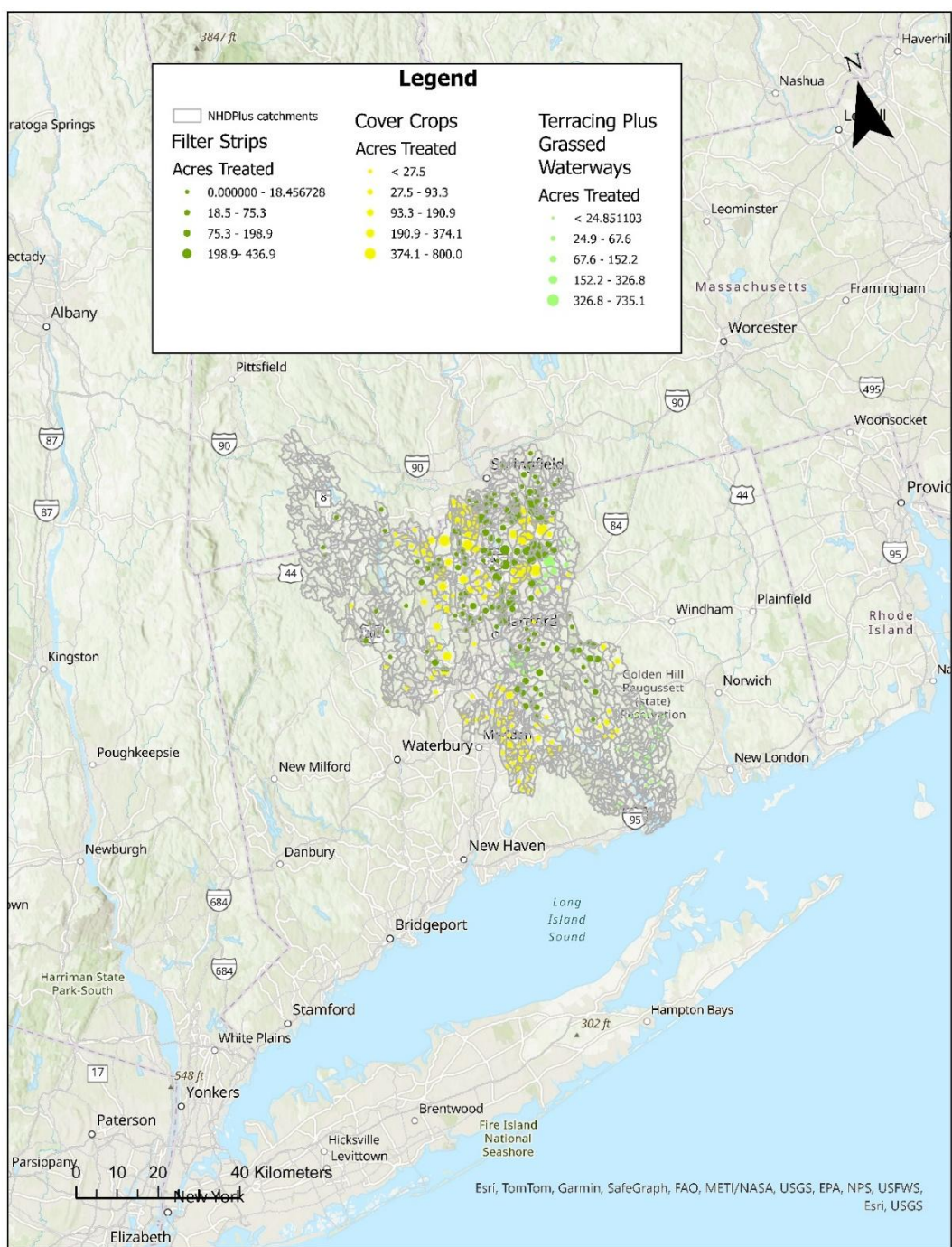


Figure D- 6. Optimal placement of agricultural conservation practices for Lower Connecticut River segment to achieve a 10 percent reduction in nitrogen loading to Long Island Sound for 2003, a wet year, including cover crops, filter strips and terracing with grassed waterways. Symbol size is proportional to agricultural acres treated although size range was restricted to ensure readability.

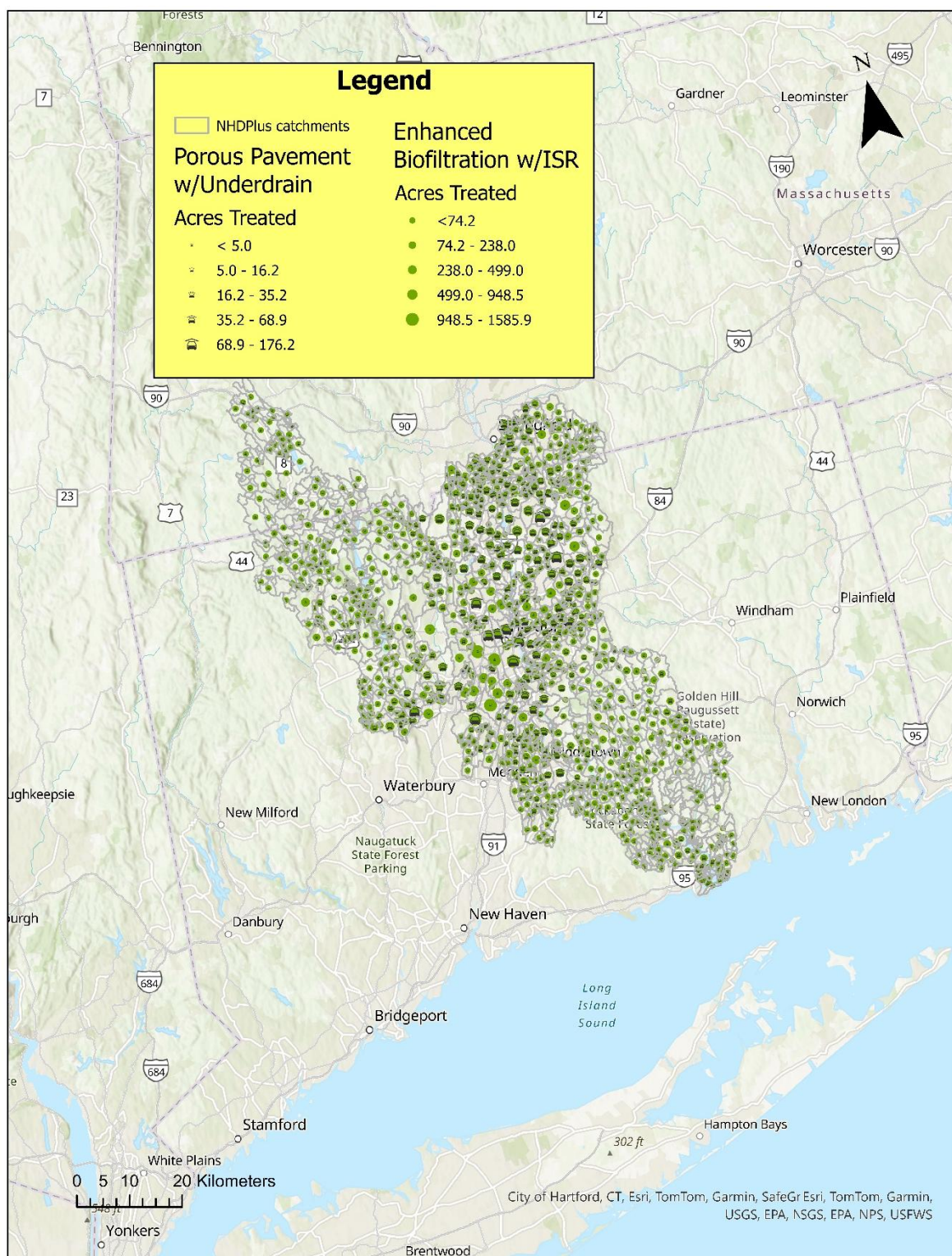


Figure D- 7. Optimal placement of urban stormwater BMPs for Lower Connecticut River segment to achieve a 10 percent reduction in nitrogen loading to Long Island Sound for 2003, a wet year, including porous pavement with underdrains (with a cap of 10% impervious area treated) and enhanced biofiltration with internal storage reservoir (ISR).

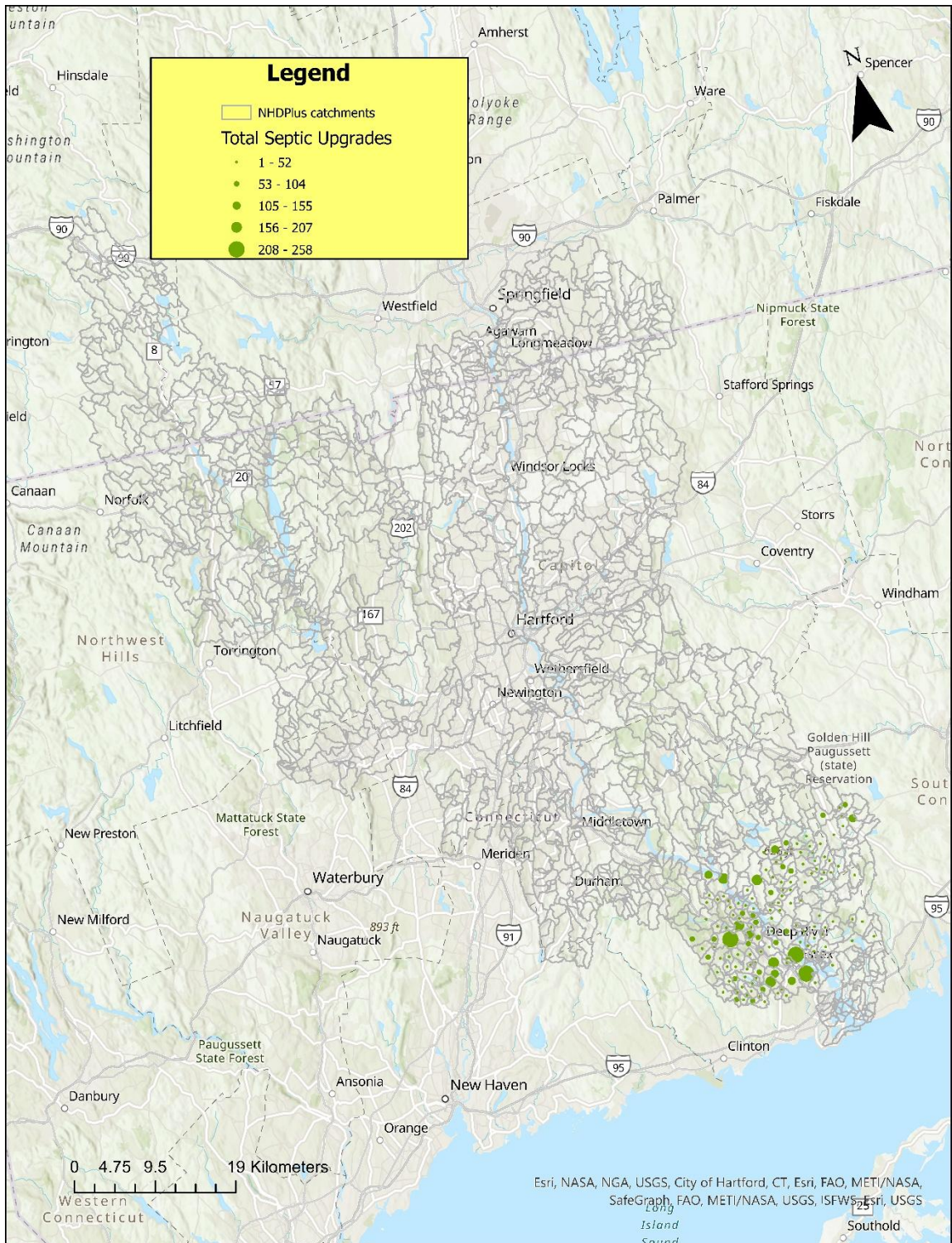


Figure D- 8. Optimal catchments for septic upgrades for Lower Connecticut River segment to achieve a 10 percent reduction in nitrogen loading to Long Island Sound for 2003, a wet year. Size of circles is proportional to number of septic upgrades per catchment.

Appendix E – Procedures for Updating Cost Estimates

In the future users may wish to update cost estimates for different management practices to a more recent time frame. The process varies for each of the four major cost categories: agricultural conservation practices, urban stormwater BMPs, septic upgrades and wastewater treatment plant upgrades.

The inflation factor for wastewater treatment plant upgrades was based on average changes in four producer price indices, Water Treating Compounds, Net Inputs to Construction Industries, Industrial Equipment and Piping and Tubing between December 2019 and December 2025, yielding an inflation factor of 1.46. The inflation factor for septic system upgrades between December 2019 and December 2025 was based on the PPI for Net Inputs to Residential Construction, Goods, or 1.43.

The updated cost of agricultural conservation practices was based on the 2026 NRCS Payment Schedules for EQIP, divided by 0.75, the fraction of costs paid by NRCS to farmers (per communications with state NRCS staff). There were some shifts in table structure and units between schedules published in 2019 and 2026, so we based costs on the updated 2026 structure. Some corrections had to be made for previous cost estimation practices due to the need to normalize costs to total agricultural acres treated rather than acres of a given ACP. Implementation units have to be converted to agricultural acres treated for use in RBEROST and this requires some assumptions (outlined in TableE1.xlsx in the Inputs directory).

The updated cost of stormwater BMPs was based on methods developed for the US EPA Stormwater Calculator that uses a weighted average of the following consumer price indices (available at <https://www.bls.gov/cpi/> from the Bureau of Labor Statistics): Concrete/Cement, Tractor Shovel Loader, Energy and Fuels/Utilities (Rossman and Bernagros, 2019). The calculated inflation factor between December 2019 and 2025 was 1.5.



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EPA/105/B-26/001I August 2026 I

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